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CS215: Data Analysis and Interpretation

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1

Variance

- Let's do a tiny experiment.
  - Please compute the mean and variance
  - a=[0, 0]
  - b=[0, 2]
  - c=[0, 4]

2

Sample Variance

Option	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x-\mu)^2}{n-1}$	Python (numpy)
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4

3

Variance

- Let's continue the tiny experiment.
  - Please compute the mean and variance
  - a=[0, 0]
  - b=[0, 2]
  - c=[0, 4]
  - d = [2 0] e = [2 2] f = [2 4]
  - g = [4 0] h = [4 2] i = [4 4]

4

Sample Variance

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\mu)^2}{n-1}$	$\frac{\sum(x-\mu)^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

5

Sample Variance

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\mu)^2}{n-1}$	$\frac{\sum(x-\mu)^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $\overline{s}^2 = \frac{12}{9} = \frac{4}{3}$
  - Dividing by (n-1) results in $\overline{s}^2 = \frac{24}{9} = \frac{8}{3}$
  - Division by n results in a $\frac{1}{2} \times \frac{8}{3}$

6

Example: Pulse-rate of 50 students

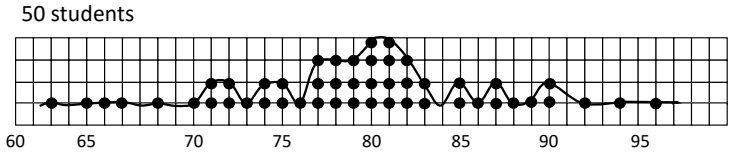
Can calculate statistic such as sample mean (79.1) and sample standard deviation (7.6) but we can do more!

Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let's plot these numbers

7

Plot of pulse rate

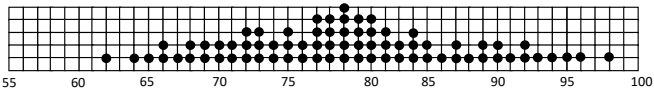


Example from Derek Rowntree

8

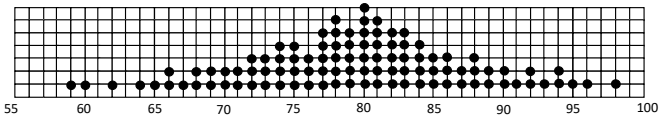
Plot of pulse rate as number increases

75 students



How do we know this?

100 students

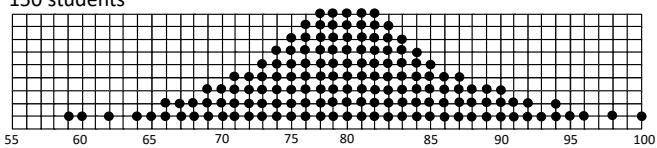


9

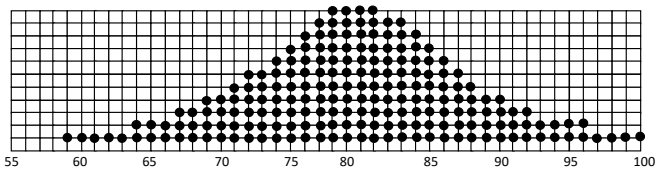
Plot is informally a distribution

The distribution approaches a symmetrical distribution

150 students



200 students



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And LLMs!!

1

Two questions from July 27

1. Could not understand what the course is about from yesterday's lecture.
2. Potential confusion on the example provided

2

First my checklist

- Announce piazza and Task00 due.
- Mentimeter results

3

What is the course about?

1. Could not understand what the course is about from yesterday's lecture.
2. I could give you the list of topics … (next slide)

4

Topics

- Distributions and their properties
- Linear Regression
- Parameter Estimation
- Hypothesis Testing, Confidence Intervals
- Sampling, Multivariate Distributions
- Visualization?
- Others

5

What is the course about?

1. I could give you the list of topics … (next slide)
2. But I decided to create a separate set of slides
    1. A sample!

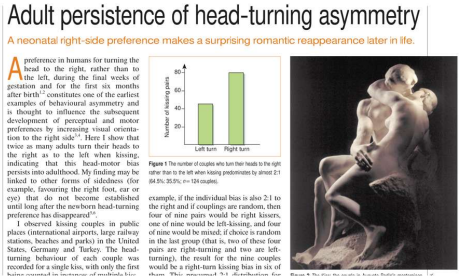
6

Nature: The scientific publication

- From Wikipedia
  - ***Nature*** is a weekly international scientific journal publishing peer-reviewed research across the natural sciences, including biology, physics, chemistry, the earth sciences, and related interdisciplinary fields. *Nature* had one of the highest impact factors among multidisciplinary science journals (50.5), reflecting its strong citation influence within the scientific literature;[1] some commentators also regard it as **among the most influential** scientific journals worldwide.[2][3][4]
  - In 2007, *Nature* (together with *Science*) received the Prince of Asturias Award
  - As of 2012, it claimed an online readership of about three million unique readers per month.[7]

7

What did Nature publish in 2003?



Adult persistence of head-turning asymmetry

A neonatal right-side preference makes a surprising romantic reappearance later in life.

A preference in humans for turning the head to the right, rather than to the left, during the final weeks of gestation and for the first six months after birth[1,2] constitutes one of the earliest examples of behavioural asymmetry and is thought to influence the subsequent development of perceptual and motor preferences by increasing visual orientation to the right side[3,4]. Here I show that twice as many adults turn their heads to the right as to the left when kissing, indicating that this head-motor bias persists into adulthood. My finding may be linked to other forms of sidedness (for example, favouring the right foot, ear or eye) that do not become established until long after the newborn head-turning preference has disappeared[5,6].

I observed kissing couples in public places (international airports, large railway stations, beaches and parks) in the United States, Germany and Turkey. The head-turning behaviour of each couple was recorded for a single kiss, with only the first

Figure 1 The number of couples who turn their heads to the right rather than to the left when kissing predominates by almost 2:1 (64.5%; 95.5%; $n = 124$ couples).

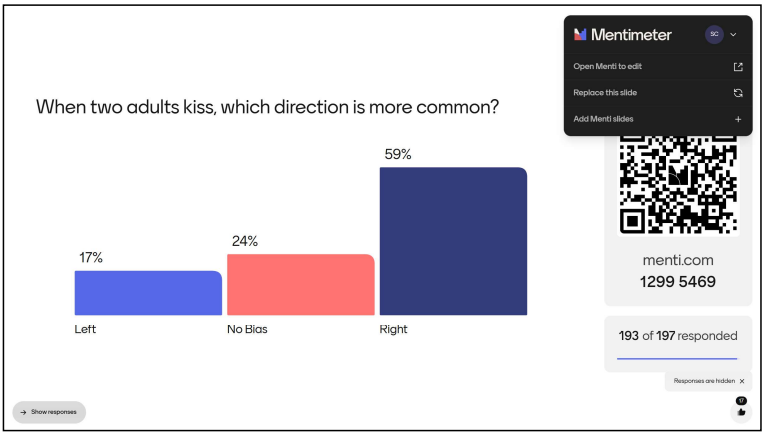
example, if the individual bias is also 2:1 to the right and if couplings are random, then four of nine pairs would be right kissers, one of nine would be left-kissing, and four of nine would be mixed; if choice is random in the last group (that is, two of these four pairs are right-turning and two are left-turning), the result for the nine couples would be a right-turn kissing bias in six of

8

Audience poll

- You do not need experience to answer
- Just gut feeling is enough

9



10

Someone actually watched

- Onur Gunturkun observed kissing couples in public places

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Data != conclusion

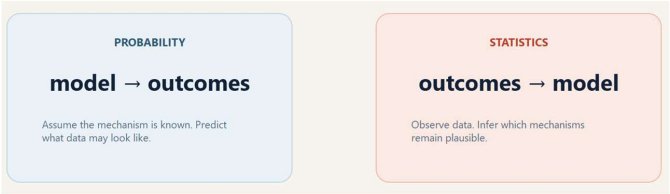
- We expect 50%
  - Is that a signal?
  - Or chance?
- This course goes backward: data -> plausible truth

WHAT WE SAW	WHAT WE WANT
$\hat{p} = 0.645$	$p = ?$
One sample of 124 observed couples	The underlying long-run tendency

12

Statistics and Probability

- Same bridge. Opposite directions



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Binomial distribution

- Right =1
- Want $\hat{p}$ = average of the 0s and 1s = proportion turning right
- Proportion is a sample mean in disguise

14

Inference has a price tag

- The model
  - Binary outcome
- Common tendency
  - Each observation shares on parameter p
- Independence
  - One observed couple does not determine the other

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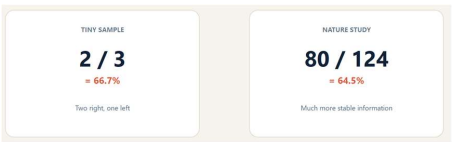
Point estimate

- A point estimate can be reliable
  - Or not
- Question for this course
  - How large is large enough?



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Quantify uncertainty

- How surprising is 80 out of 124?
- Key starting point: No-bias benchmark of p=0.5

IF THERE WERE NO POPULATION-LEVEL BIAS

Expected right turns = 62

Observed = 80 • difference = 18

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What is the course about?

Welcome to CS215!

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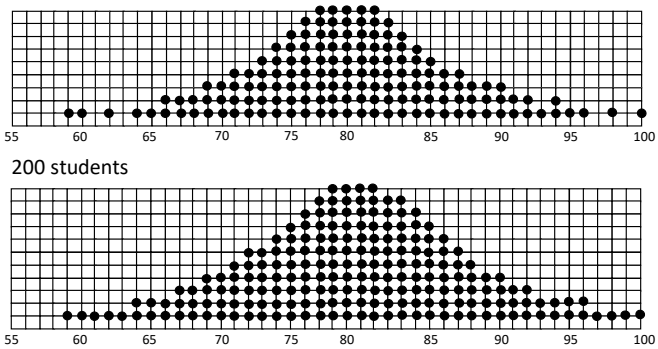
Agenda

- Selected slides from this week
  - More traditional description
- Time permitting: Grading, Evaluation, Academic Honesty, Deadline extension, “what if I missed”, …

1

Pulse Rate Plot

The distribution approaches a symmetrical distribution



2

FOUNDATIONS

Population and Observations

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

observe

SAMPLE

Observed values: $X_1, \ldots, X_n$

Statistics: $\bar{x}$ and S^2
Change from sample to sample

$$\overline{x} = \frac{\sum x_i}{n} \qquad \frac{\sum (x_i - \mu)^2}{n-1}$$

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

3

Probability Space

- A set S of all possible outcomes of a random experiment
- Event: A subset of the sample space ("an observation")
- Example:
  - An experiment whose outcome is not known in advance: A coin toss
  - We do know the complete set of possible outcomes – in this case Heads or Tails.
  - If E = {H}, then E is the event that the coin produced a heads.

4

Probability Space

- A set S of all possible outcomes of a random experiment
- Event: A subset of the sample space ("an observation")
- Probability:
  - A non-negative number associated with an event X
  - For any sequence of mutually exclusive events $X_1, X_2, \ldots$ and the union Y, $P(Y)$ = sum of individual probabilities
  - P(S) =1
- Space: The triple $\{S, E, P\}$
  - P is often written as $\Pr$

Pro Tip: Mutually Exclusive Events != Independent Events

5

Random Variables

- In many random experiments we care not about the observed values themselves, but about some numerical quantity
  - Example: the sum of two dice throws, or the number of heads in n consecutive coin tosses
- Any such quantity determined by the result of a random experiment is a *random variable*
  - Technically: A random variable is a function defined on a probability space $\{S, E, P\}$
  - $X: S \rightarrow V$ where V is a set of numerical values, e.g., set of integers or real numbers
- Random variables can be discrete or continuous

6

6

Distribution function

- A random variable is *synonymous* with a cumulative distribution function, or simply a distribution function
  - We define the cumulative distribution function (cdf) $F_X(x) = \Pr\{X \leq x\}$ (x is real)
  - Specifying the cdf is like implicitly defining the random variable
- Notation: $X \sim F_X$
- All probability questions about X can be answered in terms of F
  - Example to follow

7

Discrete Random Variables

- Random variables whose values form a finite or countably infinite sequence are discrete random variables.
  - Example: the outcome of a coin toss or a dice throw.
- The probability mass function $p_X(x) = P(X = x)$
- Example: Sum of two dice throws

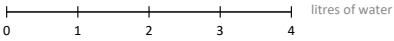
x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

8

Continuous Random Variable

- Random variables that take values across a continuum are continuous random variables.
  - Ex: Amount of water in a 4-litre jar: continuous random variable on [0, 4]
- Unlike a discrete random variable, the probability of any **particular** value in the continuum is **zero**.
  - There are infinitely many values in [0, 4], and each is equally likely.

Basic "first order"concept



9

9

Continuous Random Variable

- *Formally*, X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x)\, dx$$

Pro Tip: This seemingly roundabout way is to get over the "difficulty" mentioned in the previous slide

10

Continuous random variable: Example

Consider a cdf of the form

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x^2}, & x > 0 \end{cases}$$

The probability that *X* exceeds 1

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1) = e^{-1}$$

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11

Continuous Random Variable

- *Formally*, X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x)\, dx$$

- The function $f(x)$ is called the probability density function (pdf)

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Continuous Random Variable

- *Formally*, X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

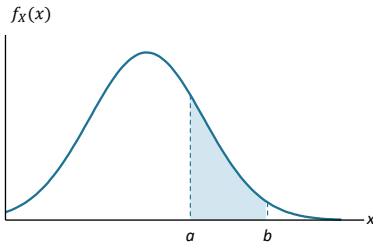
$$\Pr\{X \in C\} = \int_{x \in C} f(x)\,dx$$

- The function $f(x)$ is called the probability density function (pdf)
- The relationship between f and F:

$$F(a) = \Pr\{X \in (-\infty, a]\} = \int_{-\infty}^{a} f(x)\,dx$$

13

pdf



The shaded area between *x* = *a* and *x* = *b* is the cumulative interval measure

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Over an infinitesimal width *dx*, the quantity

$$f_X(a)\,dx$$

is the probability that *X* lies between *a* and *a* + *dx*.

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14

Probability density function (pdf)

The pdf of *X* at *x* is the **derivative** of its cdf at *x*.

$$f_X(x) = \frac{d}{dx} F_X(x)$$

It is a non-negative function such that for any set *B* of real numbers

$$P\{X \in B\} = \int_B f_X(x)\, dx$$

15

15

Probability Density

- The probability density function (pdf) is the derivative of the cumulative distribution function (cdf) at that value x

$$f_X(a) = \lim_{\epsilon \to 0} \frac{\Pr\left\{a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}\right\}}{\epsilon}$$

$$\Pr\left\{a - \tfrac{\epsilon}{2} \le X \le a + \tfrac{\epsilon}{2}\right\} \approx \ \epsilon f(a) = \int_{a-\frac{\epsilon}{2}}^{a+\frac{\epsilon}{2}} f_X(x)\, dx$$

Pro Tip: Notice how we get probability of a particular value=0 as desired

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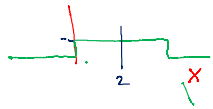
Expectation



For a **discrete** random variable *X*:

$$E(X) = \sum_i x_i \, P(X = x_i)$$

For a **continuous** random variable *X*:

$$E(X) = \int_{-\infty}^{+\infty} x \, f_X(x) \, dx$$

Expected value of X is also called *mean* and represented by the symbol μ

17

17

Expected value of a function of a random variable

For a function *g*(*X*) of a **discrete** random variable $E(g(X)) = \sum_i g(x_i) P(X = x_i)$

For a **continuous** random variable $E(g(X)) = \int_{-\infty}^{+\infty} g(x) f_X(x) \, dx$

Expected value of $(X - \mu)^2$ is called *variance* and represented by the symbol σ^2

18

18

FOUNDATIONS

Population and variance

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

- Suppose someone (an "oracle") told you that values are
  - Only ($X = 0$ or $X = 2$ or $X = 4$)
  - What's the mean and variance?
  - Expected value: $\mu = 2$
  - Variance: $\sigma^2 = 8/3$

1

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TWO DIFFERENT CALCULATIONS

Oracle knowledge vs sample estimation

QUESTION	ORACLE: POPULATION KNOWN	STATISTICIAN: ONLY A SAMPLE
Information	P(X=0)=P(X=2)=P(X=4)=1/3	Observed sample: $X_1, \ldots, X_n$
Center used	True mean μ = 2	Sample mean $\bar{x}$
Quantity	$\sigma^2 = E[(X-\mu)^2]$	$S^2 = \Sigma(X_i-\bar{x})^2/(n-1)$
Nature	A fixed population parameter	A random estimator; it varies
Result here	$\sigma^2 = (4+0+4)/3 = 8/3$ exactly	$E[S^2] = 8/3$ across repeated samples

Same number, different roles: 8/3 is the oracle target and the estimator long-run average.

POPULATION + SAMPLE + INFERENCE

2

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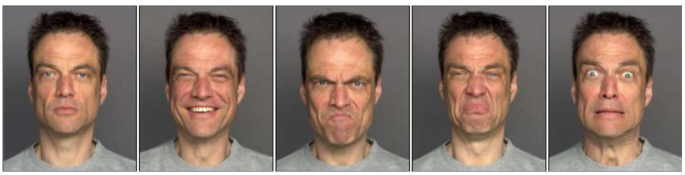
1

Agenda

- Last time
  - Random variables
  - *Expectation* of a random variable
  - Expectation of a function of a random variable
- Today
  - More on random variables
  - Time permitting: Grading, Evaluation, Academic Honesty, Deadline extension, "what if I missed", …

2

Why should this work?



Control of the complicated pdf of a face using an artificial neural network

3

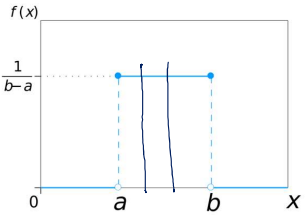
Uniform

Canonical

$p_X + f_Y = p(x+y)$?

0.3
0.9

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$



If X is uniform between 0 and 1, and Y is identical, what is the distribution of X+Y?

4

4

Random variable: notation

- A random variable is denoted by an upper case letter. $P(X = x)$
- The individual values it can take are denoted by the matching lower case letter.
  - The probability that the random variable X takes the value x

5

5

Expectation

For a **discrete** random variable *X*:

$$E(X) = \sum_i x_i \, P(X = x_i)$$

For a **continuous** random variable *X*:

$$E(X) = \int_{-\infty}^{+\infty} x \, f_X(x) \, dx$$

Expected value of X is also called mean and represented by the symbol μ

- The expected value when you throw a die is 1/6 (1+2+3+4+5+6) = 3.5
- We will never get 3.5 when we throw a dice

6

6

Function of a random variable

For a function $g(X)$ of a **discrete** random variable $E(g(X)) = \sum_i g(x_i)P(X = x_i)$

For a **continuous** random variable $E(g(X)) = \int_{-\infty}^{+\infty} g(x) f_X(x)\, dx$

7

7

Illustration (Discrete)

$X = 0, 1, 2$

$0.2 \quad 0.5 \quad 0.3$

$Y = X^2$

$0, 1, 4$

$P_Y = 0.2 \quad 0.5 \quad 0.3$

$E[Y] = \sum g(x) P_{x_i} = 0 \cdot 0.2 + 1 \cdot 0.5 + 4 \cdot 0.3$

$= \sum y P_{y_i} =$

Continuous

$Z = X^3$

$E[Z] = \int z f_Z \, dz$

8

Illustration (Continuous)

$P_Z = P_X \left|\frac{dx}{dz}\right|$ $\quad Z = X^3$ $\quad \int z p_z dz = \int g(w) p_x dx$

$$F_Z(w) = P[Z \le w] = P[X^3 \le w] = P[X \le w^{1/3}]$$

$$= w^{1/3}$$

$pdf_Z(w) = p(w)$

$$= \frac{1}{3} w^{-2/3}$$

$$\therefore E[Z] = \int w \cdot \frac{1}{3} w^{-2/3} dw = \frac{1}{4}$$

$$E[Z] = \int z \, pdf_x \, dx$$

$$= \int x^3 dx$$

$$= 1/4$$

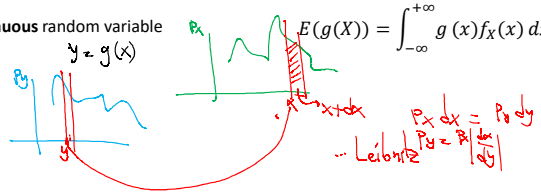
9

Function of a random variable

$Y = g(X)$ $\quad E[Y] = \sum y p_y$

For a function $g(X)$ of a **discrete** random variable $\quad E(g(X)) = \sum_i g(x_i) P(X = x_i)$

For a **continuous** random variable $\quad E(g(X)) = \int_{-\infty}^{+\infty} g(x) f_X(x) \, dx$



$p_x dx = p_y dy$

$\therefore$ Leibniz $p_y = p_x \left|\frac{dx}{dy}\right|$

10

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Expected value

Kalshi

- Roulette: A ball settles in one of 38 numbered pockets
  - Guess the right number and you win INR 35 (probability 1/38);
  - Otherwise you lose INR 1 (probability 37/38).
- Expected earnings per trial
  - (−1)(37/38) + (35)(1/38) = −0.0526
- The house edge is exactly this negative expectation.

13

13

Expected Value May Be Technically Undefined

$$P(X = x) = k/x^2 \text{ for } x \geq 1,\ x \in \mathbb{Z}^+$$

$$\sum_{x=1}^{\infty} P\,(X = x) = \sum_{x=1}^{\infty} k\,/x^2 = 1 \text{ if } k = 6/\pi^2$$

$$E(X) = \sum_{x=1}^{\infty} x\,P(X = x) = k \sum_{x=1}^{\infty} 1\,/x = \infty$$

14

14

CS215: Data Analysis and Interpretation

www.cse.iitb.ac.in/~sharat

(Many slides obtained from Prof. Ajit and Prof. Sunita and gratefully acknowledged. Fair use of internet resources as available to professors under copyright act)

1

Agenda

- Last time
  - Expectation of a function of a random variable
  - Leibnitz rule
  - Pareto principle
  - Course logistics
- Today
  - Linearity of expectations
  - Joint distributions

2

Expected Value May Be Technically Undefined

$$P(X = x) = k/x^2 \text{ for } x \geq 1,\ x \in \mathbb{Z}^+$$

$$\sum_{x=1}^{\infty} P\,(X = x) = \sum_{x=1}^{\infty} k\,/x^2 = 1 \text{ if } k = 6/\pi^2$$

$$E(X) = \sum_{x=1}^{\infty} x\,P(X = x) = k \sum_{x=1}^{\infty} 1\,/x = \infty$$

3

3

Pareto principle

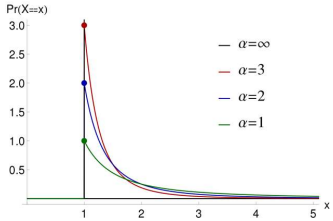
- The Pareto rule of thumb
  - 20% of pea pods contained 80% of the peas
  - 1992 United Nations Development Program Report
    - Richest 20% of the world's population receives 82.7% of the world's income
- A small fraction of causes often produces a large fraction of effects
  - Many things in life are not distributed evenly

4

4

Pareto distribution

- Alpha = 1.16 exhibits the Pareto principle
- The type I Pareto distribution is characterized by a scale parameter x_m and a shape parameter α, which is known as the *tail index*.
- The expectation is technically undefined in some cases



$$f_X(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}} & x \geq x_m, \\ 0 & x < x_m. \end{cases}$$

$$\overline{F}(x) = \Pr(X > x) = \begin{cases} \left(\frac{x_m}{x}\right)^\alpha & x \geq x_m, \\ 1 & x < x_m, \end{cases}$$

$$\mathrm{E}(X) = \begin{cases} \infty & \alpha \leq 1, \\ \frac{\alpha x_m}{\alpha - 1} & \alpha > 1. \end{cases}$$

5

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Linearity of Expectations

- A function f is linear if f(ax) = af(x) and f(x+y) = f(x) + f(y)
- If f(x) = af(x) + b, then the function is affine, but commonly termed as linear
- Expectation is not a function but an operator — it takes a function as input — and it is linear in the same sense.

Typo
f(x) = ax+b
Aug 06 has correct form

$$E(a\,g(X) + b) = \int_{-\infty}^{+\infty} (a\,g(x) + b) f_X(x)\,dx$$

Why is this useful?

$$= \int_{-\infty}^{+\infty} a\,g(x) f_X(x)\,dx + \int_{-\infty}^{+\infty} b\,f_X(x)\,dx$$

$$= a\,E(g(X)) + b$$

6

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Mean squared prediction error

- You want to predict the value of a random variable X
  - Suppose we predict it as c
  - What is the error? Maybe $(X - c)$?
- Define the error to be the mean squared error
- Equality holds exactly when c = μ
  - The expected value is the predictor with least mean squared error.

Recall "sample mean"

$$E((X-c)^2) = E((X-\mu+\mu-c)^2)$$
$$= E((X-\mu)^2 + (\mu-c)^2 + 2(X-\mu)(\mu-c))$$
$$\geq E((X-\mu)^2)$$

7

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Absolute value

- What minimizes the quantity $J(c)$
  - The median
  - The median need not be unique

$$J(c) = \int_{-\infty}^{+\infty} |x-c|\, f_X(x)\, dx$$

8

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Variance

- Variance tells you how much the values of X deviates from the mean, on the average.
- The positive square root of the variance is the standard deviation
  - Low-variance densities concentrate around a point
  - High-variance densities are spread out

$$\mathrm{Var}(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx$$

$$\mathrm{Var}(X) = E[(X-\mu)^2] = \sum_i (x_i-\mu)^2 P(X=x_i)$$

9

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Variance may be technically undefined

- For some distributions, the defining integral has no finite value, so the variance is undefined.
  - Example: the Pareto distribution with $\alpha < 2$.
- In some cases, the mean is defined but the variance is not
- In some cases, both are undefined.

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Hand Calculation of Variance

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2 + \mu^2 - 2X\mu] = E[X^2] + \mu^2 - 2E[X]\mu = E[X^2] + \mu^2 - 2\mu^2$$
$$= E[X^2] - \mu^2 = E[X^2] - (E[X])^2$$

- The middle step uses $E[X] = \mu$
  - This form — mean of the square minus square of the mean — is usually the easier one to compute with.
- Variance when you throw a fair dice
  - Earlier: expected value was 7/2
  - Option 1: $[(1-7/2)^2 + (2-7/2)^2 + \cdots (6-7/2)^2]\ (1/6)$
  - Option 2: $E[X^2] = 1/6(1+4+9+16+25+36) = 91/6$, Var = $91/6 - 49/4$

Recall "sample variance"

11

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Variance

- Shifting a random variable does not change its variance
- Scaling it scales the variance by the square.

$$\mathrm{Var}(aX+b) = E[(aX+b-E(aX+b))^2] = E[(aX+b-(a\mu+b))^2]$$
$$= E[a^2(X-\mu)^2] = a^2E[(X-\mu)^2] = a^2\mathrm{Var}(X)$$

12

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Jointly distributed random variables

- In machine learning one often needs to model relationships between two or more random variables.
  - CPI at IITB and the salary offered during placements
  - Hours spent on Instagram each day and hours of sleep at night.
  - Literacy level and crime rate
  - Price of a product, peer feedback and probability that a user clicks "Buy Now"
- If we observe multiple paired instances of X and Y, then some combinations of outcomes are more likely than others
  - Random variables rarely live alone. In real data, quantities interact, move together, or constrain one another.
  - Joint distributions captures in a mathematical way these relationships

13

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Joint Distribution function

- Recall: A random variable is synonymous with a cumulative distribution function, or simply a distribution function
  - We define the cumulative distribution function (cdf) $F_X(x) = \Pr\{X \leq x\}$
- The relationship between two random variables is defined by the joint cumulative probability distribution function

$$\mathrm{G}(x, \mathrm{y}) = \Pr\{X \leq x, \mathrm{Y} \leq \mathrm{y}\}$$

- All probability questions about X and Y can be answered in terms of G, for example

$$F_X(x) = \Pr\{X \leq x\} = \Pr\{X \leq x, Y < \infty\} = G(x, \infty)$$

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Joint pmfs

For discrete random variables *X* and *Y*, the **joint probability mass function** is

$$p_{X,Y}(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

The **pmf** of X follows by summing out the other

$$P\{X = x_i\} = P\left(\bigcup_j \{X = x_i,\ Y = y_j\}\right)$$

$$= \sum_j P\{X = x_i,\ Y = y_j\}$$

$$= \sum_j p(x_i, y_j)$$

This is called marginalizing

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three (and that's it)
  - Male and female children are equally likely and independent.
  - Let B (G) denote random variables for boys (girls)
  - What is the joint pmf p(B, G)
- No children? P(B = 0, G = 0) = 0.15
- One girl, no boy? P(B = 0, G = 1) = 0.2 × 0.5 = 0.1

16

16

Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three. What is the joint pmf p(B, G)
- Two boys and one girl? P(B = 2, G = 1) = 0.3 × (1/8) × 3 = 0.1125
  - All 8 combinations of three children are equally likely; three of them have two boys and one girl.

$i \backslash j$	0	1	2	3	Row Sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column Sum $= P\{G = j\}$	.3750	.3875	.2000	.0375	

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Linearity of Expectations

- A function f is linear if f(ax) = af(x) and f(x+y) = f(x) + f(y)
- If y = ax + b, then the function is affine, but commonly termed as linear
- Expectation is not a function but an operator — it takes a function as input — and it is linear in the same sense.

$$E(a\,g(X)+b) = \int_{-\infty}^{+\infty} (a\,g(x)+b) f_X(x)\,dx$$

$$= \int_{-\infty}^{+\infty} a\;g(x) f_X(x)\,dx + \int_{-\infty}^{+\infty} b\;f_X(x)\,dx$$

Must know

$$= a\,E(g(X)) + b$$

3

3

Variance

- Shifting a random variable does not change its variance
- Scaling it scales the variance by the square.

$$\mathrm{Var}(aX+b) = E[(aX+b-E(aX+b))^2] = E[(aX+b-(a\mu+b))^2]$$

$$= E[a^2(X-\mu)^2] = a^2 E[(X-\mu)^2] = a^2\mathrm{Var}(X)$$

Must know

4

4

Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three (and that's it)
  - Male and female children are equally likely and independent.
  - Let B (G) denote random variables for boys (girls)
  - What is the P(B = 0) ?

5

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Example

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  - Male and female children are equally likely and independent.
  - Let B (G) denote random variables for boys (girls)
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6

6

Example

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3	.0375	0	0

$i\backslash j$	0	1	2	3	Row Sum $= P\{B = i\}$
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3	.0375	0	0	0	.0375
Column Sum $= P\{G = j\}$	.3750	.3875	.2000	.0375	

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Continuous Random Variables

- Recall: X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x)\,dx$$

- We say that X and Y are jointly continuous if there exists $g(x, y)$ having the property that for any set C of pairs of real numbers

$$P\{(X, Y) \in C\} = \iint_{(x,y) \in C} g_{X,Y}(x, y)\,dx\,dy$$

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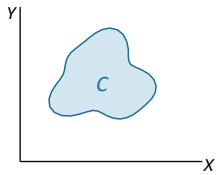
Continuous Random Variables

- The function $g(x, y)$ is called the joint probability density function (pdf)
- The joint probability that (X, Y) falls in any region of the plane is the integral of the joint pdf over that region.



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Example

- A student has four hours of free time in the evening.
- $X =$ hours spent gaming, $Y =$ hours spent streaming videos
- The feasible values satisfy $X \geq 0, \quad Y \geq 0, \quad X + Y \leq 4$
- Suppose all feasible pairs are equally likely
  - $f_{X,Y}(x, y) = \begin{cases} \frac{1}{8}, & x \geq 0, y \geq 0, x + y \leq 4 \end{cases}$
  - $f_X(x) = \int_0^{4-x} f_{X,Y}(x, y) dy = \int_0^{4-x} \frac{1}{8} dy = \frac{4-x}{8}, \quad 0 \leq x \leq 4$
  - Marginalization adds the probability density along a vertical slice of the joint-density region

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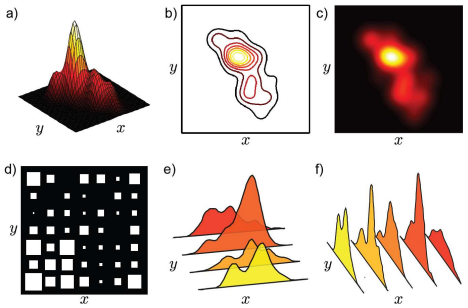
Visualizing Joint Probability



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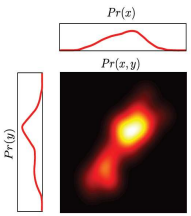
Marginalization

- We can recover probability distribution of any variable in a joint distribution by integrating (or summing) over the other variables
- The marginal pdf of one variable is obtained by integrating the joint pdf over the other:

$$Pr(x) = \int Pr(x,y)\, dy$$

$$Pr(y) = \int Pr(x,y)\, dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy$$



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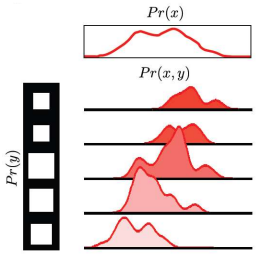
Marginalization

$$Pr(x) = \sum_{y} Pr(x,y)$$
$$Pr(y) = \int Pr(x,y)\, dx$$



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Marginalization

- Works in higher dimensions as well
  - leaves joint distribution between whatever variables are left

$$Pr(x,y) = \sum_{w} \int Pr(w,x,y,z)\, dz$$

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Expected Value

- Expected values of sums of random variable
  - E[X] + E[Y] = E [X+Y]
- Example: Expected value of the sums of two (fair) dice
  - Option 1: pmf: (1/36)[1 2 3 4 5 6 5 4 3 2 1]
    - Expectations: 2*1/36 + 3*2/36 + … + 12(1/36) = 252/36 = 7
  - Option 2: Name the dice as X_1 and X_2
- This is a very useful result, particularly because it makes no assumptions on X and Y
- This is a consequence of the general result

$$E\big(g(X,\mathrm{Y})\big) = \int \int_{-\infty}^{+\infty} g\,(x,y) f_{X,\mathrm{Y}}(x,\mathrm{y})\,\mathrm{dy}\,dx$$

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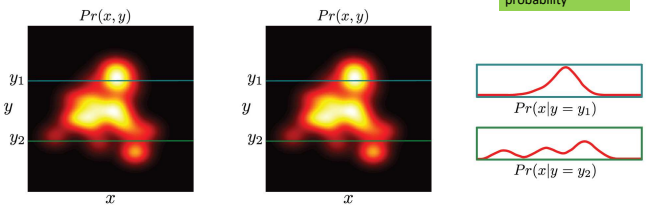
Slice of the joint pdf

- Earlier we take the sum of the probabilities to recover the marginal distribution
- What happens if we take a slice of the joint pdf as is?

Proportional, not a probability



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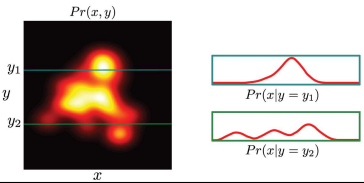
Conditional Probability

- Conditional probability can be extracted from joint probability
- Extract appropriate slice and also normalize

$$Pr(x|y=y^*) = \frac{Pr(x, y=y^*)}{\int Pr(x, y=y^*)dx} = \frac{Pr(x, y=y^*)}{Pr(y=y^*)}$$



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Conditional Probability

- More usually written in compact form

$$Pr(x|y) = \frac{Pr(x,y)}{Pr(y)}$$

- Can be re-arranged

$$Pr(x,y) = Pr(x|y)Pr(y)$$
$$Pr(x,y) = Pr(y|x)Pr(x)$$

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Conditional Probability

- This idea can be extended to more than two variables

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$\begin{aligned} Pr(w, x, y, z) &= Pr(w, x, y|z)Pr(z) \\ &= Pr(w, x|y, z)Pr(y|z)Pr(z) \\ &= Pr(w|x, y, z)Pr(x|y, z)Pr(y|z)Pr(z) \end{aligned}$$

20

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Reorganizing

$$Pr(x, y) = Pr(x|y)Pr(y)$$
$$Pr(x, y) = Pr(y|x)Pr(x)$$

Combining:

$$Pr(y|x)Pr(x) = Pr(x|y)Pr(y)$$

Re-arranging:

$$\begin{aligned} Pr(y|x) &= \frac{Pr(x|y)Pr(y)}{Pr(x)} \\ &= \frac{Pr(x|y)Pr(y)}{\int Pr(x, y)\, dy} \\ &= \frac{Pr(x|y)Pr(y)}{\int Pr(x|y)Pr(y)\, dy} \end{aligned}$$

21

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Bayes' Rule Terminology

Likelihood – propensity for observing a certain value of x given a certain value of y

Prior – what we know about y before seeing x

$$Pr(y|x) = \frac{Pr(x|y)Pr(y)}{\int Pr(x|y)Pr(y)\ dy}$$

Posterior – what we know about y after seeing x

Evidence –a constant to ensure that the left hand side is a valid distribution

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Agenda

- Last time
  - Expected values of sums of random variables
  - Joint distributions
  - Conditional probability
- Today
  - Independence
  - Covariance

i.i.d

P.C.A

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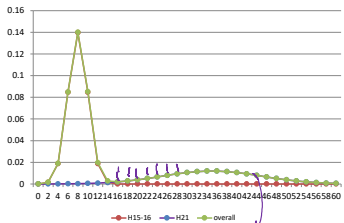
Mixture

- Discrete:
  - Z = 0: Student residence is H7-8-21
  - Z = 1: Student residence is H15-16
- Continuous: X: travel time to (New) SAC (or 9Tea7 Shach)
- $f_{X,Z}(x,z) = P(Z = z) f_{X|Z}(x \mid z)$



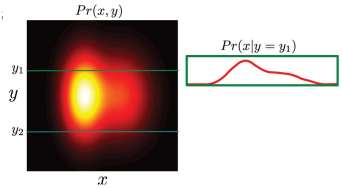
How do we interpret this graph?

5

Independence

- If two variables X and Y are independent, then variable X tells us nothing about variable Y (and vice-versa)

$$Pr(x|y) = Pr(x)$$
$$Pr(y|x) = Pr(y)$$



What will the second distribution look like at $y = y_2$?

8

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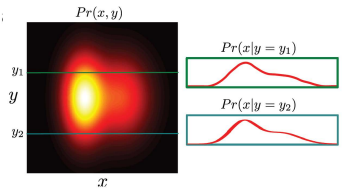
Independence

- If two variables X and Y are independent, then variable X tells us nothing about variable Y (and vice-versa)

$$Pr(x|y) = Pr(x)$$
$$Pr(y|x) = Pr(y)$$



Second distribution at $y = y_2$ is identical

9

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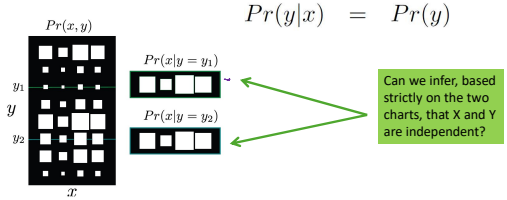
Independence

- If two variables x and y are independent then variable x tells us nothing about variable y (and vice-versa)

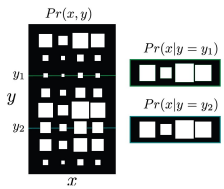
$$Pr(x|y) = Pr(x)$$
$$Pr(y|x) = Pr(y)$$



Independence

- When variables are independent, the joint factorizes into a product of the marginals:

$$Pr(x,y) = Pr(x|y)Pr(y)$$
$$= Pr(x)Pr(y)$$



Independence

- The joint pdf is the product of the marginal pdfs.
- For continuous independent random variables the joint cdf also factorizes

$$\forall x,\ \forall y,\quad f_{XY}(x,y) = f_X(x)\, f_Y(y)$$

$$F_{XY}(x,y) = F_X(x)\, F_Y(y)$$

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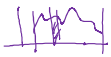
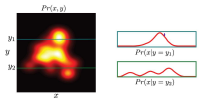
Quick Summary



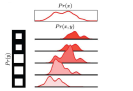
$$F_{XY}(a,b) = \int_{-\infty}^{a}\int_{-\infty}^{b} f_{XY}(x,y)\, dx\, dy$$

$$f_{XY}(a,b) = \frac{\partial^2}{\partial x\, \partial y} F_{XY}(x,y)|_{x=a,\, y=b}$$

- Rules of probability are compact and simple
- Concepts of joint, marginal, and conditional probability, and expectation underpin most of the machine learning models



Conditional probability



Marginalization

Probability = area of density surface

13

Expectation Rules

Must know

- Expected value of a constant is the constant

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$\mathrm{E}[\kappa] = \kappa$$

14

14

Expectation Rules

Must know

- Expected value of constant times function is constant times expected value of function

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$\mathrm{E}[kf[x]] = k\mathrm{E}[f[x]]$$

15

15

Expectation Rules

Must know

- Expectation of sum of functions is sum of expectation of functions

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$\mathrm{E}[f[x] + g[x]] = \mathrm{E}[f[x]] + \mathrm{E}[g[x]]$$

16

16

Expectation Rules

Must know

- Expectation of product of functions in variables x and y is product of expectations of functions if x and y are independent

$$E[f[X]] = \int f[x]Pr(X = x)dx$$

$$\mathrm{E}[f[x]g[y]] = \mathrm{E}[f[x]]\mathrm{E}[g[y]] \qquad \text{if } x, y \text{ independent}$$

17

17

Conditional Probability and Independence

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$\begin{aligned} Pr(w, x, y, z) &= Pr(w, x, y|z)Pr(z) \\ &= Pr(w, x|y, z)Pr(y|z)Pr(z) \\ &= Pr(w|x, y, z)Pr(x|y, z)Pr(y|z)Pr(z) \end{aligned}$$

How will this look like if the variables are independent?

18

18

Mutual independence

This definition for n variables is hard to comprehend, see example next

- n continuous random variables are mutually independent iff
  - for every finite subset of indices $i_1, i_2, \dots, i_k$
  - And every choice of real numbers $x_i, \dots, x_k$
  - $P(X_{i1} \le x_1, \dots, X_{ik} \le x_k) = \prod_{j=1}^{k} P\,(X_{i_j} \le x_j)$
- Pairwise independence is the much weaker condition when k=2

$$\forall x_1, x_2, \dots, x_n, \quad f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

$$\forall (x_i, x_j),\ 1 \le i \le n,\ 1 \le j \le n,\ i \ne j: \quad f_{X_i X_j}(x_i, x_j) = f_{X_i}(x_i) f_{X_j}(x_j)$$

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Example

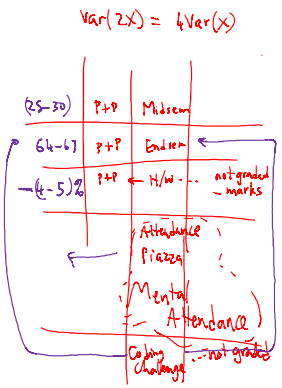
- Take the sample space {1, 2, 3, 4}, each singleton equally likely.
  - P({1}) = ¼
- Let A = {1, 2}, i.e., either the event {1} or the event {2}
  - P(A) = ½
  - Similarly, B = {1, 3}, C = {1, 4} and P(A) = P(B) = P(C) = ½ .
- Consider the intersection event AB = {1}. P(AB) = ¼
  - A and B are considered independent since ½ * ½ = ¼
- However, the intersection event ABC
  - Has P(ABC) = P({1}) = ¼ ≠ P(A)P(B)P(C)=1/8, so A, B, C are not mutually independent.

20

20

Evaluation

- "coding project"
  - Demonstrate Creativity
  - Demonstrate visualization tools
  - More details later
  - Groups of4
- Percentages are approximate
  - Your cooperation is critical



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Agenda

- Last time
  - Independence, Pair-wise independence
- Today
  - Covariance
  - Inequalities

2

Mutual independence

This definition for n variables is hard to comprehend, see example next

- n continuous random variables are mutually independent iff
  - for every finite subset of indices $i_1, i_2, \dots, i_k$
  - and every choice of real numbers $x_1, \dots, x_k$
  - $P(X_{i1} \le x_1, \dots, X_{ik} \le x_k) = \prod_{j=1}^{k} P\,(X_{i_j} \le x_j)$
- Pairwise independence is the much weaker condition when k=2

$$\forall x_1, x_2, \dots, x_n, \quad f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

$$\forall (x_i, x_j),\ 1 \le i \le n,\ 1 \le j \le n,\ i \ne j: \quad f_{X_i X_j}(x_i, x_j) = f_{X_i}(x_i) f_{X_j}(x_j)$$

3

3

Example

- Take the sample space {1, 2, 3, 4}, each singleton equally likely.
  - P({1}) = ¼
- Let A = {1, 2}, i.e., either the event {1} or the event {2}
  - P(A) = ½
  - Similarly, B = {1, 3}, C = {1, 4} and P(A) = P(B) = P(C) = ½ .
- Consider the intersection event AB = {1}. P(AB) = ¼
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- However, the intersection event ABC
  - Has P(ABC) = P({1}) = ¼ ≠ P(A)P(B)P(C)=1/8, so A, B, C are not mutually independent.

4

4

Events in the Example

- Define indicator random variables
  - $X_A = \mathbf{1}_A$, $X_B = \mathbf{1}_B$, $X_C = \mathbf{1}_C$, where, for example,
  - $X_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$
- Every pair is independent. For example,
  - $P(X_A = 1, X_B = 1) = \frac{1}{4} = P(X_A = 1)P(X_B = 1).$
- But the three variables are not mutually independent because
  - $P(X_A = 1, X_B = 1, X_C = 1) = \frac{1}{4} \neq \left(\frac{1}{2}\right)^{3.}$

5

5

Lack of Transitivity Example

- Independence is not transitive
- Suppose a system contains two physically independent components.
  - Event A: Component 1 fails today.
  - Event B: Component 2 fails today.
  - Event C: Component 1 works today.

6

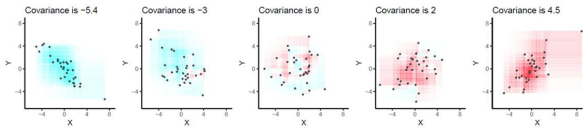
6

Linearity and Covariance

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

- Expected values of sums: E[X] + E[Y] = E [X+Y]
- However, Var[X] + Var[Y] != Var [X+Y]
  - Consider Y = X
  - We need the concept of covariance
  - Think of pairs (x, y) to define red and blue
  - A point is red if the points "agree"

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$



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Correlation Coefficient

- The strength of the relationship is measured by normalization
  - Unit free, bounded and easy to compare

8

Correlation Coefficient

- The strength of the relationship is measured by normalization
  - Unit free, bounded and easy to compare
- Sample correlation coefficient r
  - $r > 0$ positively correlated
  - r < 0 negatively correlated
  - r close to 0 uncorrelated

$$r(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y}$$

$$r = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{b_x})(y_i - \overline{b_y})}{\sigma_x\, \sigma_y}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

9

Correlation Is Not Causation

- Correlation does not necessarily imply causation
  - Consider: *High correlation between tobacco usage and lung cancer occurrence does not imply that smoking causes lung cancer*
- Sensitive to outliers

Let's go back to linearity

10

Alternate Formula

$E[(X-\mu_X)(Y-\mu_Y)]$

$Var(X) = E[X^2] - (E[X])^2$

- A formula similar to that of the mean: $Cov(X,Y)=E[XY]-E[X]E[Y]$

11

11

Covariance and variance

- Symmetric: Cov(X,Y) = Cov (Y,X)
- "Bilinear": Linear in each argument separately
  - Cov(aX,Y) = a Cov(x,y)
    - Recall: Var(aX) = a*a Var(X)
  - Cov(X+Z, Y) = Cov(X,Y) + Cov(Z,Y)
- Cov(X,X) = Var(X)
- Var(X+Y) =Var(X)+Var(Y)+2Cov(X,Y)

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

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Agenda

- Last time
  - Indicator random variables
  - Standardization of data

2

Standardization in one dimension

Raw deviation → **Standardized deviation**

$$X - \mu \qquad Z = (X - \mu) / \sigma$$

unit-dependent

unit-free: “2 standard deviations above mean”

3

Agenda

- Last time
  - Indicator random variables
  - Standardization of data
  - Correlation coefficient
  - Correlation is not causation
  - "Bi-Linearity" of covariance
- Today
  - Inequalities
  - Weak Law of "large numbers"

Cauchy–Schwarz inequality

For vectors, $|\langle x,y\rangle| \le \|x\|\|y\|$.

Inner product cannot be arbitrarily large relative to the lengths of the vectors

4

$E[(X-\mu_X)(Y-\mu_Y)]$

$\mathrm{Var}(X) = E[X^2] - (E[X])^2$

A formula similar to that of the mean: $\mathrm{Cov}(X,Y)=E[XY]-E[X]E[Y]$

$E[(X-\mu_X)(Y-\mu_Y)]$

$= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y]$

5

5

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Covariance and variance

- Symmetric: Cov(X,Y) = Cov (Y,X)
- "Bilinear": Linear in each argument separately
  - Cov(aX,Y) = a Cov(x,y)
    - Recall: Var(aX) = a*a Var(X)
  - Cov(X+Z, Y) = Cov(X,Y) + Cov(Z,Y)
- Cov(X,X) = Var(X)
- Var(X+Y) =Var(X)+Var(Y)+2Cov(X,Y) ← Must know

6

Independence $\Longrightarrow$ zero covariance

- A formula similar to that of the mean: Cov(X,Y)=E[XY]−E[X]E[Y]

$$E[XY] = \sum_i \sum_j x_i\, y_j P\{X = x_i,\, Y = y_j\} = \sum_i \sum_j x_i\, y_j P\{X = x_i\} P\{Y = y_j\}$$
$$= \sum_i x_i\, P\{X = x_i\} \sum_j y_j\, P\{Y = y_j\} = E[X]E[Y]$$

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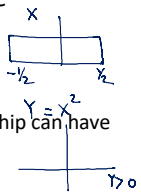
Zero covariance ⇏ Independence

- Covariance only measures **linear** association.
  - Try $Y = X^2$ with X symmetric about 0
- A perfectly deterministic but non-monotonic relationship can have zero covariance.



$$Cov(X, Y) = Cov(X, X^2) = E[X \cdot X^2] - E[X] \cdot E[Y]$$
$$E[X^3] \quad 0$$

8

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Probabilistic inequalities

- Sometimes we know only the mean or variance of a random variable, and want the probability that it takes on a certain value.
  - The exact probability cannot be computed without assumptions
- But we can obtain upper or lower bounds on that probability, and those bounds are often enough to make a decision

9

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Markov's inequality

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3
- What is the probability that a randomly chosen student gets an offer of 110×10^3, or more?
- Let *X* be a random variable that takes only **non-negative** values. For any *a* > 0:

$$P\{X \geq a\} \leq E[X]/a$$

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Markov's Inequality

$$E[X] = \int_0^\infty x\, f_X(x)\, dx = \int_0^a x\, f_X(x)\, dx + \int_a^\infty x\, f_X(x)\, dx$$

$$\geq \int_a^\infty x\, f_X(x)\, dx$$

$$\geq \int_a^\infty a\, f_X(x)\, dx$$

$$= a \int_a^\infty f_X(x)\, dx = a\, P\{X \geq a\}$$

$$\therefore\ P\{X \geq a\} \leq E[X]/a$$

Similar proof for discrete case also

11

11

Markov's inequality

$P\{X \geq a\} \leq E[X]/a$

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3
  - What is the probability that a randomly chosen student gets an offer of 110×10^3, or more?
- If we assume that the first statement means
  - Choose a random student
  - Let X = salary offered to the student. E[X] = 100×10^3
  - Then the probability a randomly chosen student gets 110×10^3 is equal to or less than 90.9%

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Chebyshev's inequality

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3 and the variance is 50×10^3
  - What is the probability that a student's salary lies between 90×10^3 and 110×10^3?
- For a random variable X with mean μ and variance σ², and any k > 0

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

$$P\{|X - 100 \times 10^3| \geq 10 \times 10^3\} \leq \frac{50 \times 10^3}{10 \times 10 \times 10^6} = 0.0005 \approx 0.05\%$$

$$\therefore P\{|X - 100 \times 10^3| < 10 \times 10^3\} = 1 - 0.05\% = 99.5\%$$

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Typo inequality needed. [≥]

Chebyshev's Inequality

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- For a random variable X with mean μ and variance σ^2, and any $k > 0$:
- Apply Markov's inequality to $(X-\mu)^2$ since it is non-negative

$$P\{(X-\mu)^2 \geq k^2\} \leq E[(X-\mu)^2]/k^2 = \sigma^2/k^2$$

$$\therefore\ P\{|X-\mu| \geq k\} \leq \sigma^2/k^2$$

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Chebyshev's Inequality

- For X with mean μ and variance σ^2, and any $k > 0$:

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- Replacing k by $k\sigma$ gives the form that is usually quoted — the bound now reads in units of standard deviations:

$$P\{|X-\mu| \geq k\sigma\} \leq \frac{1}{k^2}$$

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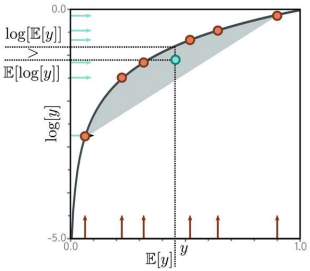
Jensen's inequality

Illustration for log function

- For any concave function
  - $f[E[y]] \geq E[f[y]]$
  - Draw a straight line between any two points, it lies under the curve
- Any weighted sum of the six points must line in the gray region
- Sample mean on the x-axis is shown



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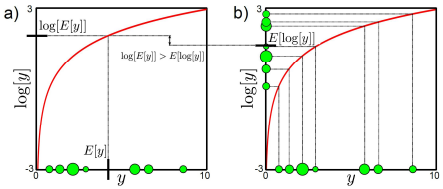
Jensen's Inequality



$$E[\log[y]] \leq \log[E[y]] \quad \text{or...} \quad \sum_y q(y)\log[y] \leq \log\left[\sum_y q(y)y\right]$$

similarly $$\int q(y)\log[y]dy \leq \int \log[q(y)]dy$$

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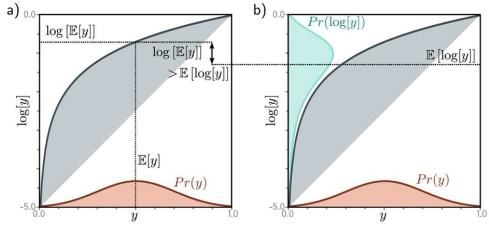
Continuous case



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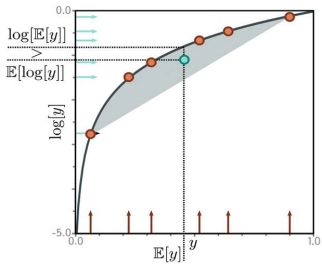
Agenda

- Last time
  - Inequalities (Markov, Chebyshev, Jensen)
- Today

Illustration for log function



$$E\left[\log[y]\right] \leq \log\left[E[y]\right]$$

2

Markov's Inequality Discrete Case

$\Pr[x>a]$

$\leq \frac{E[X]}{a}$

$$a\Pr(X > a)$$

$$= \; a\Pr(X = a+1) + a\Pr(X = a+2) + a\Pr(X = a+3) + \cdots$$

$$\leq \; a\Pr(X = a) + (a+1)\Pr(X = a+1) + (a+2)\Pr(X = a+2) + (a+3)\Pr(X = a+3) + \cdots$$

$$\leq \; \Pr(X = 1) + 2\Pr(X = 2) + 3\Pr(X = 3) + \cdots +$$
$$a\Pr(X = a) + (a+1)\Pr(X = a+1) + (a+2)\Pr(X = a+2) + (a+3)\Pr(X = a+3) + \cdots$$

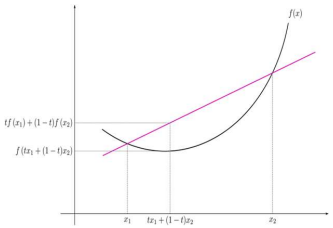
$$= \; E(X)$$

3

3

Jensen's inequality

- For any concave function and random variable Y
  - $f[E[Y]] \geq E[f[Y]]$
  - For a convex function
    - it's the other way around



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Example

- $R = \begin{cases} 1.1 & \text{with probability } \frac{1}{2} \\ 0.9 & \text{with probability } \frac{1}{2} \end{cases}$ Interpret R as return on investment
  - E[R] = 1 suggests that there is no gain or loss in investment
- However, net return $N = R_1 R_2 R_3$
  - To find expected value of wealth, without assumption of independence, use
    - $\log N = \log R_1 + \log R_2 + \cdots$
    - $E[\log N] = E[\log R_1] + E[\log R_2] + \ldots$
    - From Jensen's formula on each term, LHS < log E[R_1] + log E[R_2] + ..
      - If each return was, on an average 1, then LHS < n log E[R] = n log 1 = 0
      - Turning to the actual wealth $\exp\{E[\log N]\} < e^0 = 1$ return factor is negative
- Conclusion: It's a losing proposition, not a no-op proposition

Used in English sense

-ve ⇒ not a good situation.

Mathematically < 1 not < 0

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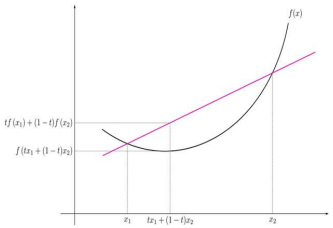
Agenda

- Last time
  - Inequalities
- Today
  - Weak Law of “large numbers”
  - Popular distribution functions



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Expected value

- When we say the expected value of a die roll is 3.5, what does that mean?
  - If you throw the die n times and find the sample mean, you should get a value close to 3.5 — provided n is large.
  - It says nothing about small n.
  - As n increases, the sample mean should move closer and closer to 3.5.
- That is the intuition. Let’s state it more formally

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The weak law of large numbers

- Let X_1, X_2, …, X_n be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

Must know

8

8

The weak law

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- From Chebyshev's inequality applied to the sample mean

$$E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \mu, \quad \text{Var}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\therefore P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \leq \frac{\sigma^2}{n\varepsilon^2}$$

- Letting n → ∞ the right-hand side vanishes, so

$$\therefore \lim_{n\to\infty} P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} = 0$$

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Comments on the weak law

Nice to know

- The identically distributed assumption is not strictly necessary
  - The random variables just need the same mean μ.
  - We assume a common variance earlier
    - If not, the $n\sigma^2$ term becomes n times the expected variance
    - The proof still goes through.
- Independence is strictly not necessary
  - Pairwise independence suffice
- The law *does* assume a well-defined expected value and variance
  - Otherwise, it may not hold.

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Law of averages

- We tend to believe that if something has been going wrong for a while, it will suddenly turn right — by the "law of averages"
  - This supposed law is a fallacy
  - A gambler tosses an unbiased coin 20 times and gets a head every time
    - He invokes the "law of averages" and concludes the next toss is more likely to be a tail
    - The probability of getting 21 heads in a row is $(1/2)^{21}$
    - The probability of getting 20 heads followed by 1 tail is also $(1/2)^{21}$
  - The coin has no memory: Events are independent with identical distribution
    - However, if he keeps tossing the coin eventually he will see tails appearing, but the tails may not "compensate" for the first 20 heads
    - Relative frequency stabilize, past imbalance need not be cancelled

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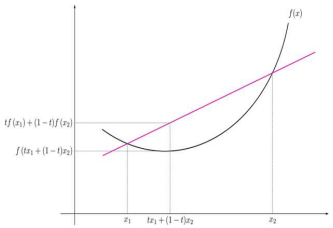
Agenda

- Last time
  - Weak Law of “large numbers”
  - Popular distribution functions



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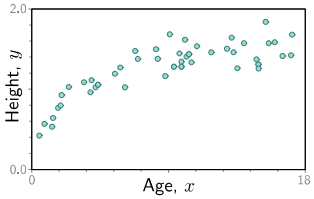
Topic Overview

- We wish to study several useful families of random variables
  - Discrete random variables: Bernoulli, Categorical, Binomial, Poisson, hypergeometric
  - Continuous random variables: Uniform, Gaussian, exponential, chi-square
- This may become tedious, so why?

3

Finding the trend

- You may have studied how to find the line fitting given points in your earlier classes



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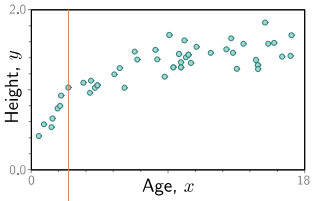
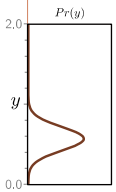




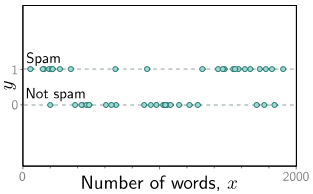
However, the output is often not a specific line (a y-coordinate for a given x) but a distribution.

What is the nature of the distribution?

5

Text classification

- Is the email spam or not?
- We may want a probabilistic interpretation of the output



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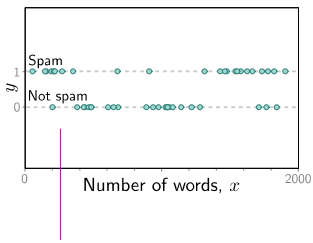
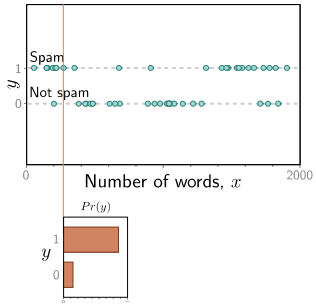


Text Classification

What is the nature of the output distribution?

menti.com
5245 0775

7

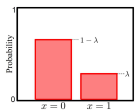


8

Bernoulli Distribution

- Example: Biased coin producing a heads with probability λ
- Outcome is either a success or a failure
- Single parameter
- Properties
  - Expectation: p
  - Variance: p(1-p)

Must know



$$Pr(x=0) = 1-\lambda$$
$$Pr(x=1) = \lambda.$$

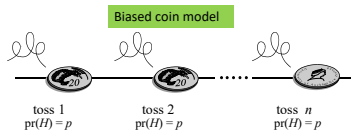
$$Pr(x) = \text{Bern}_x[\lambda]$$

4

9

Binomial distribution

- Physical model for situations of a series of trials
  - Each trial has only two outcomes: success and failure;
  - p = pr(success) is the same for every trial; and
  - trials are independent



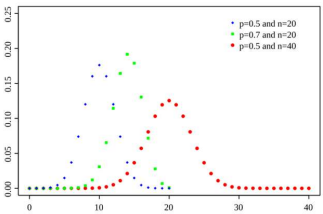
$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$\sum_i P(X = i) = \sum_i \binom{n}{i} p^i (1-p)^{n-i} = (p + 1 - p)^n$$

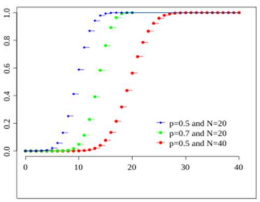
Binomial Theorem

12

Binomial distribution



pmf



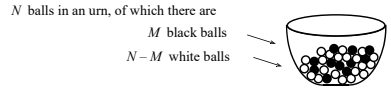
cdf

13

Urn model

- If we sample *Y* individuals at random from a finite population and
- Count X, the number of individuals with a characteristic of interest
  - Example: IITB is good about training students for the next generation. Agree or Disagree?
- What can we say about the characteristic?
- Is the distribution of X binomial?
  - The distribution of X is approximately Binomial(Y, p=M/N).
  - Operating Rule: Y/N < 0.1

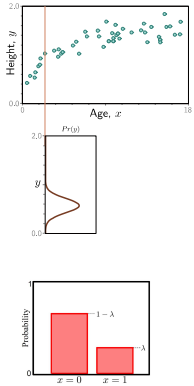
Useful approximation



14

Last Time

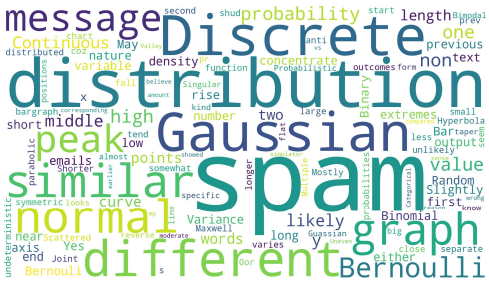
- Some distributions used over and over again
- Therefore, useful to know the concept
  - If we know the mean and variance, we can understand these better
  - However, do not memorize the pmf (pdf), means and variance, we can always look them up



2

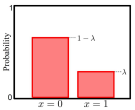
Engages with Question

What is the nature of the output distribution for text classification?

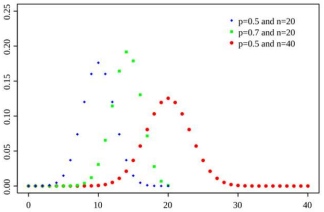


3

Agenda

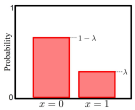


- Last time
  - Some distributions used over and over again
    - If we know the mean and variance, we can understand these better
  - Bernoulli
- Today
  - Binomial
  - Geometric
  - Hypergeometric
  - Moment generation functions

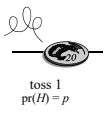


5

Bernoulli



- Some distributions used over and over again
- Last time
  - W ~ Bern(λ)
    - Expectation: λ, Variance: $\lambda(1-\lambda)$
- Physical analogy
  - Biased coin

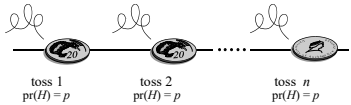


6

Binomial

- Physical model for situations of a series of trials
  - Each trial has only two outcomes: success and failure;
  - p = pr(success) is the same for every trial; and
  - trials are independent
- Expectation and Variance?

Biased coin model



$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

7

Example: US Presidents

- The Economist magazine reported that US presidents had 61 daughters and ~90 sons
- Expected value is approximately 75
- What is the probability that there are 89 or more boys?
  - (1 - binocdf (89, 150, 0.5)) in octave
- Probability theory calculates that the probability of having 89 or more sons in 151 'experiments' is 0.008799 (~0.8%), a very small number

octave convention
use 88

Were US presidents special?

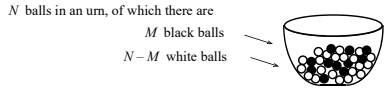
Overview of Presidential Children

- **Total Recorded (Historical baseline up to Clinton):** 89 boys and 61 girls (150 total)

8

Hypergeometric

- Urn model is actually "hypergeometric" distribution
- We chose Y balls *without replacement* and ask how many are black
- However, the binomial approximation to the urn Model Approximation is very useful
  - Y items sampled, count X as the number of people which have a characteristic
  - Approximation: X ~ Bin(Y, M/N)



9

Example: Doctors and Exercise

- A 2018 study followed a nationwide cohort of Norwegian doctors. In its 2014 survey, 307 of 526 respondents reported that they asked patients about exercise habits
  - You are choosing a doctor for your pre-diabetic uncle
  - You have a choice of three doctors in your clinic
  - What is the probability that none of the three doctors will ask about exercise?
- This problem fails the binomial requirement of "sampling with replacement" and independence of opinion
  - But we will use the binomial approximation since 3/526 is very small.
  - P(X<=0) = binocdf(0, 3, q=219/526) = 19.8%

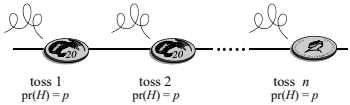
Is the clinic a good one?

10

Answers a different question than what is posed (as in class lecture probably)

Use $\frac{307}{526}$

Binomial distribution



- Expectation
- Variance

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

13

13

Binomial Distribution

- Defining indicator variables, a binomial random variable *X* is the *sum* of indicator random

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$X_i = \begin{cases} 1, & \text{trial } i \text{ succeeds (probability } p) \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = np$$

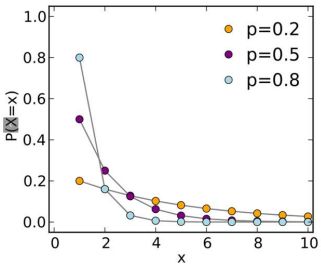
$$\mathrm{Var}(X) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = np(1-p)$$

14

Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

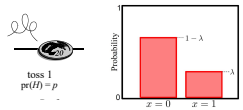
- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- "Memoryless"
- E[X] = ?
- E[X] = 1.p + (1-p)[1 +E[X]]
- Var(X) = $(1-p)/p^2$

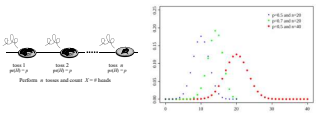


15

Agenda

- Last time
  - Some distributions used over and over again
  - Bernoulli, Binomial, Geometric
- We capture the mean and variance of these distributions
  - Example of US presidents and the variance
- Today
  - Moment generation functions
  - Hypergeometric





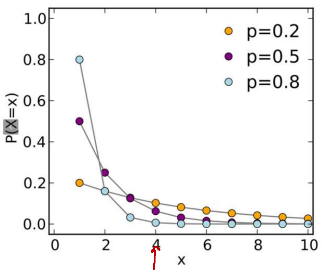
2

Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

X ~ k

- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- "Memoryless"



- Octave: geopdf(3, 0.2)
- from scipy.stats import geom
  geom.pmf(4,0.2)

3

Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- E[X] = ?
  - E[X] = 1.p + (1-p)[1 +E[X]]
  - E[X] = 1/p
- Var(X) = (1-p)/p^2

4

Moment generating function

- The moment generating function (MGF) of X is

$$\phi_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x) \quad \text{(discrete r.v.)}$$

5

5

Why MGF?

- Observe differentiating the MGF with respect to t

$$\phi_X(t) = E(e^{tX})$$

$$\phi_X'(t) = \frac{d}{dt}(E(e^{tX})) = E\left(\frac{d}{dt}(e^{tX})\right) = E(Xe^{tX})$$

$$\phi_X'(0) = E(X)$$

$$\phi_X^{(2)}(t) = \frac{d}{dt}(E(Xe^{tX})) = E(X^2e^{tX})$$

$$\phi_X^{(2)}(0) = E(X^2),\ \ldots,\ \phi_X^{(n)}(0) = E(X^n)$$

6

6

Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$M_X(t) = E[e^{tX}] = \sum_{k=1}^{\infty} e^{tk} P_X(k)$$

$$M_X(t) = \sum_{k=1}^{\infty} e^{tk} p q^{k-1}$$

$$q = 1 - p$$

$$M_X(t) = pe^t \sum_{k=1}^{\infty} (qe^t)^{k-1}$$

$$= pe^t \sum_{j=0}^{\infty} (qe^t)^j$$

$$= \frac{pe^t}{1 - qe^t}.$$

For $0 < p < 1$, $t < -\log q$

$$M_X'(t) = \frac{pe^t(1 - qe^t) - pe^t(-qe^t)}{(1 - qe^t)^2}$$

$$= \frac{pe^t - pqe^{2t} + pqe^{2t}}{(1 - qe^t)^2}$$

$$= \frac{pe^t}{(1 - qe^t)^2}.$$

$$E[X] = M_X'(0)$$

$$= \frac{p}{(1 - q)^2}$$

$$= \frac{1}{p}.$$

7

Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$q = 1 - p$$

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \sum_{k=1}^{\infty} e^{tk} p q^{k-1} \\ &= pe^t \sum_{k=1}^{\infty} (qe^t)^{k-1} \\ &= pe^t \sum_{j=0}^{\infty} (qe^t)^j \\ &= \frac{pe^t}{1 - qe^t}. \end{aligned}$$

$$\begin{aligned} M_X''(t) &= pe^t(1 - qe^t)^{-2} + pe^t \left[-2(1 - qe^t)^{-3}(-qe^t)\right] \\ &= \frac{pe^t}{(1 - qe^t)^2} + \frac{2pqe^{2t}}{(1 - qe^t)^3}. \end{aligned}$$

$$\begin{aligned} E[X^2] &= M_X''(0) \\ &= \frac{p}{(1-q)^2} + \frac{2pq}{(1-q)^3} \\ &= \frac{1}{p} + \frac{2q}{p^2}. \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= E[X^2] - E[X]^2 \\ &= \frac{1}{p} + \frac{2q}{p^2} - \frac{1}{p^2} \\ &= \frac{p + 2q - 1}{p^2} \\ &= \frac{q}{p^2} \\ &= \frac{1 - p}{p^2}. \end{aligned}$$

8

Moment

- The moment generating function (MGF) of X is
- The moment of X of order n is $m_n = E(X^n),\ n \geq 1$
  - The first moment is the mean
- The central moment of X is E[(X – E[X])n]
  - The second central moment is the variance

$$\phi_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x) \quad \text{(discrete r.v.)}$$

$$= \int_{-\infty}^{\infty} f_X(x)\, e^{tx}\, dx \quad \text{(continuous r.v.)}$$

9

9

Why MGF?

$$m_{\mathrm{n}} = E(X^{\mathrm{n}}),\ \mathrm{n} \geq 1$$

- Because in the series expansion, every "moment" m_i appears as a coefficient
  - The function generates the moments.

$$e^{tX} = 1 + tX + (tX)^2/2! + (tX)^3/3! + \cdots$$

$$\phi_X(t) = E(e^{tX}) = 1 + t\,m_1 + t^2 m_2/2! + t^3 m_3/3! + \cdots$$

10

10

Uniqueness

- If two discrete random variables X and Y have MGFs $\phi_x(t)$ and $\phi_y(t)$ that both exist and agree for all t, then X and Y have the same pmf
- Same holds for continuous random variables also

11

11

Other properties

- If Y = aX + b
- If X and Y are independent
- If Z equals X with probability p and Y with probability 1 – p:

$$\phi_Y(t) = e^{tb}\,\phi_X(at)$$

$$\phi_{X+Y}(t) = \phi_X(t)\,\phi_Y(t)$$

$$\phi_Z(t) = p\,\phi_X(t) + (1-p)\,\phi_Y(t)$$

13

13

Sampling with and without replacement

- Suppose there are k objects each of a different type
- When you sample two different objects with replacement,
  - You pick a particular object with probability 1/k, and
  - You place it back (replace it)
  - You pick the second object. The probability of picking an object of another type is again 1/k.
- When you sample without replacement,
  - probability that your first object was of so and so type is 1/k
  - the probability that your second object was of so and so type is now 1/(k-1) because you didn't put the first object back

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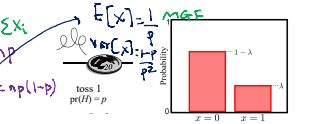
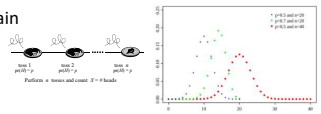
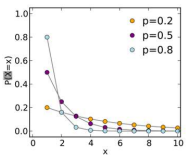
14

Agenda

- Last time
  - Some distributions used over and over again
  - Bernoulli, Binomial, Geometric
- Today
  - Hypergeometric
  - Continuous distribution







2

Hypergeometric Distribution

- If we sample *Y* balls at random from a finite population and
  - Count X, the number of black balls
- X ~ Hypergeometric(M, N-M, Y) (black, white, Y)

Cannot make all Y white

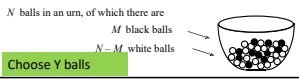
$$\text{Range} = \{\max(0, Y-(N-M)), \{\max(0, Y-(N-M))+1, \ldots, \min(Y, M)\}$$

$$P_X(x) = \begin{cases} \frac{\binom{M}{x}\binom{N-M}{Y-x}}{\binom{N}{Y}} & \text{for } x \in \text{Range}(X) \\ 0 & \text{otherwise} \end{cases}$$

N=10,M=7,Y=5, then max(0,5−3)=2

Cannot get more black balls than in the bowl

Octave: hygepdf(x, N, M , Y)



3

Hypergeometric Alternate notation

- N black balls, M white balls, N+M total balls
- n items chosen at random

$$P(X = i) = \frac{\binom{N}{i}\binom{M}{n-i}}{\binom{N+M}{n}}, \quad 0 \le i \le \min(N, n)$$

54

4

Expectation

- Consider random variable X_i which has value 1 if the ith trial produces a black ball and 0 otherwise
- We want to compute $E[\Sigma X_i]$

$$P(X_1 = 1) = \frac{N}{N+M}$$

$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1)P(X_1 = 1) + P(X_2 = 1 \mid X_1 = 0)P(X_1 = 0)$$

$$= \frac{N-1}{N+M-1}\frac{N}{N+M} + \frac{N}{N+M-1}\frac{M}{N+M} = \frac{N}{N+M}$$

$$\text{In general, } P(X_i = 1) = \frac{N}{N+M}$$

55

5

Expectation

- Consider random variable X_i which has value 1 if the ith trial produces a black ball and 0 otherwise
- We want to compute $E[\Sigma X_i]$

$$P(X_1 = 1) = \frac{N}{N+M}$$
$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1)P(X_1 = 1) + P(X_2 = 1 \mid X_1 = 0)P(X_1 = 0)$$
$$= \frac{N-1}{N+M-1}\frac{N}{N+M} + \frac{N}{N+M-1}\frac{M}{N+M} = \frac{N}{N+M}$$
$$\text{In general, } P(X_i = 1) = \frac{N}{N+M}$$

55

6

Expectation

- Each X_i is a Bernoulli random variable with parameter $p = \frac{N}{N+M}$

$$X = \sum_{i=1}^{n} X_i$$

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = \frac{nN}{N+M}$$

56

8

Variance

- Variance of each individual event can be computed easily, since Each X_i is a Bernoulli random variable with parameter $p = \frac{N}{N+M}$

$$\mathrm{Var}(X_i) = P(X_i = 1)(1 - P(X_i = 1)) = \frac{NM}{(N+M)^2}$$

- But are these variables independent?
  - For example, X_1 and X_2 are not independent
- So we rely on

$$\mathrm{Var}(X) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$$

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Covariance

$$\mathrm{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$

$$E(X_i X_j) = P(X_i = 1, X_j = 1) = \frac{N-1}{N+M-1}\frac{N}{N+M}$$

$$\mathrm{Cov}(X_i, X_j) = \frac{N(N-1)}{(N+M)(N+M-1)} - \left(\frac{N}{N+M}\right)^2 = \frac{-NM}{(N+M)^2(N+M-1)}$$

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Variance

- X_i and X_j are not independent

$$\mathrm{Var}(X) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$$

$$\frac{n(n-1)}{2} \qquad \frac{-NM}{(N+M)^2(N+M-1)}$$

$$\mathrm{Var}(X) = \frac{nNM}{(N+M)^2}\left(1 - \frac{n-1}{N+M-1}\right)$$

$$= np(1-p)\left(1 - \frac{n-1}{N+M-1}\right)$$

$$\mathrm{Var}(X) \approx np(1-p) \quad \text{when } N \text{ or } M \text{ is large}$$

56

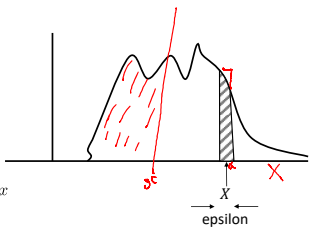
12

Continuous Distributions

- The probability density function (pdf) is the derivative of the cumulative distribution function (cdf) at that value x

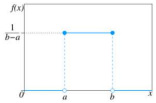
$$f_X(a) = \lim_{\epsilon \to 0} \frac{\Pr\left\{a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}\right\}}{\epsilon}$$

$$\Pr\left\{a - \tfrac{\epsilon}{2} \le X \le a + \tfrac{\epsilon}{2}\right\} \approx \epsilon f(a) = \int_{a-\frac{\epsilon}{2}}^{a+\frac{\epsilon}{2}} f_X(x)\,dx$$



13

Uniform distribution



- A uniform random variable over the interval [a,b]
  - valid pdf – non-negative and integrates to 1
  - Mean (and median) equal to $(b+a)/2$

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

E[x]

$$E(X) = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2}$$

M(t) $E(e^{tX}) = \int_a^b \frac{e^{tx}}{b-a} dx = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)}, & t \neq 0 \\ 1, & t = 0 \end{cases}$

$$E(X^2) = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2+ab+b^2}{3}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

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Why uniform?

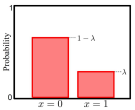
Seed .. 73456

- Uniform random variables, especially over the [0,1] interval are very important, in developing programs to draw samples from other distributions
- That's the theory
  - In practice, we assume that we have access to a generator that can generate a canonical random variable
    - The random number generator is actually a pseudo-random number generated

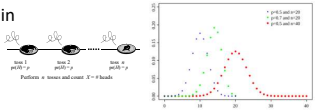
119

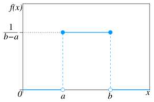
16

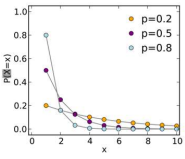
Agenda



- Last time
  - Some distributions used over and over again
  - Hypergeometric
  - Continuous distribution
- Today
  - PRNG
  - Uniform distribution
  - Random subset







2

Hypergeometric

$$P(X = i) = \frac{\binom{N}{i}\binom{M}{n-i}}{\binom{N+M}{n}}$$

- If we sample *n* balls at random from a finite population without replacement, and count X, the number of black balls
- Consider X_i which has value 1 if the ith trial produces a black ball and 0 otherwise
  - Let $T = N + M$
  - Clearly, $P(X_1 = 1) = \frac{N}{T}$
  - Assume $P(X_k = 1) = \frac{N}{T}$
- Although the X_i's are **not independent**, every position in the sequence has the same marginal probability

$$\begin{aligned} P(X_{k+1} = 1) &= P(X_{k+1} = 1 \mid X_k = 1)P(X_k = 1) \\ &\quad + P(X_{k+1} = 1 \mid X_k = 0)P(X_k = 0) \\ &= \frac{N-1}{T-1}\frac{N}{T} + \frac{N}{T-1}\frac{M}{T} \\ &= \frac{N(N-1)+NM}{T(T-1)} \\ &= \frac{N(T-1)}{T(T-1)} \\ &= \frac{N}{T}. \end{aligned}$$

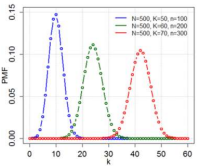
N black balls
M white balls
Choose n balls
How many black?

3

Hypergeometric

$$P(X = i) = \frac{\binom{N}{i}\binom{M}{n-i}}{\binom{N+M}{n}}$$

- If we sample n balls at random from a finite population and count X, the number of black balls
- How do we choose “at random” 100 people from a group of 200?

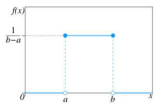


N black balls
M white balls
Choose n balls
How many black?

4

Uniform distribution

- Uniform random variables, especially over the [0,1] interval are very important, in developing programs to draw samples from other distributions
- That’s the theory
  - In practice, we assume that we have access to a generator that can generate a canonical random variable
    - The random number generator is actually a pseudo-random number generated



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5

Pseudorandom Numbers

- Concept of a seed
- For any initial seed, subsequent numbers appear to be random, cannot be predicted …

This is desirable

6

Demo

7

Random Numbers: Produce integers

- rand() is often used

What's Wrong With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

Adapted from rand() considered harmful (Stephan T. Lavavej)

9

Random Numbers (API)

What's Right With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

All required headers are included!
All included headers are required!
Headers are sorted!
One True Brace Style!
%d is correct for int!

10

Random Numbers (API)

- rand() is often used

What's Wrong With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

Frequency: 1 Hz!

32-bit seed!

warning C4244: 'argument' : conversion from 'time_t' to 'unsigned int', possible loss of data

Range: [0, 32767]

Non-uniform distribution!

Linear congruential → low quality!

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Random Numbers (API)

- Assume rand() is uniform in [0, 32767]

// [0, 99] src → [0, 99] dest

// [100, 199] src → [0, 99] dest

// …

// [32700, 32767] src → [0, 67] dest

- The output is not uniform

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Hello random world

- Mersenne twister
  - Not cryptographically secure
- Automatic seed

```python
import random
//random.seed(42)
n = int(input())
a = []
for i in range(n):
    a.append(random.randint(1, 100))
for i in range(n - 1):
    min_index = i
    for j in range(i + 1, n):
        if a[j] < a[min_index]:
            min_index = j
    a[i], a[min_index] = a[min_index], a[i]
for x in a:
    print(x, end=" ")
print()
```

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Hello, "Random" World!

```cpp
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
        std::cout << dist(mt) << " ";
    }
    std::cout << std::endl;
}
```

Engine: [0, 2^{32})

Deterministic 32-bit seed

Distribution: [0, 99]

Note: [inclusive, inclusive]

- Engine: fast, seedable, reproducible
- Obsessive implementors (perfect uniformity), deterministic but may vary across platforms

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Pseudorandom Numbers

This is desirable

- Concept of a seed
- For any initial seed, subsequent numbers appear to be random, cannot be predicted …
- … until we reach the period after which the numbers reappear in the same order so
  - We need a long period
  - Need efficient pseudo random numbers generator (prng) (should be as cheap as anything)

15

Why uniform?

- Uniform random variables, especially over the [0,1] interval are very important, in developing programs to draw samples from other distributions
- *We assume the existence of a canonical random number generator*

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16

Generating a random subset

- Choosing a population for a medical trial
- Generate a random subset B_k of size k from a set $A = a_1, a_2, \ldots, a_n$ of size n
  - All the $\binom{n}{k}$ subsets must be equally likely
- Consider another problem
  - Generate a discrete random variable X having the pmf

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \cdots, \sum_j p_j = 1$$

122

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Generating a discrete pmf

- Generate a discrete random variable X having the pmf
- Example: 0.2, .15, .25, 0.4
- Algorithm
  - Generate a sequence of random numbers U that is uniformly distributed over [0, 1]

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \cdots, \sum_j p_j = 1$$

$$X = \begin{cases} x_0 & \text{If } U < p_0 \\ x_1 & \text{If } p_0 \le U < p_0 + p_1 \\ \vdots & \vdots \\ x_j & \text{If } \sum_{i=0}^{j-1} p_i \le U < \sum_{i=0}^{j} p_i \\ \vdots & \end{cases}$$

$$0 < a < b < 1 \longrightarrow P\{a \le U < b\} = b - a$$

$$P\{X = x_j\} = P\left\{\sum_{i=0}^{j-1} p_i \le U < \sum_{i=0}^{j} p_i\right\} = p_j$$

122

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Generating a random subset

- Generate a random subset B_k of size k from a set $A = a_1, a_2, \ldots, a_n$ of size n
  - All the $\binom{n}{k}$ subsets must be equally likely
- Define indicator variable

$$I_j = \begin{cases} \mathbf{1} & \{a_j \in B_k\} \\ \text{else} & 0 \end{cases}$$

122

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Generating a random subset

- $P(I_1 = 1) = k/n$
  - Denominator all outcomes: $\binom{n}{k}$ ways to pick k objects out of n
  - Numerator (favourable outcomes)
    - Item 1 is picked, there is only one way of picking it, and
    - we have to pick k-1 items out of the remaining n-1 items i.e. $\binom{(n-1)}{(k-1)}$
  - Probability: favourable divided by total.
- If I_1 is 1, then having chosen I_1, the remaining k-1 items must be chosen to be "equally probable"
  - $P(I_2 = 1 \mid I_1 = 1) = \frac{k-1}{n-1}$

123

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Generating a random subset

- $P(I_1 = 1) = k/n$
  - $P(I_2 = 1 \mid I_1 = 1) = \frac{k-1}{n-1}$
  - If I_1 is 0, $P(I_2 = 1 \mid I_1 = 0) = \frac{k}{(n-1)}$
- Therefore, $P(I_2 = 1) = \frac{(k-I_1)}{(n-1)}$
- Continuing,

← That could be better as $P(I_2 = 1 \mid I_1)$

$$P(I_j = 1 \mid I_1, \ldots, I_{j-1}) = \frac{k - \sum_{i=1}^{j-1} I_i}{n - (j-1)}$$

123

21

Generating a random subset

- Use an algorithm similar to the earlier algorithm
  - Generate at most n random numbers with the stated probabilities
  - Choose item 1 if U_1 < k/n else skip
  - Choose item 2 if U_2 < (k-1)/(n-1) else skip
  - …

$$U_j \sim \text{Uniform}(0, 1)$$

$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

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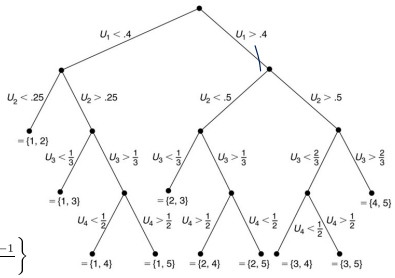
Example

k=2, n=5



$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

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Generating a random subset

- Algorithm stops
  - Step j if $I_1 + I_2 + \cdots + I_j = k$
    - Random subset B_k contains those indices whose indicator values are 1
  - Or, if the number of unfilled entries in the random subset B_k equals the number of remaining elements in A

$$U_j \sim \text{Uniform}(0, 1)$$

$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

125

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Agenda

- Last time
  - PRNG
  - Uniform distribution
  - Random subset
- Today

Hello, "Random" World!

```
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
        std::cout << dist(mt) << " ";
    }
    std::cout << std::endl;
}
```

Engine: [0, 2^{32})

Deterministic 32-bit seed

Distribution: [0, 99]

Note: [inclusive, inclusive]

2

Last Time

- Last time
  - PRNG
  - Uniform distribution
  - Random subset
- Example
  - Two midsemester question papers
  - Want to "randomly" assign papers

Hello, "Random" World!

```
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
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    }
    std::cout << std::endl;
}
```

Engine: [0, 2^{32})

Deterministic 32-bit seed

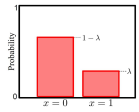
Distribution: [0, 99]

Note: [inclusive, inclusive]

3

Last Time

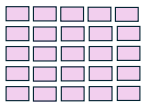
- Last time
  - PRNG, Uniform distribution
  - Random subset
- Example
  - Two midsemester question papers, want to randomly assign papers
  - Room size constraint, one room has only ~30% space



4

Last Time

- Last time
  - PRNG, Uniform, Random subset
- Example
  - Three different TAs with different capacity to grade
  - 0.2, 0.5, 0.3



5

Generating a discrete pmf

- Generate a discrete random variable X having the pmf
  - Example: X = 0, 1 and 2 and 3 with probabilities 0.2, .15, .25, 0.4
- Algorithm
  - Generate a sequence of random numbers U that is uniformly distributed over [0, 1]

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \cdots, \sum_j p_j = 1$$

$$X = \begin{cases} x_0 & \text{If } U < p_0 \\ x_1 & \text{If } p_0 \le U < p_0 + p_1 \\ \vdots & \vdots \\ x_j & \text{If } \sum_{i=0}^{j-1} p_i \le U < \sum_{i=0}^{j} p_i \\ \vdots & \end{cases}$$

$$0 < a < b < 1 \longrightarrow P\{a \le U < b\} = b - a$$

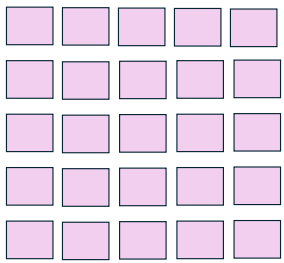
$$P\{X = x_j\} = P\left\{\sum_{i=0}^{j-1} p_i \le U < \sum_{i=0}^{j} p_i\right\} = p_j$$

122

6

Last Time

- Last time
  - PRNG, Uniform distribution
  - Random subset: Any of the $\binom{n}{k}$ possibilities should be equally likely
- Example
  - Moderation: Choose k out of n
  - 2 out of 5



7

$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

k=2, n=5

Example

- Samples
  - 0.79 0.48 0.19 0.24 0.97
    1. RHS = 2/5 Do not select Item 1 since $0.79 \not< 0.4$
    2. RHS = 2/4 = 0.5. Select item 2 since 0.48 < 0.5
    3. RHS = 1/3 = 0.33. Select 3 since 0.19 < 0.3
    4. We are done, we have selected 2 items

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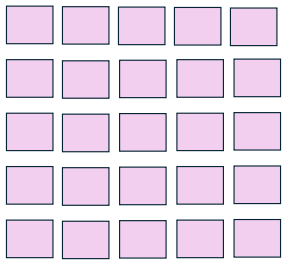
Last Time

- Last time
  - PRNG, Uniform distribution
  - Random subset
- Example
  - Moderation
  - 2 out of 5



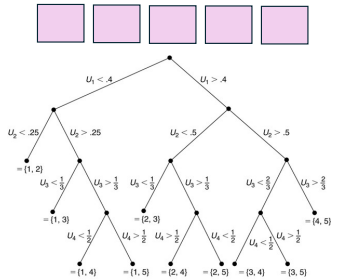
10

$$I_j = \mathbf{1}\left\{U_j < \frac{k - I_1 - \cdots - I_{j-1}}{n - j + 1}\right\}$$

k=2, n=5

Example

- Samples
  - 0.79 0.48 0.19 0.24 0.97
- Chance of getting to any of the leaf nodes is exactly any of the $\binom{n}{k}$



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Agenda

- Last time
  - PRNG
  - Uniform distribution
  - Random subset
- Today
  - Normal distribution

Hello, "Random" World!

```
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
        std::cout << dist(mt) << " ";
    }
    std::cout << std::endl;
}
```

Engine: [0, 2^{32})

Deterministic 32-bit seed

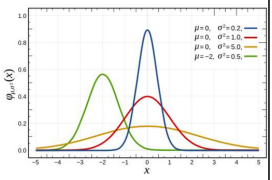
Distribution: [0, 99]

Note: [inclusive, inclusive]

14

Normal Distribution

- A continuous random variable with parameters mean μ and variance σ^2
  - This pdf is symmetric about the mean μ and has the shape of the "bell curve".
  - If μ=0 and σ=1, it is called the standard normal distribution



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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15

Example: Pulse-rate of 50 students

Can calculate statistic such as sample mean (79.1) and sample standard deviation (7.6) but we can do more!

Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let's plot these numbers

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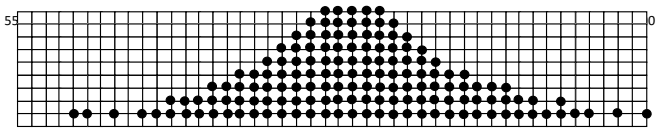
Distribution

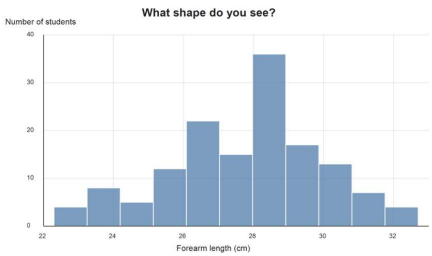
The distribution approaches a normal symmetrical distribution

150 students



17

Self-reported forearm length



~140 students (male and female)

18

Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Verifying validity of pdf
  - Consider Gaussian pdf with mean 0 and variance ½

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad X \sim N(0,1)$$

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \, dr \, d\theta = \pi$$

20

Mean and MGF

$$E(X-\mu) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu) e^{-(x-\mu)^2/(2\sigma^2)} dx = 0$$

$$\sigma \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-y^2/2} dy = -\sigma \frac{1}{\sqrt{2\pi}} \, e^{-y^2/2} \Big|_{-\infty}^{\infty}$$

$$\Phi_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

66

21

Variance

$$E[(X-\mu)^2] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu)^2 e^{-(x-\mu)^2/(2\sigma^2)} dx = \sigma^2$$

R.V. pdf

$$\int u\,dv = uv - \int v\,du$$

$$\int y e^{-y^2/2} dy = -e^{-y^2/2}$$

67

22

Properties

- Median = mean
  - Because of symmetry of the pdf about the mean
- Mode = mean
  - Set the first derivative of the pdf to 0

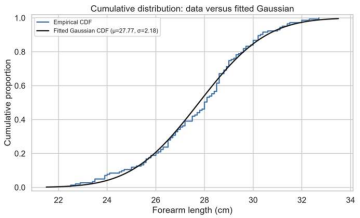
69

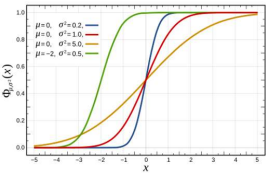
23

Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Cumulative distribution function





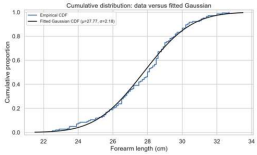
$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{x^2}{2}} dx$$

24

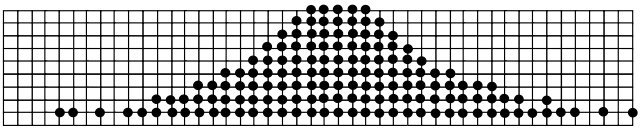
Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time
  - Normal distribution
- Today
  - More on normal distribution (why?)



Forearm length of ~150 students

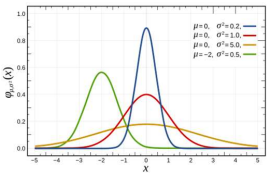


Plot of pulse rate of ~150 students

2

Normal Distribution

- A continuous random variable with parameters mean μ and variance σ^2
  - This pdf is symmetric about the mean μ and has the shape of the "bell curve".
  - If μ=0 and σ=1, we have the standard normal distribution
- We take the positive square root of the variance, conventionally called the standard deviation



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

63

3

Pulse-rate of 50 students

Can calculate statistic such as sample mean [79.1] and sample standard deviation [7.6] but we can do more!

Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let's plot these numbers

4

sample mean (79.1) and sample standard deviation (7.6)

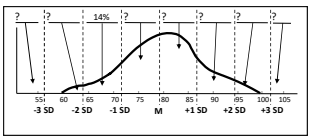
Fill in the "?" mark values!

Example

Percentages: 0, 2, 14, 34, 32, 16, 2, 0

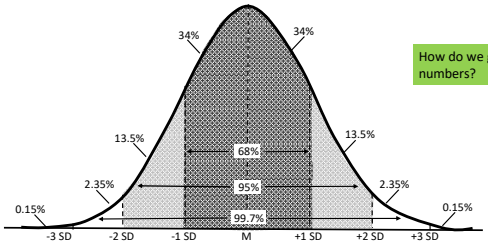
- Consider the 50 students
  - Data has been sorted
  - Underlined 7 data are not measured but computed for reference
- We can bin data based on standard deviations
  - 7 observations (out of 50) between -1 and -2
- … and classify data as, e.g., low pulse rate

56.3	62	63.9	64	65	66	68	70	71	71
71.5	72	72	73	74	74	75	75	76	77
77	77	78	78	78	79	79	79	79.1	80
80	80	80	81	81	81	81	82	82	82
83	83	85	85	86	86.7	87	87	88	89
90	90	92	94	94.3	96	101.9			



5

Normal Distribution

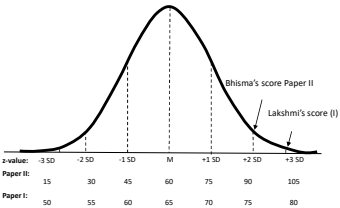


How do we get these numbers?

6

Mahalanobis distance

- Suppose two students Lakshmi and Bhisma, in JEE Mains Papers I & II, got 80% marks and 90% marks respectively
  - Which student should be ranked higher?
- The generalization of $\frac{(x-\mu)}{\sigma}$ in higher dimensions is the Mahalanobis distance
- The z-score of x is the number of standard deviations x is from the mean



z-value:	-3 SD	-2 SD	-1 SD	M	+1 SD	+2 SD	+3 SD
Paper II:	15	30	45	60	75	90	105
Paper I:	50	55	60	65	70	75	80

29 June is celebrated as **National Statistics Day** (birth anniversary of P.C. Mahalanobis)

8

Properties

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, what is pdf of Y?

Proof on board

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$

9

Properties

- Z is a standard, or unit normal distribution
- All probability statements of X can be made in terms of probabilities of Z
- Therefore, we study $F_Z(z)$

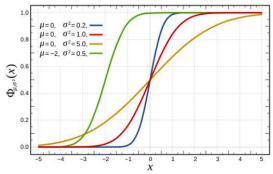
If $X \sim N(\mu, \sigma^2)$ and $Z = \dfrac{X - \mu}{\sigma}$, then $Z \sim N(0, 1)$

10

Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $Z = \frac{(X-\mu)}{\sigma}$ is a standard, or unit normal distribution
- Cumulative distribution function for the standard normal
  - Cannot get numbers like 68-95-99 without numerical integration
  - Closely related to the "erf" function

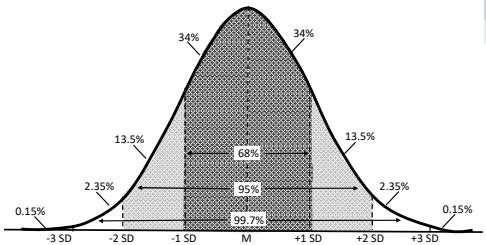


$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{x^2}{2}} dx$$

11

Normal Distribution

n	$\Phi(n)$-$\Phi(-n)$
1	68.2%
2	95.4%
3	99.7%
4	99.99366%
5	99.9999%
6	99.9999998%



13

Example

- Find the probability X < 180 when μ=174 and σ=6.57
  - In calculations involving a continuous random variable, we usually do not have to worry about whether interval endpoints are included or excluded
  - normcdf(180, 174, 6.57)
    - ans=0.8194

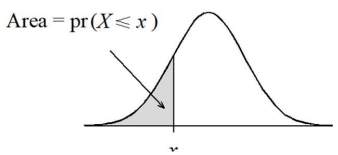


TABLE A1 *Standard Normal Distribution Function:* $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2}\, dy$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621

14

Basic method for obtaining probabilities

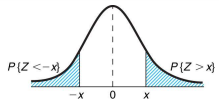
- Sketch a Normal curve, marking the mean and other values of interest.
- Shade the area under the curve that gives the desired probability.
- Devise a way of getting the desired area from lower-tail areas.
- Obtain component lower-tail probabilities from a computer program (or in the exam, provided tables)

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035

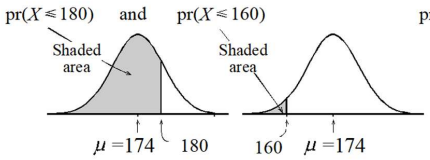
15

Example

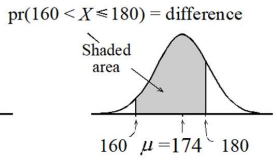


- Find the probability 160 < X < 180 when μ=174 and σ=6.57

Programs supply



We want



16

Example

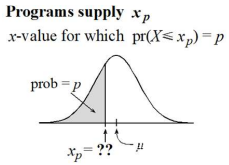
- Find $\Pr(z \leq 1.1357)$
- Correct the z-value to two decimal places z=1.14
  1. Look down the first column "z" to find 1.1. This give you the row.
  2. Look across with the second decimal place to find the required cell

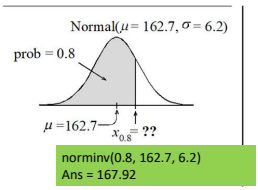
z	0	1	2	3	4	5	6	7	8	9
1.0	.841	.844	.846	.848	.851	.853	.855	.858	.860	.862
1.1	.864	.867	.869	.871	.873	.875	.877	.879	.881	.883
1.2	.885	.887	.889	.891	.893	.894	.896	.898	.900	.901
1.3	.903	.905	.907	.908	.910	.911	.913	.915	.916	.918
1.4	.919	.921	.922	.924	.925	.926	.928	.929	.931	.932

17

Inverse probability: p-Quantile

- Find the woman whose height is at the 80th percentile if the mean is 162.7 and standard deviation is 6.2





18

Inverse Probability

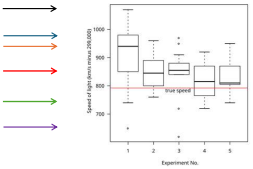
- Other than the sample mean and the sample variance, another way to summarize is the percentile
  - Toy example (each data is associated with a percentile)
    - Marks: 100, 90, 75, 70, 60
    - Percentile: 100, 80, 60,40,20
- The sample $100p$ percentile $(0 \leq p \leq 1)$ is defined as the data value y such that $100p\%$ of the data have a value less than or equal to y, and $100(1-p)$ % of the data have a larger value
  - For a data set with n sample points, the sample $100p$ percentile is that value such that at least np of the values are less than or equal to it.
  - And at least $n(1-p)$ of the values are greater than or equal to it

19

Box Plot

- The sample 25 percentile = first quartile, the sample 50 percentile = second quartile, the sample 75 percentile = third quartile
  - Length of the box is IQR: Interquartile range

- Example: 42 values
  - Low is 57, High is 70
  - First quartile: 0.25*42=10.5
  - Median
  - Third: 0.75*42=31.5



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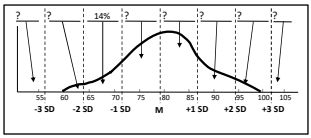
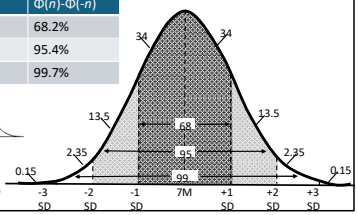
Agenda

- Last time
  - Percentile and box-plots
  - Normal distribution
    - Transformations, Leibniz rule
      - z-score and Mahalanobis distance
      - Proportions
    - Inverse-probability or p-quantile
- Today
  - More on normal distribution (why?)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



n	$\Phi(n)$-$\Phi(-n)$
1	68.2%
2	95.4%
3	99.7%



$$X \sim N(\mu, \sigma^2),\ Y = aX + b,\ Y \sim N(a\mu + b, a^2\sigma^2)$$

1

Why normal?

- So far, we have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - If each of the r.v. are identical
    - $E[Y] = nE[X_i]$
    - $Var[Y] = nVar[X_i]$
  - What about the distribution of Y?
    - Binomial: $X + Y \sim \text{Bin}(n + m, p)$

2

Sums of i.i.normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- What is the distribution of Y?
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
  - Proof via Moment Generating Functions
  - First, the MGF of a unit normal

3

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

MGF of standard normal

$$E[e^{tZ}] = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2-2tx)/2} dx$$

$$= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx$$

$$= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy$$

$$= e^{t^2/2}$$

4

MGF of Normal

- If Y = aX + b

$$\phi_Y(t) = e^{tb}\,\phi_X(at)$$

$$E[e^{tX}] = E[e^{t\mu + t\sigma Z}] = E[e^{t\mu}e^{t\sigma Z}] = e^{t\mu}E[e^{t\sigma Z}] = e^{t\mu}e^{(\sigma t)^2/2} = e^{\mu t + \sigma^2 t^2/2}$$

5

Sums of i.i.normal distribution

- If X and Y are independent
- Aligning the MGF to the MGF in the previous slide we see
  - $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$

$$\phi_{X+Y}(t) = \phi_X(t)\,\phi_Y(t)$$

$$E\left[e^{t\sum_{i=1}^{n} X_i}\right] = \prod_{i=1}^{n} e^{\mu_i t + \sigma_i^2 t^2/2}$$

$$= e^{\mu t + \sigma^2 t^2/2}$$

6

Example

- Rain in Mumbai
  - Mean annual rainfall = 231 cm, standard deviation 56 cm.
  - Find the probability that rain in 2026 falls short of 2025 by 50 cm
- If we did not know the rainfall in 2025
  - Let $X_1, X_2 \sim \mathcal{N}(231, 56^2)$
  - Z = $X_1 - X_2 \sim \mathcal{N}(0, 2(56^2))$
  - $P\left(Z > 50/\left(56\sqrt{2}\right)\right) = P(Z > 0.63) \approx .0.26$

Assumes r.v. X_1, X_2 are normal

7

Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - If each of the r.v. are identical
    - $E[Y] = nE[X_i]$
    - Var[Y] = nVar[X_i]
    - So the standard deviation of Y is related by the $\sqrt{n}$ factor
  - If each r.v. is also normal, then the distribution of Y is also normal
  - *What if the r.v. comes from identical, but from a non-normal distribution?*

9

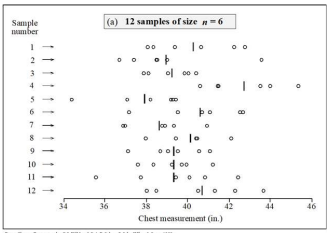
Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - What if the r.v. comes from a different distribution?
- Central limit theorem
  - For large n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
- Let's understand this theorem, and instead of looking at the sum, look at the expression $\frac{Y}{n}$, i.e., the sample mean (for convenience)

10

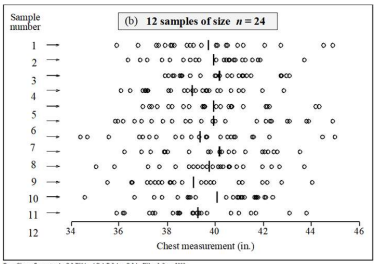
Example

- Twelve samples of size 6 from 5768 soldiers
- Mean is shown with the vertical bar
- Observation: The sample mean is not the same
  - The sample mean is a random variable
  - We will use the sample mean as a proxy for the sums of random variables from an unknown distribution



11

Example



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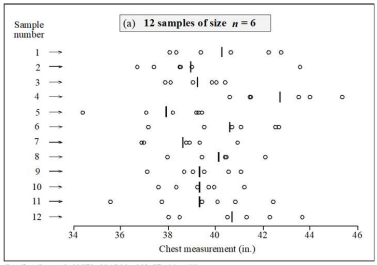

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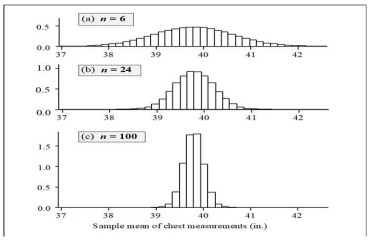
12

Example

- With twelve samples, the histogram is noisy
- Instead of twelve samples of two different sizes, we take 100,000 samples of 3 different sizes
- In each case, the distribution of the sample means is approximately normal



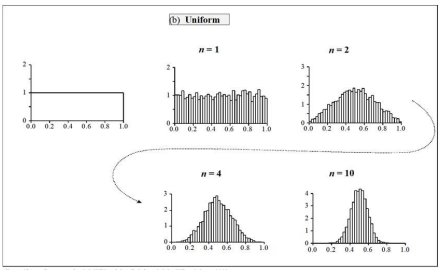
13

Why normal?

- We have seen for independent random variables, and $Y = X_1 + X_2 + \cdots X_n$
  - What if the r.v. comes from a different distribution?
- Central limit theorem
  - For large n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
- How large is n?

14

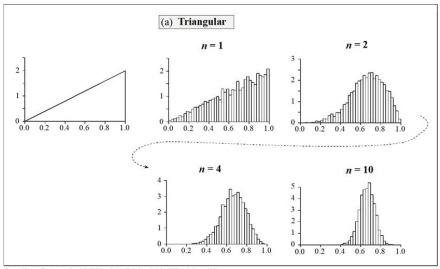
Uniform distribution



5000 observations

15

Triangular distribution



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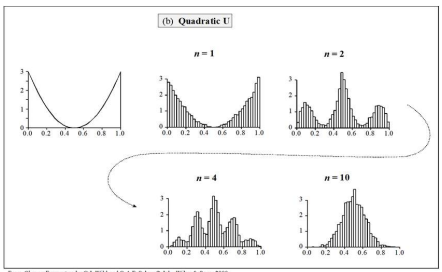
5000 observations

16

Quadratic



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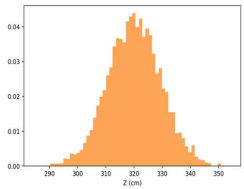
5000 observations

17

Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time: Why normal?
  - z-score and proportions
  - Sums of
    - independent but
    - not identical
    - normal distribution
  - The sum is normal
  - Central Limit Theorem



2

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and *independent*
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem,
  - What happens if measure heights X_1 of 100 selected national basketball players (normal distribution)
  - and, say, 100 kho-kho players X_2 (also normal distribution)
  - Assume independence
  - Put them all in this room
  - Distribution?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

3

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
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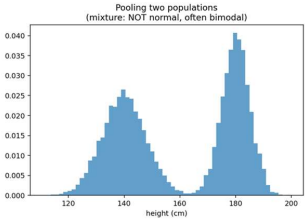
  - and, say, 100 kho-kho players X_2?
  - Put them all in this room
  - Distribution?



5

Sums of independent normal distribution

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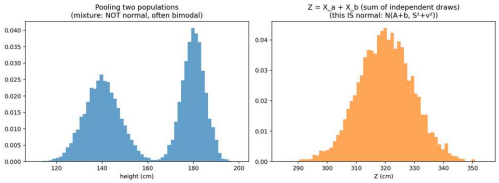
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

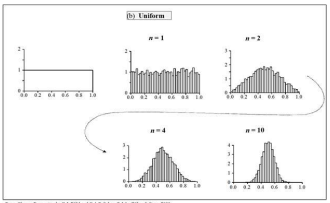


6

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Agenda

- Last time: Why normal?
  - z-score, proportions
  - Sums of independent but not identical normal distribution
  - Sums of independent identical distributions, not necessarily normal (Central Limit Theorem)



7

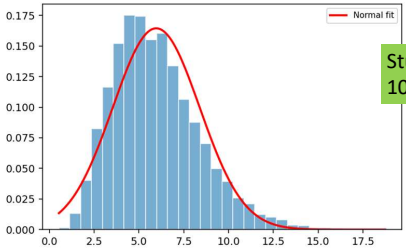
Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large n, the distribution of Y(n) is approximately normal with mean and variance as earlier seen as if the distributions were normal
- How large is n?

The statement of the theorem is not precise

8

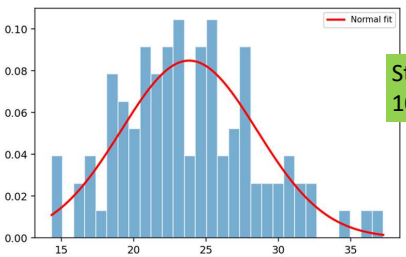
Which distribution is “more” normal?



Student A: n = 6, 10000 experiments

9

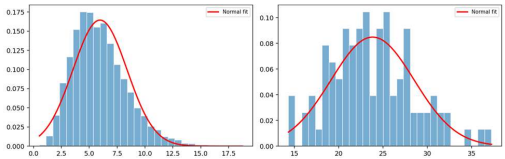
Which distribution is “more” normal?



Student B n=24, 100 experiments

10

Which distribution is “more” normal?



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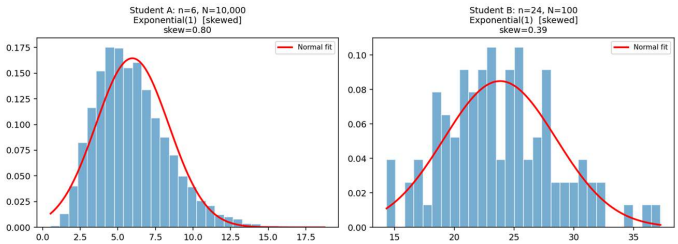
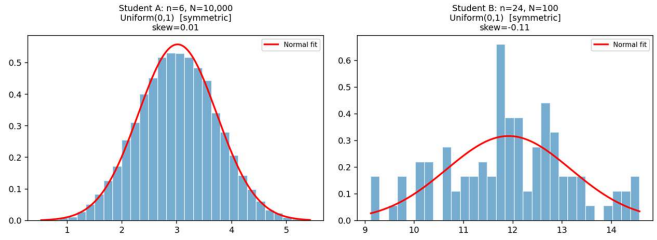
11

Which distribution is “more” normal?



13

But …



14

Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Notice that it helps to know the mean and variance of typical distributions corresponding to X_i

Takeaway message

15

Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \sim \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
- Exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{Y-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

16

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial

- Your CR is organizing "garba" in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to "overbook" the event maximizing cover fee
  - How many registrations can he accept if the probability of more than 100 showing up is 5%?
- X number of students: Bin$(n, 0.7)$
  - What is the largest n for "good case" $\longrightarrow P(X \geq 101) \leq 0.05)$

17

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial

- Your CR is organizing "garba" in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to "overbook" the event maximizing cover fee
- X number of students: Bin$(n, 0.7)$
- What is the largest n for "good case" $\longrightarrow$ $P(X \geq 101) \leq 0.05)$
- Binomial: For each n,
  - Evaluate $P(X \geq 101) = \sum_{k=101}^{n} \binom{n}{k} (0.7)^k (0.3)^{n-k}$
  - For example, with n = 131, $P(X \geq 101) = \sum_{k=101}^{131} \binom{131}{k} (0.7)^k (0.3)^{131-k} \approx 0.0441$

18

Example: Normal and Binomial

$$E(X) = np$$
$$\mathrm{Var}(X) = np(1-p)$$

- How many registrations can he accept if the probability of more than 100 showing up is 5%?
  - $P(X \geq 101) \leq 0.05)$
- Normal approximation $X \approx \mathcal{N}(0.7n, 0.21n)$
  - Continuity correction
  - Map to z-score
  - Inverse probability and one-sided tail
  - Solve quadratic equation
- Complicated binomial sum has same answer as quadratic equation

$$P(X \geq 101) \approx P(Y > 100.5),$$

$$P\left(Z > \frac{100.5 - 0.7n}{\sqrt{0.21n}}\right) \leq 0.05.$$

$$P(Z > 1.645) = 0.05,$$

$$\frac{100.5 - 0.7n}{\sqrt{0.21n}} \geq 1.645.$$

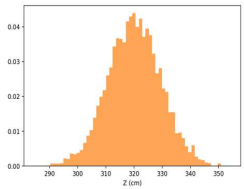
$$n \approx 131.24.$$

19

Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time:
  - z-score and proportions
  - Sums of independent but not identical normal distribution
  - Central Limit Theorem
    - "Common man" version
    - Technical version

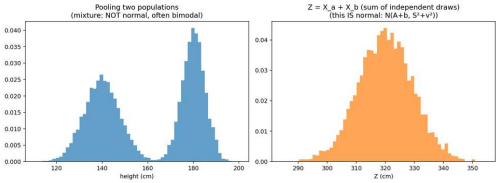


2

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture



3

Why normal?

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
    - Take home message emphasizes the division by n
    - Although Y above is actually Y_n

Take home message

4

Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \overset{\text{iid}}{\sim} \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
  - We know exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{\text{X}-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

5

Central limit theorem

More precise statement

- $X_1, X_2, \ldots, X_n$ independent and identically distributed random variables each with mean μ and variance $\sigma^2 < \infty$
  - $Y_n \triangleq X_1 + X_2 + \cdots X_n$
- As $n \to \infty$, the distribution Y_n converges to that of $\mathcal{N}(0, \sigma^2)$
  - Y_n converges in distribution to normal

The statement is more precise but a tad awkward

Ignore Typo:

See next slide

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right)$$

Convergence in distribution (see here for more)

78

6

CLT

Why the complicated statement?

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Untangling

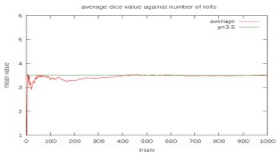
$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \;\dot{\sim}\; N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

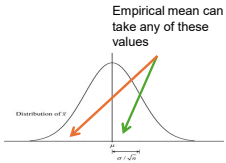
$$\frac{1}{n}\sum_{i=1}^{n} X_i \;\dot{\sim}\; N(\mu, \sigma^2/n)$$

7

CLT and LLN



- The law of large numbers says that the empirical mean calculated from a large number of samples is equal to (or very close) to the true mean μ (of the distribution from which the samples were drawn)
- The central limit theorem says that the empirical mean calculated from a large number of samples is a random variable following a Gaussian distribution with mean equal to the true mean μ



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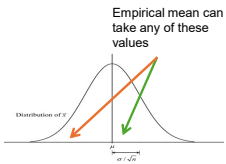
8

CLT and LLN



- Limiting argument as $n \to \infty$
  - The variance drops to zero

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \dot{\sim} N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i \dot{\sim} N(\mu, \sigma^2/n)$$

83

9

The weak law of large numbers

- Let X_1, X_2, …, X_n be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

Must know

10

10

CLT

- As stated, the random variables X_i are independent and identically distributed.
- Alternate: require independence but allow the random variables to belong to different distributions
  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
  - "take home message" … it works

- Misconceptions
  - "Lots" of random sampling "reveals" normal distribution
  - Does not apply to random sampling of any variable
    - Applies to sampling distribution of mean
  - Empirical rule of thumb (n=30) can lead to seriously flawed inferences

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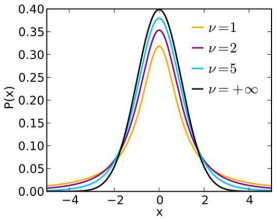
11

Agenda

- Last time
  - Central Limit Theorem
- Today
  - Estimates and Estimators
  - Means and proportions in the context of CLT
  - Standard Error and Approximations
  - The "student" distribution



1

Expectation and Variance of iid

- For identical X_i random variables, with $\mathbb{E}[X_i] = \mu$
  - $Y = X_1 + X_2 + \cdots X_n$ and $\bar{X} = Y/n$, $\mathbb{E}[\bar{X}] = \mu$
  - With independence, and finite variance σ^2, $\text{Var}(\bar{X}) = \sigma^2/n$
- Proportions
  - In the 1998 Teen survey, 1000 American teenagers (12 – 17 years) were interviewed. 27% reported having seen illegal drugs sold at school
  - Although this sort of thing is a hypergeometric distribution, we model this is W$\sim$ Bin(n, p) (why?)
    - Here W = Bin (1000, 0.27)
    - The random variable $\hat{P}$ = W/n represents proportions
  - $\mathbb{E}[\hat{P}] = \text{p}, \text{Var}(\hat{P}) = \text{p}(1-\text{p})/n$

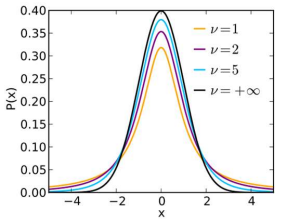
2

Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to index the distribution
- You don't observe the whole population, you get a **random sample.**
  - A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$ that does not depend on unknown parameters *before* observation
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
- After you actually observe, an ***estimate*** is a realized quantity calculated from the data to estimate an unknown parameter
  - Capital letters refer to random variables
  - Small letters refer to observed values

3

CLT Approximation



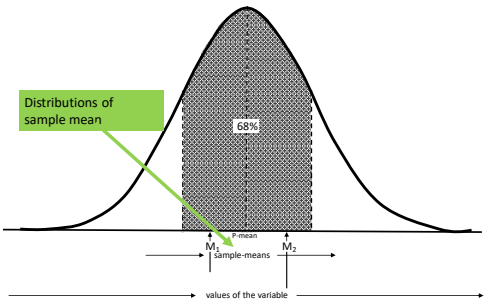
4

CLT

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
- Central limit theorem
  - For large (but finite) n, the distribution of Y is *approximately* normal with mean = $n\mu$ and variance $n\sigma^2$ (as if the distributions were normal)
  - The distribution of the sample mean $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Recall what we mean by "distribution of the sample mean"

5

Distribution of sample means



6

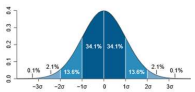
Example

- Observations $x_1, x_2, \ldots x_{100}$ [n=100] (or $x_1, x_2, \ldots x_{10}$) [n=10]
- Sample mean of observations: "estimate" denoted by $\overline{x}$
- If we want to ask questions like "what are the chances that the estimate will be …" we are abstracting and thinking about the random variable $\bar{X}$ (the estimator)
  - The CLT says that $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ regardless of where X_i came from (for large n)

89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

7

CLT Example



- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
  - This is a single sample from your friend with choice of n=10000
- Let $X_1, X_2, \ldots, X_n$ be the random variables indicating whether the coin toss was a success (head).
  - These are i.i.d. random variables whose sum is a random variable with:
  - Expected value $n\mu = 10000(0.5) = 5000$, $\sigma = \sqrt{(10000 * 0.5 * 1.5)} = 50$
  - By CLT, $\bar{X} \approx \mathcal{N}(5000, 2500)$ and the z-score corresponding to 5200 (with continuity correction) is (5199.5-5000)/50 = 3.99 (4 standard deviations)
- Your friend is joking (why?)

Typo 0.5 not 1.5

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Review

- We use both $\bar{x}$ and $\bar{X}$ to refer to a sample mean. For what purposes do we use the former and for what purposes do we use the latter?
- What is meant by "the sampling distribution of $\bar{X}$?
- How is the population mean of the sample average $\bar{X}$ related to the population mean of individual observations?
- How is the population standard deviation of $\bar{X}$ related to the population standard deviation of individual observations?
- What happens to the sampling distribution of $\bar{X}$ if the sample size is increased?

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CLT for proportions

- 1000 American teenagers were interviewed. 27% reported having seen illegal drugs sold at school
- A single sample proportion is denoted by $\hat{p}$ (Here $\hat{p} = 0.27$)
- The corresponding random variable is $\hat{P}$
- The sample proportion $\hat{p}$ is an estimate of the true population parameter p from Bin(n,p)
- CLT: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$
  - In this example, B = Bin(1000, 0.27) itself is a (good) approximation, and the CLT is a further (good) approximation before

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CLT for proportions

- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
- Let $\hat{p}$ be the sample proportion of heads
  - $E[\hat{p}] = p = 0.5,\ \ \mathrm{sd}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.25}{10000}} = 0.005$
- With continuity correction, zscore $= \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} = \frac{0.51995-0.5}{0.005} = 3.99$
  - Your friend is joking

Though not wrong, $se(\hat{p})$ is more appropriate

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Review

You want	Continuity correction on count	On proportion
$P(X \geq k)$	$k - 0.5$	$(k-0.5)/n$
$P(X \leq k)$	$k + 0.5$	$(k+0.5)/n$
$P(X = k)$	$[\mathrm{k} - 0.5, \mathrm{k} + 0.5]$	$[(k-0.5)/n, (k+0.5)/n]$

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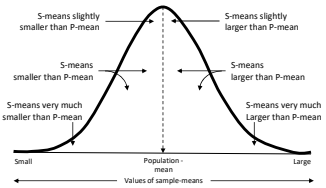
Sample Mean vs Population Mean



A single sample mean will give us a value that is different from the population mean

How bad is it?

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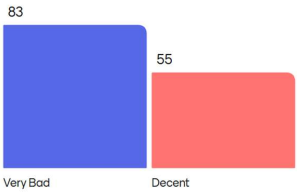
How bad is a single sample?

How bad is a single sample? **Can't say is not an option**



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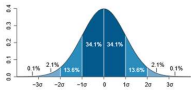
Logic of sampling



- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes, the population is normal in many cases, i.e., $\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ (note the approximation)
- We will address the second point shortly ("student distribution")
- Define: $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$ where $s_x^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$
  - This is called the standard error of the sample mean

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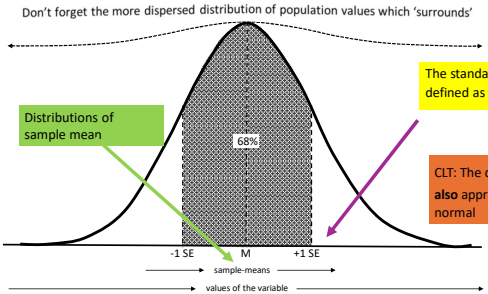
Standard Error



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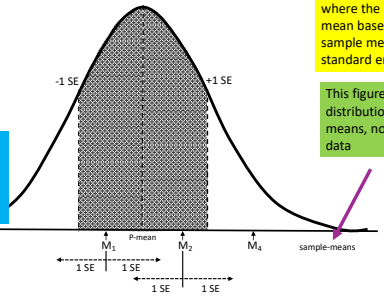
Logic of Sampling

If the estimated mean is M2, 68% of the time our population mean will be within 1 standard error of M2

If we take a sample survey of 100 students, and the estimated mean is 50, and estimated standard deviation is 15, then we can say with 68% confidence that …

And with 95% confidence that the population mean is between 47 and 53



This figure tells us a very good idea about where the population mean based on a sample mean and standard error

This figure is about the distribution of sample means, not the actual data

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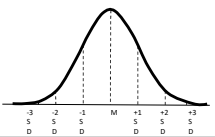
The population mean is fixed These are informal statements The exact def will appear in subsequent lectures the concept is illustrated here

From Samples To Population

- Given a single pulse rate n=1, say 90, what can we say about the population distribution?
  - With n=1, our sample mean will not be accurate (but it won't be too off)
  - If we have a mean of a number of samples, we will have a sample standard deviation which will be different from the population standard deviation
- How wrong will we be? The size of the standard error of the mean depends on
  - The standard deviation within the sample
  - The size of the sample
  - The proportion of the population covered by the sample



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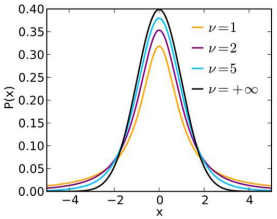
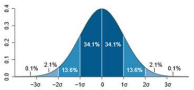
Agenda

- Last time
  - Informal approach to confidence intervals
  - Approximate distribution of the sample mean using CLT
  - Approximate distribution of the sample proportion using CLT
- Today
  - The "student" distribution



Logic of sampling



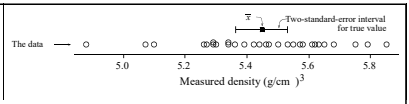
- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes,
  - The population is normal in many cases, i.e., $\overline{\text{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Further, we have an even more convenient result $\bar{X} \approx \mathcal{N}(\mu, \text{se}(\bar{x}))$

Also at sp se²(x)

This is a pedagogical development

Later in this lecture we see that the Student's t distribution is appropriate

Example



- In 1798 Cavendish measured the density of the earth
  - sample mean $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3
  - Therefore se($\bar{x}$) = 0.04102858
- From the theory, the true mean is $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- The number of digits of precision is unwarranted so we can say that Cavendish predicted true value between 5.37 and 5.53
- Current accepted value is ~ 5.517

5.50	5.61	4.88	5.07	5.26	5.55	5.36	5.29	5.58	5.65
5.57	5.53	5.62	5.29	5.44	5.34	5.79	5.10	5.27	5.39
5.42	5.47	5.63	5.34	5.46	5.30	5.75	5.68	5.85	

3

Review

- Why is the standard deviation of $\bar{X}$ not a useful measure of the precision of $\bar{X}$ as an estimator in practical applications?
- What measure of precision do we use in practice?
- How is se($\bar{x}$) related to the standard deviation of $\bar{X}$?
- When we use the formula se$(\bar{x}) = s_x/\sqrt{n}$, what is s_x and how do you obtain it?

4

Beyond Approximation

- Although a good practical tool, there are two approximations
  - The CLT approximation for large n
  - The "se($\bar{x}$)" approximation for the unknown standard deviation
- Both approximations are problematic for small n
- If the population were known to be exactly normal, then we have exact results

5

Distribution of Sample mean

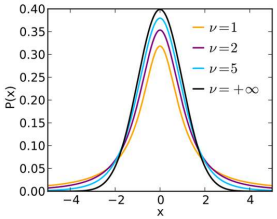
- For independent identical (iid) **normal** random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\bar{X} = Y/n$, $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
  - $\mathbb{E}[\bar{X}] = \mu, \mathrm{Var}(\bar{X}) = \sigma^2/n$
  - $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal
- $\bar{x}$ is an estimate of the sample mean $\bar{X}$
  - $\bar{x} = \sum_i x_i \,/n$
  - This formula comes from the estimator $\bar{X}$ by simply observing the values

Nothing changes except (!) that this is exact

6

The student t-distribution

- If the population were known to be exactly normal, then we have the result $T_n = (\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - This is the student t-distribution with parameter "n-1"
- Here we have introduced
  - A new statistic S (it is a statistic not an estimator)
  - A new distribution T_n (df) where "df" stands for degrees of freedom



This is new, we get to use "S" instead of σ

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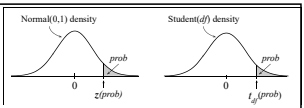
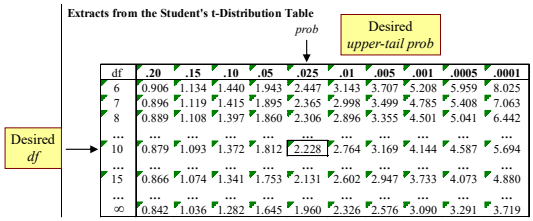
Statistic for the variance



- S is easily recognizable
  - We define the random variable $S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}$
  - S is the square root
- Thus, we have the practical result $(\bar{X} - \mu)/\mathrm{se}(\bar{X}) \sim t_{n-1}$
- For small n,
  - using t-values rather than z-values is robust even for non-normal distribution (empirical result)
  - For proportions, when the binomial assumption is valid, t is not needed
- For large n, t-distribution tends to the standard normal anyway
- In stat courses, tables for the t-distribution are available

8



df	.20	.15	.10	.05	.025	.01	.005	.001	.0005	.0001
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
...	...	...	...	...	...	...	...	...	...	...
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
...	...	...	...	...	...	...	...	...	...	...
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
...	...	...	...	...	...	...	...	...	...	...
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.719

Some results

… and we just introduced T out of the blue!

9

#1: Expectation of S^2

- We can show that $E(S^2) = \sigma^2$
  - Given $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ we can show that $S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\overline{X}^2}{n-1}$
  - $E[(n-1)S^2] = E\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right) = nE\left(X_i^2\right) - nE(\overline{X}^2)$
    - But $E(W^2) = \mathrm{Var}(W) + E(W)^2$
  - $E[(n-1)S^2] = n\,\mathrm{Var}(X_1) + nE(X_1)^2 - n\,\mathrm{Var}(\overline{X}) - nE(\overline{X})^2$
  - $E[(n-1)S^2] = n\sigma^2 + n\mu^2 - n(\sigma^2/n) - n\mu^2 = (n-1)\sigma^2$
- Note: this is not the expectation of S

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$$\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}\left[(X_i - \mu) - (\bar{X} - \mu)\right]^2$$

$$= \sum_{i=1}^{N}(X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_{i=1}^{N}(X_i - \mu) + \sum_{i=1}^{N}(\bar{X} - \mu)^2.$$

But

$$\sum_{i=1}^{N}(X_i - \mu) = \sum_{i=1}^{N}X_i - N\mu = N(\bar{X} - \mu),$$

and

$$\sum_{i=1}^{N}(\bar{X} - \mu)^2 = N(\bar{X} - \mu)^2.$$

Therefore

$$\boxed{\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}(X_i - \mu)^2 - N(\bar{X} - \mu)^2}$$

11

Bessel Correction

- If we had started with n in the denominator
- Then we have ended up with $nE(S^2) = (n-1)\,\sigma^2$
- In such the expectation would have a (n-1)/n factor
  - This is the Bessel's correction
  - For n = 2, we would have ended up with E(S^2) = ½ σ^2
  - This is exactly what we saw in the experiment on the first day of class

12

Experiment

Now focusing on the last two columns

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	((0-0)²+(0-0)²)/1 = 0	0
0 2	1	((0-1)²+(2-1)²)/1 = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

13

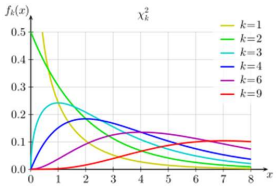
#2: Independence

- For independent identical (iid) **normal** random variables X_i with mean μ and variance σ^2
  - $\bar{X}$ is normal with mean μ and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$
- Both $\bar{X}$ and S^2 are computed from the same data, yet they are independent, this is a surprising result (not proven here)
- Further
  - $\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2$ which is the sum of standard normals from *iid* X_i
  - The rhs has a name! $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - Thus distribution of the statistic $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

15

Chi-square distribution [1/3]

- If $Z_1, Z_2, \ldots, Z_n$ are independent *standard* normal then
  - $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is said to have a chi-square distribution with n degrees of freedom and is denoted as $X \sim \chi_n^2$
  - $f_X(x) = \frac{x^{n/2-1}e^{-x/2}}{2^{n/2}\Gamma(n/2)}$
  - $\Gamma(y) = \int_0^\infty x^{y-1}\, e^{-x} dx$
- Example (n=1)
  - $Z_1 \sim N(0,1), X = Z_1^2$
  - $f_X(x) = \frac{e^{-x/2}}{\sqrt{x}\sqrt{2\pi}}$



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Chi-square distribution [2/3]

- In stat courses, tables for the chi-square distribution are available for different number of degrees of freedom, and for different values of the probability
- MGF: $\Phi_X(t) = (1-2t)^{-n/2}$ (t < ½)
- Additive property
  - If X1 and X2 are *independent* chi-square random variables with n1 and n2 degrees of freedom respectively, then X1 + X2 is also a chi-square random variable with n1 + n2 degrees of freedom

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Chi-square distribution [3/3]

- Can we use the additive property?
  - $(n-1)S^2 = \sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \sum_{i=1}^{n}(X_i-\mu)^2 - n\left(\overline{X}-\mu\right)^2$
  - $\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \frac{n(\overline{X}-\mu)^2}{\sigma^2}$
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right)^2$

Recognize standard normal

Recognize as χ_n^2

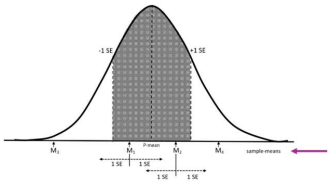
Recognize as $(n-1)/\sigma^2 S^2$

Independent

18

Agenda

- Last time
  - χ^2_n distribution
  - The S^2 statistic
  - Independence of S^2 and $\bar{X}$
- Today
  - Confidence interval



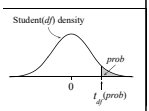
1

Approximate confidence interval

- Earlier: Cavendish conducted experiments seeking density of earth
  - sample mean of 29 observations $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3 and therefore se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Assumptions and problems
  - Experiments may be "biased"
  - Distribution of means is only approximately normal
  - Too many significant digits
- Coarse bound can be strengthened with the "student t-distribution"

3

Confidence interval with t-distribution



- Earlier: Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 se($\bar{x}$) = 0.04102858
  - True mean lies in the interval $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $\Pr\left(-t_{0.025,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{0.025,n-1}\right) = 0.95,$
  - tinv(0.975,28) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.37, 5.53]

Note 28, not 29

Typo 5.36

4

Significant digits

- Cavendish conducted experiments with 29 values
  - $\bar{x}$ = 5.447931 s_x = 0.2209457 $\mu \in 5.447931 \pm 2.0484 \times 0.04102858$
- Coarse bound can be strengthened with the "student t-distribution"
  - tinv(28, 0.975) = 2.0484, so $\bar{x} \approx 5.45$, confidence interval = [5.36, 5.53]
- Rules of thumb for significant digits
  - Intermediate calculations as accurate as possible
  - Estimate to be reported up to the first significant digit of se, se one more significant digit than the se, confidence interval maybe one more magnitude compared to the length of the interval (0.1681 in this case)

Rules of thumb!

Typo tinv(0.975,:-)

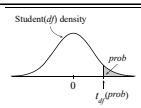
Typo ↷ read estimate instead of se"

Typo: one more "order of"

Example
$\bar{x} = 5.45$
$se(\bar{x}) \leq .041$
If we calculate
CI = [23.76, 45.12]
order of magnitude is 1 = 10^1
So present as [24, 25]

5

A 95% confidence interval



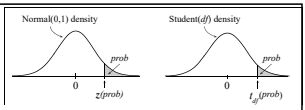
- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors

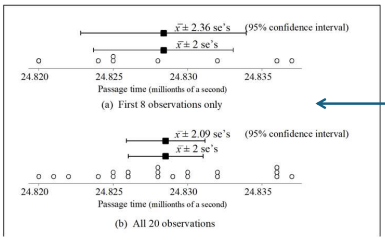
Value of the Multiplier, t, for a 95% CI

df :	7	8	9	10	11	12	13	14	15	16	17
t :	2.365	2.306	2.262	2.228	2.201	2.179	2.160	2.145	2.131	2.120	2.110
df :	18	19	20	25	30	35	40	45	50	60	∞
t :	2.101	2.093	2.086	2.060	2.042	2.030	2.021	2.014	2.009	2.000	1.960

6

Review example





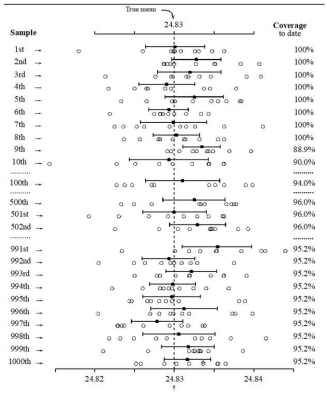
Newcomb's experiment to measure the speed of light in the year 1882

Time for a distance of 3721 m across the river Potomac

7

Example

- 1000 experiments independent samples of size 10 from $\mathcal{N}(\mu = 24.83, \sigma^2 = 0.005^2)$
- In each case the 95% confidence interval is overlaid
- What is your observation?
- Repeated sampling
- Many independent studies
- Planning a study: In my study I will capture the correct value 95% of the time



8

Review

- Effect of increasing the confidence level (df = ∞)

80% CI, $\bar{x} \pm 1.282\ \mathrm{se}(\bar{x})$

90% CI, $\bar{x} \pm 1.645\ \mathrm{se}(\bar{x})$

95% CI, $\bar{x} \pm 1.960\ \mathrm{se}(\bar{x})$

99% CI, $\bar{x} \pm 2.576\ \mathrm{se}(\bar{x})$

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Effect of increasing sample size

- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?

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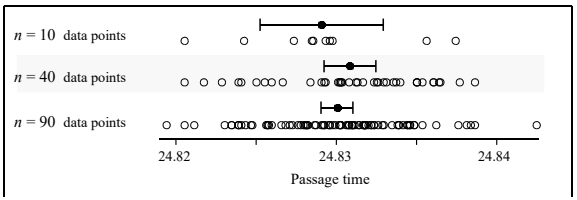
Effect of increasing sample size

- If we have n observations and we want to increase the precision (say double it), how many more observations do we need?



11

But what exactly is this t-distribution?

I can visualize the plot, I get df, I can use it to get the confidence interval, I can do significance digits but …

12

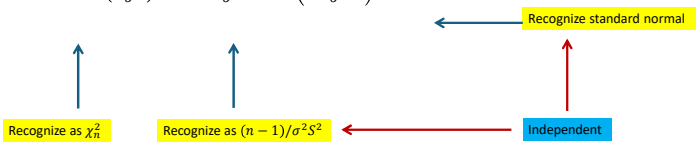
The t-distribution [1/2]

- Definition: If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_{\mathrm{n}}$ are independent then
  - $T = \frac{Z}{\sqrt{V/\mathrm{n}}}$ is said to be a t-distribution with degrees of freedom *n*
- We have been using the result that if population were known to be normal, then $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - We must therefore show that the LHS is a ratio of a standard normal to the square root of an appropriate term
  - Divide the numerator and denominator by $\sigma/\sqrt{n}$
    - To recognize that the numerator is standard normal
    - And denominator is S/σ that is S^2/σ^2

13

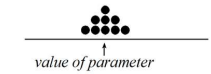
The t-distribution [2/2]

- But
  - $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - And recall the independence from below to complete the result
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2} + \left(\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right)^2$

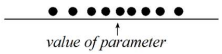


14

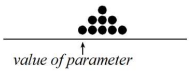
Bias and precision



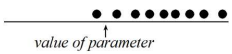
(a) No bias, high precision



(b) No bias, low precision



(c) Biased, high precision



(d) Biased, low precision

15

Unbiased Estimator

- The estimator $\bar{X}$ is an example of an unbiased estimator of the population mean because the expected value of $\bar{X}$ is μ
- The ***standard error*** of any estimate $\hat{\theta}$[denoted se($\hat{\theta}$)] estimates the variability of $\hat{\theta}$ values in repeated sampling and is a measure of the ***precision*** of $\hat{\theta}$

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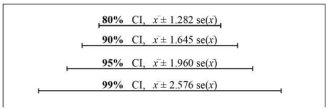
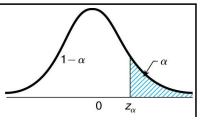
Agenda

- Last week
  - Is $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$?
    - Large N, and Replacing σ by se$(\bar{x})$
  - χ^2_n distribution, The S^2 statistic
  - Independence of S^2 and $\bar{X}$
  - The t-distribution
  - $\mathbb{E}(S^2) = \sigma^2$
  - Confidence Interval
- Today
  - Hypothesis Testing

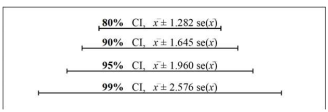


1

Context



- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds



2

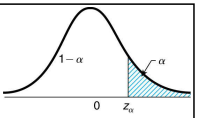
Recap for sample means

- With normal distributions, $(\bar{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$
- The standard error of the sample mean is $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$
- 95% confidence interval: A type of interval that contains the true value of a parameter for 95% of samples
  - For small values of n, and with a normal distribution, and with $\alpha = 0.05$, $\bar{x} \pm t_{1-\alpha/2}(n-1)\text{se}(\bar{x})$ indicates the confidence interval
- Practice: We use the same expression for confidence interval for
  - Small n, and with approximately normal distribution,
  - With the CLT approximation (large n) as a conservative practice

Notation inconsisten

[upper tail notation $t_{\alpha/2}$ used by Ross (survival probability) for t with subscript $P(T > Z)$]

Recap for sample proportions



- Small n: Cannot use the normal approximation for the Bin(n,p)
- Large n: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$ either via Stirling approximation or CLT applied to the Bernoulli
  - More formal: For $p \in (0,1)$ as $n \to \infty, \frac{\sqrt{n}(\hat{p}-p)}{\sqrt{p(1-p)}} \xrightarrow{d} \mathcal{N}(0,1)$
  - Large N: $np(1-p) > 10$ we use $\frac{(\hat{p}-p)}{\text{se}(\hat{p})} \xrightarrow{d} \mathcal{N}(0,1)$ where the standard error of the sample proportion is $\text{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p}}{n}}$
  - 95% confidence interval: With $\alpha = 0.05$, $\hat{p} \pm z_{1-\alpha/2}\text{se}(\hat{p})$
    - t-distribution not applicable; unlike the mean case, the variance is dependent on mean

In Ross $Z_{0.025} = 1.96$

Notation inconsistent with Ross who uses $P(Z > Z_{\alpha/2})$ for Z with subscript

Hypothesis Testing

- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds
  - We cannot conclusively prove a hypothesis
- Science is not a library of truth, it's a debugging process
  - Example: Mass is not absolute (Rest mass vs mass observed): $m_{\text{rel}} = \frac{m}{\sqrt{1-\frac{v^2}{c^2}}}$.
  - Being wrong is not a bug, it's a feature (engine?) of science

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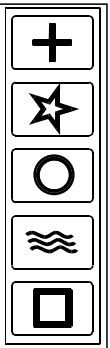
Example #1



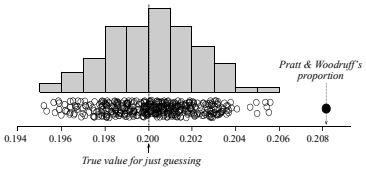
- In 1938, scientists showed 60,000 pictures
  - Students predict what is behind the card
  - If students guessed randomly there would be 12000 correct answers (proportion is 0.2)
  - … but actual result was 12489 (proportion is 0.208)
  - Is 20.8% being larger than 20% a "chance event" or
  - Hypothesis: Students can mind read correct answer in an MCQ

6

Disproving the hypothesis

- Conduct computer simulation
  - of 400 experiments
  - Generate 60,000 possible "answers" on the back of the card
  - Guess with 0.2% probability one of the 5 cards

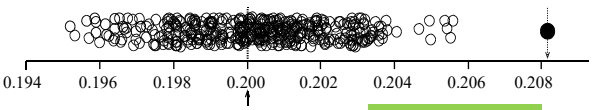
Typo: 20% probability



7

Hypothesis Testing

- We cannot "rule-in" a hypothesized value for a parameter
  - For example, no amount of computer experiments can conclusively say that the proportion is 0.201 vs 0.199
  - The graph would just shift to the right (or left)
- We can only determine whether there is evidence to rule out a hypothesized value

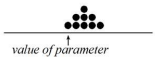


So can students mind-read?

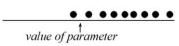
8

Example #2

- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (earlier, n=29)
    - Current accepted value 5.517
  - Hypothesis: Cavendish's experiments were biased



(c) Biased, high precision

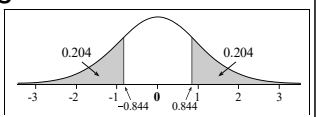


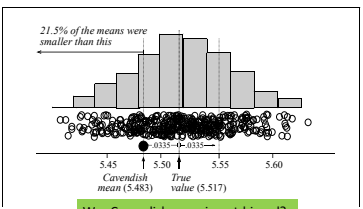
(d) Biased, low precision

9

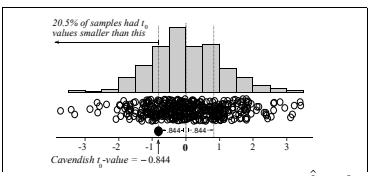
Disproving the hypothesis

- Computer simulation
  - 400 samples from an unbiased experiment





Was Cavendish experiment biased?



$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

10

Review

- What intuitive criterion do we use to determine whether the hypothesized parameter value (p=0.2 in the ESP example, and μ = 5.517 was Cavendish) was *credible* in the light of the data?
- Why was it that μ = 5.517 may be considered credible but p=0.2 was *not credible*?
- Why do we switch to measuring distance between estimate and hypothesized parameter value in terms of standard errors?
  - What do t_0-values tell us?

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Hypothesis Testing

- In settling a research hypothesis, we turn skeptical and assume the research hypothesis is false
  - The null hypothesis is H_0 (or H_o)
  - The alternate is the research hypothesis H_1 or H_a
- Is there data to reject the null hypothesis?
  - Solution:
    - Compute the test statistic
      - Is this statistic "too far" from the null hypothesis?
    - If true, then reject the null hypothesis
    - If false, cannot reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

Our goal is to reject the null hypothesis

12

The t-test

- Using $\hat{\theta}$ to test $H_0: \theta = \theta_0$ versus H_a
  1. Calculate the test statistic t_0
  2. Calculate the p-value
  3. Interpret the p-value

P	%	Translation
> 0.12	-12%	No evidence against H0
0.1	-10%	Weak evidence against H0
0.05	-5%	Some evidence against H0
0.01	-1%	Strong evidence against H0
	-	
0.001	0.10%	Very Strong evidence against H0

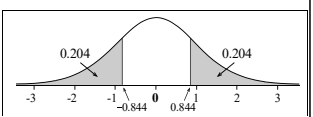
$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\theta} - \theta_0\rvert$ too large)	$P = 2\, pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

13

Example #2

- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (Accepted value 5.517)
  - Research hypothesis: His last 23 experiments resulted in biased measurement
    - Null hypothesis: H_0: His experiments were not biased
    - Compute the t-statistic: -0.844
    - Compute the p-value: 0.204
    - Interpretation: Cannot refute H_o



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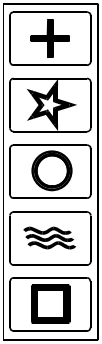
Clarification: p-value
$= 0.204 \times 2 = 0.408$

Example #1

- In 1938, scientists collected 60,000 pieces of data
  - If students guessed randomly
    - there would be 12000 correct answers, but actual result was 12489
  - Research hypothesis: Students mind read correct answer in an MCQ
  - Null hypothesis: proportion is 0.2
    - Compute the t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92
    - Compute the p-value: < 10^{-6}
    - Interpret: Can refute the null hypothesis
    - Inference: Not pure guessing (maybe mind read?)



15

Agenda

- Earlier
  - Hypothesis testing
  - Parameters and Statistic
- Today
  - Parameter estimation
  - MLE

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\theta} - \theta_0\rvert$ too large)	$P = 2\, pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

1

Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter
  - So we have done parameter estimation already! What's new?

2

Parameters

- Bernoulli: $\theta = p$ (probability of success)
- Poisson: $\theta = \lambda$ (rate parameter)
- Uniform: θ is a vector of the upper and lower bounds
- Normal: θ is a vector of μ and σ^2
- More complex distributions such as the beta or the pareto distribution might well have more parameters
- Goal: How to find the parameters

3

Context and workflow

4

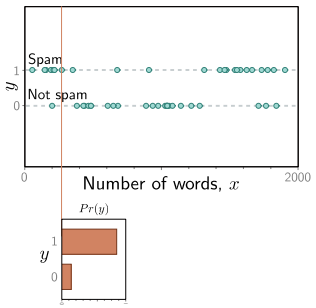
Context

- Step 1: "Real" business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - Is this email spam?
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos seen by a teacher in the next project report

5

Context

- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Is this email spam?
  - Choose Bernoulli as the model, and find out "success" probability



6

Context

- Step 1: Real business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos in project reports seen by a teacher in school
- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Bernoulli? Binomial? Gaussian? Poisson?

7

Context

- Step 3: Collect training samples
  - Number of words in an email message
  - Sample server failure data observed over (say) n=3 days
  - Stock price change over the last 10 days
  - Number of typos in past reports
- Step 4: Estimate the unknown parameters using the training sample

8

Review

- In the Cavendish experiment
  - What was the “real problem”
  - What was the model employed?
  - What kind of data collected?
  - How were the parameters reported? (Hint: use the word “interval” in your answer

9

Fitting parameters to model

10

Components of Solution

- Given: a density $f(x, \theta)$ with parameters θ
- Given: sample: $D = \{x_1, x_2, \ldots, x_I\}$
  - The i-th sample is a random variable X_i assumed to be iid as per $f(x, \theta)$
- Find θ
  - Since D is a finite sample, we cannot really know the actual θ. Best we can do is obtain an estimate of θ.
  - We will denote the estimate as $\hat{\theta}$

11

Three methods

- Maximum likelihood
  - Sample D is all you have got
  - We get a *point* estimator $\hat{\Theta}$
- Maximum a posteriori and Bayesian approach
  - We also have some "prior" information

12

Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of x_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to 1

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

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Bayes' Rule Terminology

Likelihood – propensity for observing a certain value of x given a certain value of y

Prior – what we know about y before seeing x

$$\Pr(y|x) = \frac{\Pr(x|y)\Pr(y)}{\int \Pr(x|y)\Pr(y)\,dy}$$

Posterior – what we know about y after seeing x

Evidence – a constant to ensure that the left-hand side is a valid distribution

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Maximum a posteriori (MAP) [1/2]

- As the name suggests we find the parameters which maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} \Pr(\theta|\mathbf{x}_{1\ldots I}) \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right] \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]\end{aligned}$$

Again we have assumed that data was independent

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Maximum a posteriori (MAP) [2/2]

Maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\hat{\theta} = \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]$$

Point estimate

Since the denominator doesn't depend on the parameters we can instead maximize

$$\hat{\theta} = \arg\max_{\theta} \left[\prod_{i=1}^{I}\Pr(\mathbf{x}_i \mid \theta)\Pr(\theta)\right]$$

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Bayesian Approach [1/2]

Compute the posterior *distribution* over possible parameter values using Bayes' rule:

$$\Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) = \frac{\prod_{i=1}^{I} \Pr(\mathbf{x}_i \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}$$

Not a point estimate

Principle: why pick one set of parameters? There are many values that could have explained the data. Try to capture all possibilities

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Predictive densities for 3 methods

Maximum likelihood:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with ML parameters $\Pr(\mathrm{x}^* | \hat{\boldsymbol{\theta}})$

Maximum a posteriori:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with MAP parameters $\Pr(\mathrm{x}^* | \hat{\boldsymbol{\theta}})$

Bayesian:
Calculate weighted sum of predictions from all possible values of parameters

$$\Pr(\mathbf{x}^* \mid \mathbf{x}_{1\ldots I}) = \int \Pr(\mathbf{x}^* \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) \, d\boldsymbol{\theta}$$

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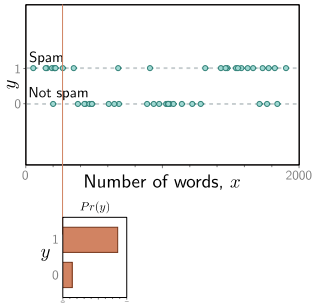
Agenda

- Earlier
  - Typical workflow of a data scientist (Four steps)
  - Estimator and estimate
  - Three methods to get an estimate
  - Point estimate
  - Likelihood vs probability
- Today
  - MLE for normal
  - Linear regression



2

Context and workflow

3

Problem Definition

- Given some data samples $D = \{x_1, x_2, \dots, x_I\}$
- Given some intended distribution $\Pr(\mathbf{x}_{1\dots I}|\boldsymbol{\theta})$
  - Bernoulli: $\theta = p$ (probability of success)
  - Poisson: $\theta = \lambda$ (rate parameter)
  - Uniform: θ is a vector of the upper and lower bounds
  - Normal: θ is a vector of μ and σ^2
- Find the value of the parameter that "optimally" fits the distribution assuming $\mathrm{X}_1, \mathrm{X}_2, \dots, \mathrm{X}_n$ are mutually independent (given θ)

4

Optimal Fitting parameters to model

Three methods

5

Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of X_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to one

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

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MLE Derivation

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Fitting normal distribution: ML

- As the name suggests we find the parameters under which the data $x_{1\ldots I}$ is most likely

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \left[\Pr(\mathbf{x}_{1\ldots I} \mid \boldsymbol{\theta}) \right]$$
$$= \arg\max_{\theta} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i \mid \theta) \right]$$

Likelihood given by pdf

$$\Pr(x \mid \mu, \sigma^2) = \text{Norm}_x[\mu, \sigma^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-0.5 \frac{(x-\mu)^2}{\sigma^2} \right]$$

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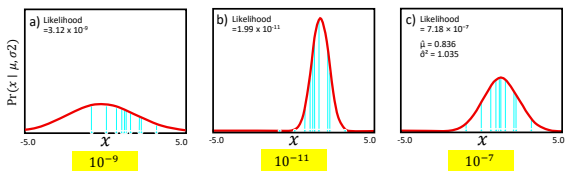
Fitting normal distribution: ML

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I} \Pr(x_i \mid \mu, \sigma^2)$$
$$= \prod_{i=1}^{I} \text{Norm}_{x_i}[\mu, \sigma^2]$$
$$= \frac{1}{(2\pi\sigma^2)^{I/2}} \exp\left[-0.5 \sum_{i=1}^{I} \frac{(x_i-\mu)^2}{\sigma^2} \right]$$

Ten data points



10^{-9} 10^{-11} 10^{-7}

9

Fitting normal distribution: ML

Algebraically: $\hat{\mu}, \hat{\sigma}^2 = \arg\max_{\mu,\sigma^2} \left[\Pr(x_{1\ldots I} \mid \mu, \sigma^2) \right]$

where: $\Pr(x \mid \mu, \sigma^2) = \text{Norm}_x[\mu, \sigma^2]$

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I} \text{Norm}_{x_i}[\mu, \sigma^2] = \frac{1}{(2\pi\sigma^2)^{I/2}} \exp\left[-0.5 \sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2}\right]$$

or alternatively, we can maximize the **logarithm**

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2]\right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log[2\pi] - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]\end{aligned}$$

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Fitting normal distribution

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2]\right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log(2\pi) - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right] \\ \frac{\partial L}{\partial \mu} &= \sum_{i=1}^{I} \frac{(x_i - \mu)}{\sigma^2} \\ &= \frac{\sum_{i=1}^{I} x_i}{\sigma^2} - \frac{I\mu}{\sigma^2} = 0\end{aligned}$$

Solution: $\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$

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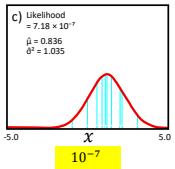
Fitting normal distribution: ML

Maximum likelihood solution:

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$



Looks familiar?!

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MLE

- Bernoulli: $\hat{p}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Poisson: $\hat{\lambda}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Uniform: $\hat{v} = X_{(1)} = \min_i X_i$, $\quad \hat{\theta} = X_{(n)} = \max_i X_i$

Bernoulli has 1 parameter "p"
uniform has 2 parameters the lower and upper bounds

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Linear Regression

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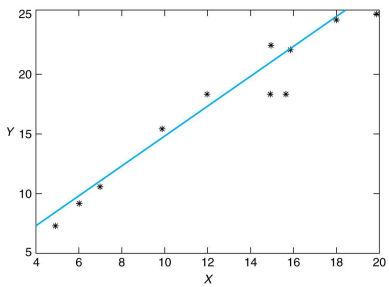
Linear Regression

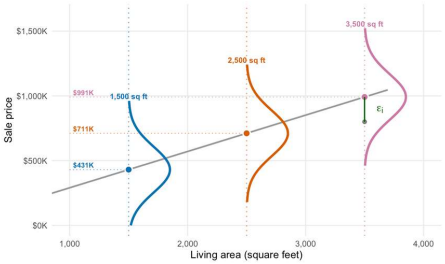


Most common case:
k=1

15

15

Linear Regression



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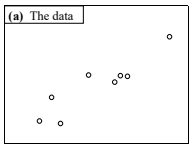
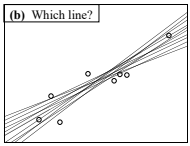
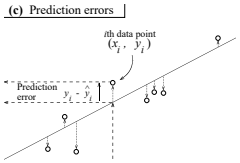
Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated line: A + Bx
  - Minimize sum of squared vertical residuals





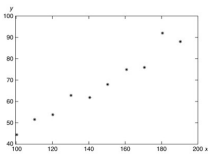


One (set of) outputs, one (set of) inputs

3

Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated regression line: A + Bx
  - Minimize sum of squared vertical residuals
- In this form, there is no mention of uncertainty
- Solution via linear algebra: minimize norm$(Up - v)$
  - $U_{n\times 2}$: [**1** x_i]
  - $p_{2\times 1}$ = $(A,B)^T$ (estimators)
  - $v_{n\times 1}$: Y_i

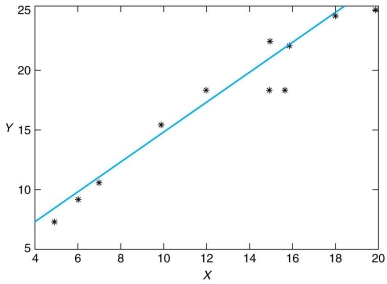


One output, one input

4

4

Fitting points to a line



$$B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

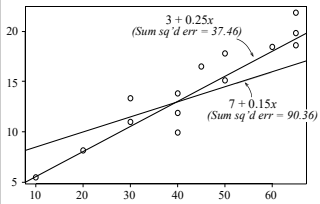
$$A = \bar{Y} - B\bar{x}$$

Proof on board

7

7

Review



Prediction Errors

		3 + 0.25x		7 + 0.15x	
x	y	$\hat{y}$	$y-\hat{y}$	$\hat{y}$	$y-\hat{y}$
40	9.90	13.00	-3.10	13.00	-3.10
50	17.80	15.50	2.30	14.50	3.30
60	18.40	18.00	0.40	16.00	2.40
45	16.50	14.25	2.25	13.75	2.75
40	11.90	13.00	-1.10	13.00	-1.10
10	5.50	5.50	0.00	8.50	-3.00
30	11.00	10.50	0.50	11.50	-0.50
20	8.10	8.00	0.10	10.00	-1.90
50	15.10	15.50	-0.40	14.50	0.60
30	13.30	10.50	2.80	11.50	1.80
65	21.80	19.25	2.55	16.75	5.05
40	13.80	13.00	0.80	13.00	0.80
65	18.60	19.25	-0.65	16.75	1.85
65	19.80	19.25	0.55	16.75	3.05
	Sum of squared errors		37.46		90.36

8

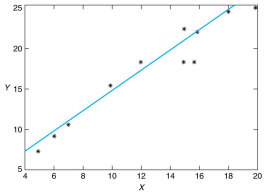
fixed => "not-random"

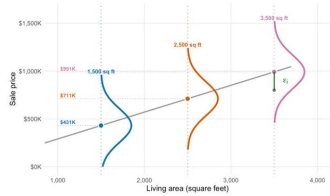
Linear Regression

"fixed" rather than constant

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$

β, α, x_i are assumed to be constant. We don't know β and α





$Y_i \sim \mathcal{N}(\beta x_i + \alpha, \sigma^2)$

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Least-squares fitting as MLE [1/2]

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- Maximizing likelihood gives the least-squares rule; applied to Y_i, it results in the normal equations

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I} | \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i | \boldsymbol{\theta})\right]$$

normal equations

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$
$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

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Least-squares fitting as MLE [2/2]

- Consider the linear model: $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
- Solving the "normal" equations results in
- $\boxed{B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \qquad \boxed{A = \bar{Y} - B\bar{x}},$
- where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\qquad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$

$$b_1 := \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := \hat{\beta}_0 = \bar{y} - b_1\bar{x}$$

Formula: observed values of our intercept and slope estimators A,B.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

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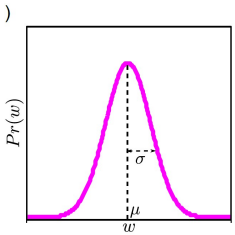
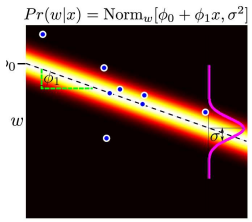
Linear Regression Review





Most common case:
k=1

$$Pr(w_i|\mathbf{x}_i, \boldsymbol{\theta}) = \text{Norm}_{w_i}\left[\phi_0 + \boldsymbol{\phi}^T\mathbf{x}_i, \sigma^2\right]$$

Image notation: w is our response Y; ϕ_0=α, ϕ_1=β; ϕ is the coefficient vector.

12

12

Regression Terminology

- Y is called output (response, dependent) variable
  - Also outcome, target, label, explained, predicted, regressand
- $(x_1, \dots, x_k)$ are called input or covariate or independent variables
  - Also predictor, feature, regressor, explanatory, attribute, factor, design, …
- We want to estimate the conditional density: $f(Y|\, x_1, \dots, x_k)$
  - $E = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_\mathrm{n}, y_\mathrm{n})\} = \{(x_{i_1}, x_{i_2}, \dots x_{i_k}, y_i) : i = 1, \dots, \mathrm{n}\}$

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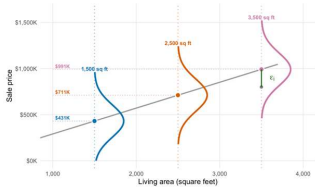
Agenda

- Earlier
  - Simple Line Fitting
  - Linear algebra approach
  - MLE approach
- Today
  - Distribution of the estimators
  - Unknown variance
  - Multiple Linear regression



$$Y_i \sim \mathcal{N}(\alpha + \beta x_i, \sigma^2)$$

2

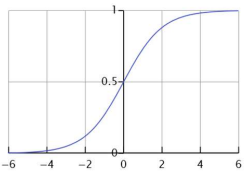
Regression asides

OLS, Logistic, polynomial, multiple

3

Logistic Regression

- Consider $\eta = \beta_1 x_1 + \cdots + \beta_k x_k + \alpha$
- Normal linear regression: $\mathbb{E}[Y|x] = \eta$
- Logistic regression: Probability of Y =1 where $\Pr(Y = 1|x) = \sigma(\eta)$
  - $\sigma(\eta) = \frac{1}{1+e^{-\eta}}$
  - Parameter estimates use iterative methods.



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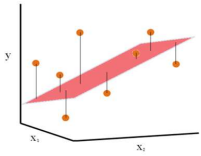
Polynomial Regression

- This is still linear in the parameters
- $Y = \beta_1 x + \beta_2 x^2 + \cdots + \beta_r x^r + \alpha + \epsilon$
- OLS treats x, x^2, …, x^r as the predictor columns.

5

Multiple Linear Regression



- $Y \mid x_1, \ldots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
- $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
- Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
- $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix}$
- $Y = X\beta + \varepsilon$

6

Review: Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,\ldots,X_n$
- **Estimator**: A statistic $T(X_1,\ldots,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter

Oct06

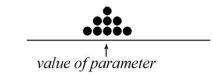
7

Understanding the derived parameters

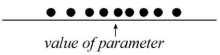
A look at bias, variance and mean-squared error of an estimator

8

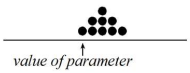
Review: Bias and precision



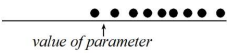
(a) No bias, high precision



(b) No bias, low precision



(c) Biased, high precision



(d) Biased, low precision

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Estimators are random variables

- The sample mean estimator is a random variable
- For independent and identically distributed (iid) random variables with a population mean μ, and finite variance σ^2
  - Distribution only for normal: $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - Sample mean: *unbiased* estimator of population mean because $\mathbb{E}[\bar{X}] = \mu$
  - We also defined the estimator $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$ ← Note the denominator
    - Distribution only for normal: $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$
  - For i.i.d. samples with finite variance: Unbiased estimator of the variance because $\mathbb{E}[S^2] = \sigma^2$

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ML Estimators

- Maximum likelihood estimation requires a specified statistical model.
  - The model supplies the likelihood used to estimate its parameters
  - If you have no parametric model in mind, then the MLE recipe is probably irrelevant
- The Maximum Likelihood estimator is also a random variable
  - Why? Because its value is a function of the samples from some underlying distribution.
  - The ML estimator has its own sampling distribution, mean and variance.
  - An ML estimate is the observed value of the ML estimator.

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Estimator bias

- The expected value of the estimator, i.e. E($\hat{\theta}$) may not be equal to the true parameter θ, i.e., it may be biased.
  - Define: Bias of the estimator: $E(\hat{\theta}) - \theta$
- Earlier: when given samples are modeled as coming from a normal
  - ML estimator for the mean of a Gaussian is unbiased
  - ML estimator for the variance is biased
  - For large n, the bias is not a major issue

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

Observed $\hat{\mu}=\bar{x}$, with I=n (our usual notation). The random mean estimator is $\bar{x}$.

Note the denominator

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$

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Estimator variance

- Merely knowing the estimator bias does not give us a complete idea of the quality of the estimator.
  - What if different samples of X gave wildly different estimates of θ for the same estimator?
- The variance of an estimator is (as expected!):

$$\mathrm{Var}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right]$$

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Evaluating a point estimator

- Let X= $(X_1, X_2, \dots, X_n)$ be a set of n i.i.d. random variables from a given distribution with parameter θ.
- Suppose $\hat{\theta}$ is an estimator of θ
  - $\hat{\theta} = g(X_1, \dots, X_n)$
  - How does one conclude whether a given estimator is good?
- One metric for this is to evaluate $\left(\hat{\theta} - \theta\right)^2$
  - This is a random variable
  - Mean squared error: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right]$
- We desire estimators with low mean-squared error

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Bias and Variance

- Algebraic identity: $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right] + \left(\mathbb{E}[\hat{\theta}] - \theta\right)^2$

  variance — Square of bias

- Suppose θ is the population mean we wish to estimate.
  - An estimator that outputs a constant c has a bias $c - \theta$ (zero variance; unbiased only if θ equals c)
  - An estimator that outputs X_i (for some i) will have no bias, but it will inherit the variance of X_i and its variance does not decrease as the sample size increases

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Slope estimator B $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

Intercept estimator A $= \bar{Y} - B\bar{x}$

Linear Regression Estimator

Notation from book: The A and B are estimators of parameters α and β in $Y = \alpha + \beta x + \epsilon$. They are random variables

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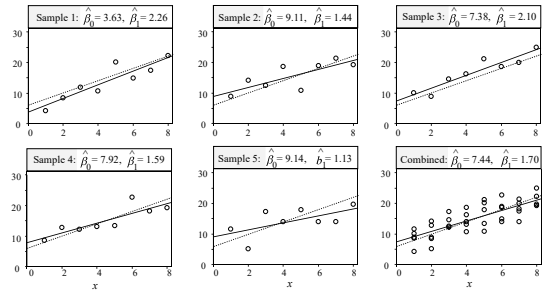
Example: Sampling variation



Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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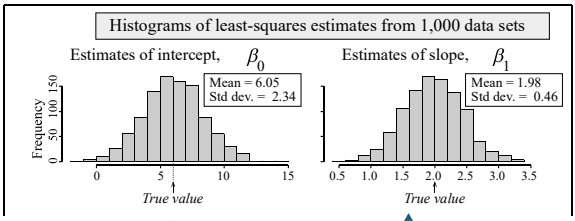
Example: Distribution of estimator



Notation:
$\hat{\beta}_0 = A$ and $\hat{\beta}_1 = B$
(from Ross book)

intercept

slope

Data generated from $Y = 6 + 2x + \epsilon, \epsilon \sim \mathcal{N}(0,9)$

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Linear regression Parameters

- estimate $= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}, Y_i = \alpha + \beta x_i + \varepsilon_i, \varepsilon_i \sim N(0,\sigma^2)$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \Longrightarrow B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_{j=1}^{n}(x_j-\bar{x})^2}$ ← Note the denominator
  - But $\sum_{i=1}^{n}(x_i-\bar{x})(Y_i-\bar{Y}) = \sum_{i=1}^{n}(x_i-\bar{x})Y_i - \bar{Y}\sum_{i=1}^{n}(x_i-\bar{x})$
- $B = \frac{\sum_{i=1}^{n}(x_i-\bar{x})Y_i}{\sum_{j=1}^{n}(x_j-\bar{x})^2} = \sum_{i=1}^{n}\frac{(x_i-\bar{x})}{\sum_{j=1}^{n}x_j-\bar{x})^2}Y_i$ (a linear combination of Y_i)
- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$

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Linear regression Parameters

- $B \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_i x_i^2 - n\bar{x}^2}\right)$
  - Unbiased estimator: $\mathbb{E}[B] = \beta$
- $A = \bar{Y} - B\bar{x}, A \sim \mathcal{N}(\alpha, \mathrm{Var}(A))$
  - Unbiased estimator: $\mathbb{E}[A] = \alpha$ and $\mathrm{Var}(A) = \frac{\sigma^2 \sum_i x_i^2}{n(\sum_i x_i^2 - n\bar{x}^2)}$

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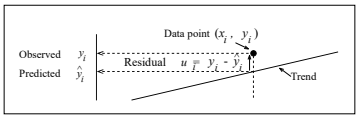
Linear regression Parameters

- Define $\mathrm{SSR} = \sum_{i=1}^{n} (Y_i - A - B\mathrm{x}_i)^2$
  - Then $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}, \Longrightarrow \mathbb{E}[\frac{SS_R}{\sigma^2}] = n - 2$
  - Unbiased: $\mathbb{E}\left[\frac{\mathrm{SSR}}{n-2}\right] = \sigma^2$

Note the denominator



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Least-squares fitting as MLE

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
  - Minimizing the negative log likelihood results in
    - Same expression for A and B as in the ordinary least square (linear algebra)
    - Different expression for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$

Note the denominator

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Review

- OLS and Gaussian MLE give the same coefficient estimators A and B.
- σ^2 is estimated from the residuals in both cases. They differ only in the divisor: n (MLE) or n−2 (unbiased).
  - The n−2 is the degrees of freedom left after fitting 2 parameters, the direct generalization of n−1.
- Normality buys exact sampling distributions and inference, not the point estimates.
- For large n the divisor difference is small. Use S^2=SSR/(n−2); software reports its observed value s^2.

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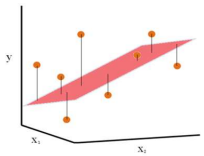
Agenda

- Earlier
  - Distribution of the estimators
  - Unknown variance
- Today
  - Multiple Linear regression



2

Least-squares optimization

- Simple linear model: $Y = \alpha + \beta x + \epsilon$, ϵ a random variable $\mathbb{E}[\epsilon] = 0$
  - Training data: $\{(x_i, y_i): i = 1 \dots n\}$
  - α and β unknown, estimated using estimators A and B
- Want to minimize $\sum_{i=1}^{n}(Y_i - Bx_i - A)^2$ which is done either via calculus or linear algebra
- Result in the normal equations (estimator form)
- Observed-data form of solution
- One can show the estimators are *unbiased*

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$

$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

$$b_1 := b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := a = \bar{y} - b_1\bar{x}$$

3

Least-squares via normality assumption

- To find how good the estimator is we make additional assumptions on the error
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - Likelihood: $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta\, x_i - \alpha)^2}{2\sigma^2}\right]$
  - Log likelihood $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- We reach the same equation as before for estimating α and β
- With $S_{xx} = \sum_i \mathrm{x}_i^2 - n\bar{\mathrm{x}}^2$, the distributions are obtained:
  - $B \sim \mathcal{N}(\beta, \frac{\sigma^2}{S_{xx}})$ and $A \sim \mathcal{N}(\alpha, \frac{\sigma^2 \sum_i \mathrm{x}_i^2}{n S_{xx}})$

4

Variance

- Define $\mathrm{SSR} = \sum_{i=1}^{n}(Y_i - A - B\mathrm{x}_i)^2$
- With $S^2 = \frac{\mathrm{SSR}}{n-2}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased) (requires uncorrelated ϵ_i)
- With additional assumption of normality
  - biased MLE estimator for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$
  - Distribution $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}$

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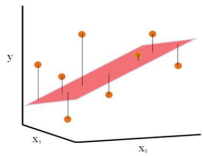
Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i) : i = 1 \dots n\}$
  - $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
  - $B=(B_0,\dots,B_k)^T$ estimates $\beta=(\beta_0,\dots,\beta_k)^T$



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Multiple Linear Regression via normality

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i) : i = 1 \dots n\}$
  - $Y \mid x_1, \dots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
    - $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $L(\beta_0, \dots, \beta_k, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2}{2\sigma^2}\right]$
- $\ell = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$
- Applying the maximizing rule to random responses gives the estimator $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$

8

Normal Equations

$$\ell(\beta_0, \ldots, \beta_k, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{i1} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_r} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{ir} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0, \quad r = 1, \ldots, k$$

Replacing the observed responses y_i by the random responses Y_i makes the minimizing coefficients the estimators B_0 … B_k (next slide)

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Normal Equations

$$\sum_{i=1}^{n} Y_i = B \sum_{i=1}^{n} x_i + nA$$

$$\sum_{i=1}^{n} x_i Y_i = B \sum_{i=1}^{n} x_i^2 + A \sum_{i=1}^{n} x_i$$

Earlier simple linear regression

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} x_{i1} + B_2 \sum_{i=1}^{n} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1} Y_i = B_0 \sum_{i=1}^{n} x_{i1} + B_1 \sum_{i=1}^{n} x_{i1}^2 + B_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{i1} x_{ik}$$

$$\sum_{i=1}^{n} x_{ik} Y_i = B_0 \sum_{i=1}^{n} x_{ik} + B_1 \sum_{i=1}^{n} x_{ik} x_{i1} + B_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}^2$$

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Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- Rewrite all the n equations in matrix form

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$Y \in \mathbb{R}^{n\times 1}, \quad X \in \mathbb{R}^{n\times p}, \quad \beta \in \mathbb{R}^{p\times 1}, \quad \epsilon \in \mathbb{R}^{n\times 1} \qquad p = k + 1$$

$$Y = X\beta + \epsilon$$

This is now a matrix notation

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Assemble Matrices

$$X^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1k} & x_{2k} & \cdots & x_{nk} \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X^T Y = \begin{bmatrix} \sum_i Y_i \\ \sum_i x_{i1} Y_i \\ \vdots \\ \sum_i x_{ik} Y_i \end{bmatrix}$$

$$X^T X = \begin{bmatrix} n & \sum_i x_{i1} & \sum_i x_{i2} & \cdots & \sum_i x_{ik} \\ \sum_i x_{i1} & \sum_i x_{i1}^2 & \sum_i x_{i1} x_{i2} & \cdots & \sum_i x_{i1} x_{ik} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i x_{ik} & \sum_i x_{ik} x_{i1} & \sum_i x_{ik} x_{i2} & \cdots & \sum_i x_{ik}^2 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

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Normal Equations

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1\sum_{i=1}^{n} x_{i1} + B_2\sum_{i=1}^{n} x_{i2} + \cdots + B_k\sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1}Y_i = B_0\sum_{i=1}^{n} x_{i1} + B_1\sum_{i=1}^{n} x_{i1}^2 + B_2\sum_{i=1}^{n} x_{i1}x_{i2} + \cdots + B_k\sum_{i=1}^{n} x_{i1}x_{ik}$$

$$\sum_{i=1}^{n} x_{ir}Y_i = B_0\sum_{i=1}^{n} x_{ir} + \sum_{j=1}^{k} B_j\sum_{i=1}^{n} x_{ir}x_{ij}, \quad r = 1, \ldots, k$$

$$\sum_{i=1}^{n} x_{ik}Y_i = B_0\sum_{i=1}^{n} x_{ik} + B_1\sum_{i=1}^{n} x_{ik}x_{i1} + B_2\sum_{i=1}^{n} x_{ik}x_{i2} + \cdots + B_k\sum_{i=1}^{n} x_{ik}^2$$

$$X^TY = X^TXB$$

$$B = (X^TX)^{-1}X^TY$$

Assumes invertibility

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Example

- Diameter of a tree is measured at a standardized point and is a function of the age, elevation, rainfall and specific gravity
- We can, for instance, ignore the specific gravity and compute the estimated linear regression model to be $y = 11.54873 + 0.05728x_1 + 0.08712x_2 + 7.33231x_3$

	Age (years)	Elevation (1,000 ft)	Rainfall (inches)	Specific Gravity	
1	44	1.3	250	.63	18.1
2	33	2.2	115	.59	19.6
3	33	2.2	75	.56	16.6
4	32	2.6	85	.55	16.4
5	34	2.0	100	.54	16.9
6	31	1.8	75	.59	17.0
7	33	2.2	85	.56	20.0
8	30	3.6	75	.46	16.6
9	34	1.6	225	.63	16.2
10	34	1.5	250	.60	18.5
11	33	2.2	255	.63	18.7
12	36	1.7	175	.58	19.4
13	33	2.2	75	.55	17.6
14	34	1.3	85	.57	18.3
15	37	2.6	90	.62	18.8

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Estimators are unbiased

$$
\begin{aligned}
E[\mathbf{B}] &= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}] \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta}+\mathbf{e})] \quad \text{since } \mathbf{Y}=\mathbf{X}\boldsymbol{\beta}+\mathbf{e} \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= E[\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= \boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top E[\mathbf{e}] \\
&= \boldsymbol{\beta}
\end{aligned}
$$

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Variance

- Define SSR as before as the sum of the squares of the residuals
  - $\Sigma_i(Y_i - B_0 - \Sigma_j B_j x_{ij})^2$
- With $S^2 = \frac{\text{SSR}}{n-k-1}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased)

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Multivariate Normal Distributions

- The least square estimators $B_0, B_1, \ldots B_k$ which are the elements of the matrix B are linear combinations of the independent normal random variables Y_i
- The set of independent random variables form an interesting object
  - The multivariate normal distribution
- We say the estimator vector B has a multivariate normal distribution

$$B = (X^T X)^{-1} X^T Y$$

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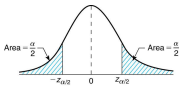
Agenda

- Earlier
  - Confidence intervals

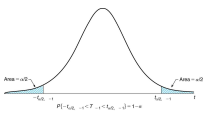
$$\Pr\left(-z_{0.025} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



1

Review: 95% confidence interval

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors
- This works exactly if the distribution is normal, and
  - "quite well" when the distribution is unknown
  - $(\bar{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $\Pr\left(-t_{0.025,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{0.025,n-1}\right) = 0.95,$



2

Review

TABLE 7.1 *100(1 −α) Percent Confidence Intervals*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S = \sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)}$$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}\right)$	$\left(\overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\overline{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$	$\left(-\infty, \overline{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}\right)$	$\left(\overline{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}, \infty\right)$

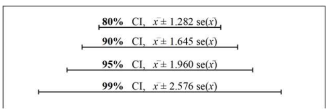
We use the empirical standard deviation in practice

Lower bound: at least as large

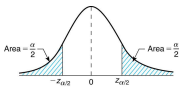
3

Agenda

- Earlier
  - Confidence intervals
- Today
  - Comparing two populations or treatments



$$\Pr\left(-z_{0.025} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



4

Course details from IITB for CS215
- The role of statistics.
- Graphical and numerical methods for describing and summarizing data.
- Probability.
- Population distributions.
- Sampling variability and sampling distributions.
- Inference based on a single sample.
- Hypothesis testing a single sample.
- Comparing two populations or treatments.
- Simple linear regression and correlation.
- Case studies.

Example: Comparing two populations

- Rheumatoid arthritis (RA) is a debilitating condition
- We measuring sulfhydryl concentration (mmol) in blood samples
- Statistical question:
  - What is the difference between concentration in those without and with RA?
  - Let's compute confidence intervals

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

5

Example: Comparing two populations

- Statistical question: what is the difference between those without RA and those with rheumatoid arthritis?
  - n = 7, df = 6, t(0.025, 6) = 2.447 SE = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
  - n = 6, df = 5, t(0.025, 5) = 2.571 SE = 0.44049/√6 = 0.17983, margin = 2.571 × 0.17983 = 0.4623 CI = 2.42 ± 0.4623 = (1.958, 2.882)
- The intervals overlap.
  - This might indicate that, for the without RA condition, the true mean is more than 1.958 (lower end of the with-RA condition)
  - This is not the way we compare two populations

6

How different are the two population?

- Confidence interval for a difference between population means
  - $(\mu_1 - \mu_2)$: $(\bar{x}_1 - \bar{x}_2) \pm t\ \mathrm{se}(\bar{x}_1 - \bar{x}_2)$ ← Two means
- When two samples are independent the variance adds up, so

  $\mathrm{se}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
- Conservative degree of freedom approximation for the t-distribution:
  - $\min(n_1 - 1, n_2 - 1)$ ← Employed in practice
- In the RA blood sample example:
  - Difference between means = 2.42 – 1.92143 = 0.49857, se = sqrt(0.07559²/7 + 0.44049²/6) = 0.1821, t(0.025, 5) = 2.571, margin=0.4681, CI = (0.030, 0.967) does not include zero, the with-RA always exceeds the without-RA

7

Confidence intervals for difference

- Previous discussions assume empirical approximations
- If population is normal and means are estimated,
  - Assume: variances are known, then CI uses the known variances, they are added, and we work with exact normal distribution (see next slide)
  - Assume variances are unknown, and we use the empirical values
    - If they are different, then CI is complicated (o-o-s).
    - If they are same, then CI uses the pooled estimator and df = (n+m-2)
  - Note: Many cases variances are unknown and different!

8

Review

TABLE 7.2 *100(1 − α) Percent Confidence Intervals for $\mu_1 - \mu_2$*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$

$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2,\, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Assumption	Lower Confidence Interval
σ_1, σ_2 known	$(-\infty, \overline{X} - \overline{Y} + z_\alpha\sqrt{\sigma_1^2/n + \sigma_2^2/m})$
σ_1, σ_2 unknown but equal	$\left(-\infty, \overline{X} - \overline{Y} + t_{\alpha,\, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}\right)$

Note: Upper confidence intervals for $\mu_1 - \mu_2$ are obtained from lower confidence intervals for $\mu_2 - \mu_1$.

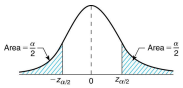
Pooled estimator

9

Agenda

- Earlier
  - Confidence intervals for proportions
  - Comparing means for two populations or treatments
- Today
  - Confidence intervals for proportions from two different populations

$$\Pr\left(-z_\alpha \leq \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \leq z_\alpha\right)$$



1

Review

2

Example: Comparing two populations

- We measuring sulfhydryl concentration (mmol) in blood samples of those without and with rheumatoid arthritis (RA)
- Statistical question:
  - What is the difference in the concentration for those without and with RA?
  - We look for the confidence interval in the difference in means using the provided data

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

Course details from IITB for CS215

- The role of statisitcs.
- Graphical and numerical methods for describing and summarising data.
- Probability.
- Population distributions.
- Sampling variability and sampling distributions.
- Estimation using a single sample.
- Hypothesis testing a single sample.
- *Comparing two populations or treatments.*
- Simple linear regression and correlation.
- Case studies.

3

Review: Confidence interval for difference

TABLE 7.3 *100(1 − α) Percent Confidence Intervals for $\mu_1 - \mu_2$*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$

$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Pooled estimator

If we do not know the variances, and whether the variances are equal, then, as an empirical measure, we use a conservative estimate with df=min(n-1, m-1)

4

CI is not a protection against fabricated or sloppy data. We assume random sampling with independent observations, and an unbiased instrument to measure the data

Review: Proportions

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm$ z standard errors applied to *proportions*
- This works well if the distribution is normal, and
  - "quite well" when the distribution is unknown provided n is "sufficiently" large
  - Example: np(1-p)>10

$$\Pr\left(-z_{0.025} \leq \frac{\hat{p}-p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq z_{0.025}\right) \approx 0.95$$

5

Example: Comparing two populations

- 68% (n=870) non-smokers get "good" grades compared to 41% (n=130) smokers.
- What can we conclude?
- Contrast a single population where we ask
  - Does the fraction 0.68 (for non-smokers) represent the true proportion in the population

Get good grades

percent

Characteristics by Smoking and Drinking Status

(Table entry = % of group saying yes)		Smoker	Nonsmoker
Get mostly A's or B's?		41	68
Read 1 or more hours/day?		54	72
Get drunk at least once a month?		63	10
Have smoked Marijuana?		79	14
Likely to try illegal drug in future?		42	14
	n	130	870

Smoked cigarettes in last 30 days.

Consumed more than a few sips of alcohol in the last 30 days.

Source: "Back to School Teen Survey"

6

Review: Single population

$$\Pr\left(-z_\alpha \le \frac{\hat{P}-p}{\sqrt{\frac{p(1-p)}{n}}} \le z_\alpha\right)$$

- Does the fraction 0.68 (for non-smokers) represent the true proportion in the population?
  1. Is the normal approximation valid? 870*0.68*(1-0.68) > 10, so yes. Empirical rule
  2. $\Pr\left(-z_{0.025} \le \frac{\hat{P}-p}{se(\hat{P})} \le z_{0.025}\right) = 0.95$ and $z_{0.025} = 1.96$,
  3. $se(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.01582$
  4. Therefore $CI = \hat{p} \pm 1.96\,\hat{se}(\hat{p})$ $= 0.68 \pm 1.96(0.01582)$ $= 0.68 \pm 0.03100$
  5. It is not unreasonable to expect 65% of non-smokers getting good grades in the population

Non-causality: Non-smoking is not a shortcut to getting good grades

7

Review: How big should my sample be?

- The margin is $z \times \sqrt{(}p(1-p)/n$
- If m is the margin of error, then
  - want $z \times \sqrt{(}p(1-p)/n \le m$
  - Implies if we guess a proportion p^*want $n \approx \frac{z^2}{m} p^*(1-p^*)$
- Examples
  - Margin is 10%, p=0.3, n=81
  - Margin is 5%, p=0.3, n=323

8

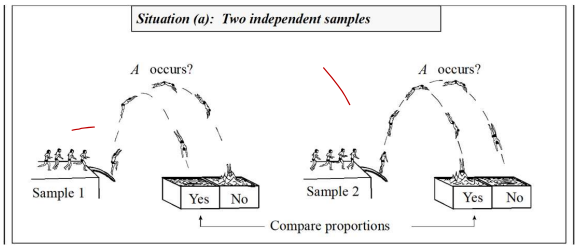
Example: Comparing two populations

- Good grades: 68% (n=870) non-smokers, 41% (n=130) smokers
- What can we conclude about the difference between the two populations?
  - Is the normal approximation tenable?
    - 130*0.41*(1-0.41) > 10? 870*0.68*(1-0.68) > 10, so yes.
  - Point estimate of the difference = 0.68-0.41, standard error = 0.045944
  - CI = $0.27 \pm 1.96 * 0.045944$= [0.18, 0.36]
  - Not unreasonable that non-smokers outperform smokers, the difference is 18 percentage points (with 95% confidence)

$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

9

Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Variance adds up

10

Example #2

- 18% difference in the level of support for Clinton. What's the CI?
- Level of support for Clinton in NJ and NY: may be considered independent
- Level of support for Clinton vs Dole in New Jersey: same population is being asked to make a choice
  - What's the difference?

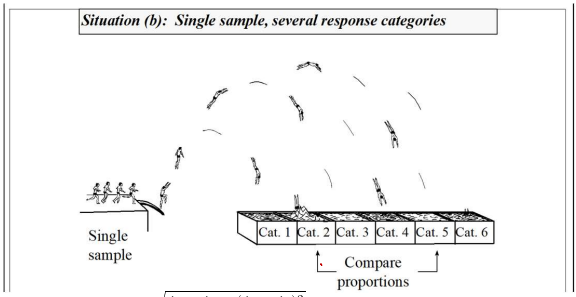
percent

		Pre-election Polls				Election Results		
State	n	Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996.

11

Proportions



$$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 + \hat{p}_2 - (\hat{p}_1 - \hat{p}_2)^2}{n}}$$

12

Example

- Level of support: for Clinton: 51 and for Dole: 33, difference is 18%.
- What is the confidence interval?
  - Answer: margin = 0.056, CI = [0.12, 0.24]

State	n	Pre-election Polls				Election Results		
		Clinton	Dole	Perot	Other/Undecided	Clinton	Dole	Perot
New Jersey	1,000	51	33	8	8	53	36	9
New York	1,000	59	25	7	9	59	31	8
Connecticutt	1,000	51	29	11	9	52	35	10

Source: Quinnipiac College Polling Institute press release, 8 Nov. 1996.

14

Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?

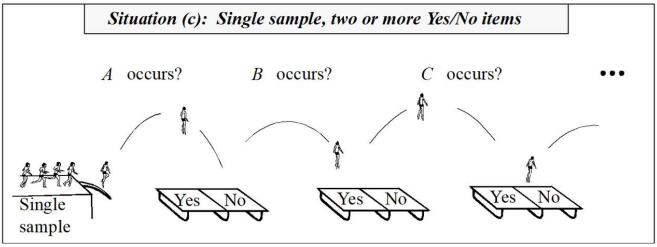
Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

Smoked cigarettes in last 30 days.
Consumed more than a few sips of alcohol in the last 30 days.
Source: "Back to School Teen Survey"

15

Proportions from independent samples



$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\min(\hat{p}_1 + \hat{p}_2, \hat{q}_1 + \hat{q}_2) - (\hat{p}_1 - \hat{p}_2)^2}{n}} \qquad \hat{q}_1 = 1 - p_1, \hat{q}_2 = 1 - p_2$$

16

Example #3

- Consider smokers who admit to getting drunk at least once a month (63%) vs smoked marijuana (79%)
  - Are smokers more likely to try marijuana compared to drinking?
- Point estimate = 0.79 – 0.63
- n = 130 is large enough
- se = 0.065, [0.03, 0.29]

Characteristics by Smoking and Drinking Status

(Table entry is % of group saying yes)	Smoker	Nonsmoker
Get mostly A's or B's?	41	68
Read 1 or more hours/day?	54	72
Get drunk at least once a month?	63	10
Have smoked Marijuana?	79	14
Likely to try illegal drug in future?	42	14
n	130	870

Smoked cigarettes in last 30 days.
Consumed more than a few sips of alcohol in the last 30 days.
Source: "Back to School Teen Survey"

17

Review

1. First verify if the normal approximation can be made
2. Check which case applies
   - Proportions from two independent samples
   - One sample of size n, several response categories
   - One sample of size n, many yes/no items

○ Calculate the standard error based on the formula

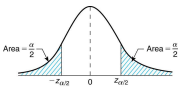
○ Use the normal approximation with the appropriate confidence level

18

Agenda

- Earlier
  - Confidence intervals for two populations or treatments
- Today
  - Hypothesis testing (two populations)

In India, Gallup puts approval at 36%, exceeding disapproval by at least 10 percentage points. Pew found that 40% of Indians viewed UN favourably, and only 18% unfavourably

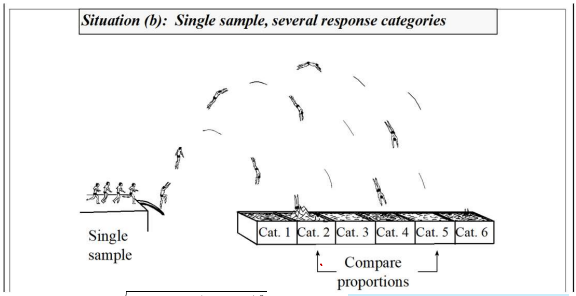


1

Review

2

Proportions



$$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 + \hat{p}_2 - (\hat{p}_1 - \hat{p}_2)^2}{n}}$$

Want to estimate true difference in the proportions $\widehat{P_1} - \widehat{P_2}$

3

Proportions for Case (b)

- Sample n independent people with given proportions
- Observe two mutually exclusive indicators for each person
  - $X_i = \mathbf{1}\{\text{person i chooses characteristic 1}\}$, and $Y_i = \mathbf{1}\{\text{person i chooses characteristic 2}\}$
  - $\mathbb{E}[X_i] = p_1$ (also probability that person i will choose characteristic 1)
  - Define: Person level difference as $D_i = X_i - Y_i$
  - $D_i = 1$ with probability p_1, -1 with probability p_2 and 0 with probability $1 - p_1 - p_2$
  - $E[D_i] = p_1 - p_2, E[D_i^2] = p_1 + p_2, \text{Var}(D_i) = p_1 + p_2 - (p_1 - p_2)^2$
- For the entire population,
  - Define $D = \sum_i^n D_i$, then difference P between the sample proportions is D/n
  - Applying the CLT approximation we have $P \overset{\text{approx}}{\sim} \mathcal{N}\left(p_1 - p_2, \frac{p_1 + p_2 - (p_1 - p_2)^2}{n}\right)$

4

Significance Testing via Hypothesis

Warning: concepts coming up, not proofs

5

Review: Hypothesis Testing

- We cannot "rule-in" a hypothesized value for a parameter
- We can only determine whether there is evidence to rule out a hypothesized value

6

Review: Hypothesis Testing

- In settling a research hypothesis, we turn skeptical and assume the research hypothesis is false
  - The null hypothesis is H_0 (or H_o)
  - The alternate is the research hypothesis H_1 or H_a
- Is there data to reject the null hypothesis?
  - Solution:
    - Compute the test statistic
      - Is this statistic "too far" from the null hypothesis?
    - If true, then reject the null hypothesis
    - If false, cannot reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

7

Review: The t-test

- Using $\hat{\theta}$ to test $H_0: \theta = \theta_0$ versus H_a
  1. Calculate the test statistic t_0
  2. Calculate the p-value
  3. Interpret the p-value

These numbers are "rule of thumb"

P	%	Translation
> 0.12	-12%	No evidence against H0
0.1	-10%	Weak evidence against H0
0.05	-5%	Some evidence against H0
0.01	-1%	Strong evidence against H0
0.001	-0.10%	Very Strong evidence against H0

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

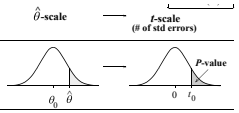
Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\theta} - \theta_0\rvert$ too large)	$P = 2\, pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

8

P-value from t-tests

- The P-value is the probability that, if the hypothesis was true, sampling variation would produce an estimate that is further away from the hypothesized value than our data-estimate
- The P-value measures the strength of the evidence **against** H_0
- The **smaller** the P-value, the **stronger** the evidence against H_0
  - The second and third points are true for significance tests generally, and not just for t-tests

Alternative Hypothesis	Evidence against H_0: $\theta = \theta_0$ provided by
H_1: $\theta > \theta_0$	$\hat{\theta}$ too much bigger than θ_0

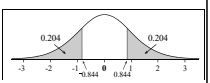


Example #2 Piazza post 242

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

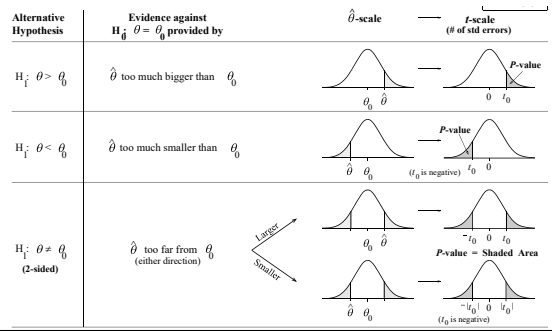
- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (Accepted value 5.517)
  - Research hypothesis: His last 23 experiments resulted in biased measurement
    - Null hypothesis: H_0: His experiments were not biased, the data followed the normal distribution
      - We do not know the mean and variance of the distribution corresponding to H_0
      - But we have sample variance, and sample standard deviation
    - Compute the t-statistic: -0.844, Compute the p-value: 0.204
    - Interpretation: Cannot refute H_o

Piazza post 242: Why in the cavendish example did we take t distribution t



Clarification: p-value is not 0.204 but 0.408

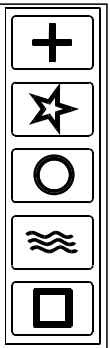
The t-test



11

Example #1: Piazza post 154

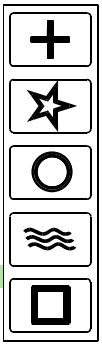
- In 1938, scientists showed 60,000 pictures
  - Students predict what is behind the card
  - If students guessed randomly there would be 12000 correct answers (proportion is 0.2)
  - … but actual result was 12489 (proportion is 0.208)
  - Is 20.8% being larger than 20% a "chance event" or
  - Hypothesis: Students can mind read correct answer in an MCQ



12

Example #1: Piazza post 154

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$



- In 1938, scientists collected 60,000 pieces of data
  - If students guessed randomly
    - there would be 12000 correct answers, but actual result was 12489
  - Research hypothesis: Students mind read correct answer in an MCQ
  - H_0: the experiment is Bernoulli (0.2)
    - For n = 60000, we apply the normal approximation to the proportion
    - We know the mean and variance (piazza post 154)
      - t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001633, t=4.99
      - OR t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92

Piazza post 154: Take this

    - Either way: p-value: < 10^{-6}
    - Interpret: Can refute the null hypothesis
    - Inference: Not pure guessing (maybe mind read?)

13

p-value for proportions

Piazza post 154: Take this

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

- Which form to use for the p-value?
  - t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001633, t=4.99
  - OR t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92
- There are (at least) 3 ways all asymptotically equivalent
  1. Lagrange multiplier test or Score test (mostly taught in intro stats, more principled in this case, piazza post 154)
  2. Likelihood Ratio Test
  3. Wald test (Oct05 class): *corresponds to the confidence interval test*
  - (One can of course use the exact test: just use the binomial)

14

Type I Errors

- If the null hypothesis is true, and we decide to reject
  - at the 95% confidence level, we will make an error 5% of the time
  - At the 99% level, we would be less wrong
  - Never reject, we would never be wrong

Decision Making

	Actual situation
Decision made	H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error

15

Tests and Intervals

- Two different methods for coping with the uncertainty
  - True value of a parameter caused by the sampling variation in estimates
  - Confidence interval: A fixed level of confidence is chosen. We determine a range of possible values for the parameter that are consistent with the data
  - Significance test: Only one possible value for the parameter, called the hypothesized value, is tested. We determine the strength of the evidence provided by the data against the proposition that the hypothesized value is the true value
- The significance level is the probability of making a Type-I error
  - A two-sided test of $H_0: \theta = \theta_0$ is significant at the 5% level if and only if θ_0 lies outside a 95% confidence interval for θ

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Tests and Intervals

Must know

- Statistical significance relates to the existence of an effect.
- The practical significance of an effect depends on its size.
- A small P-value provides evidence that the effect exists but says nothing at all about the size of the effect.
  - Example: Risk of death for smokers prescribed beta-blockers
    - P-value = 0.048 (Cannot reject null hypothesis "beta-blockers have no effect on death")
    - Confidence interval (factor) [1.004, 2.5]
- To estimate the size of an effect, compute a confidence interval
  - Never quote a P-value about the existence of an effect without also providing a confidence interval estimating its size.

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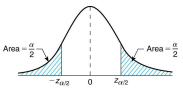
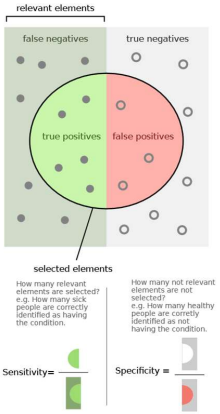
Clarification: The sentence in quotes is H_0. p-value at $\alpha = 0.05$ fails to reject H_0

Agenda

- Earlier
  - Significance testing
- Today
  - Hypothesis testing
  - Type II errors
  - Qualitative variables and goodness-of-fit





2

Type I Errors

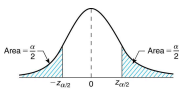
- If the null hypothesis is true, and we decide to reject
  - at the 90% confidence level, we will make an error 5% of the time
  - At the 95% level, we would be less wrong

Decision Making

Decision made	Actual situation: H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error

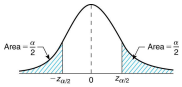
Typo 10%

Also see slide 12



3

Review



- 5 signals (possibly corrupted) are received by a transmitter on a particular day
  - mean value of $\bar{X} = 9.5$
- Null hypothesis: signal has mean 8, noise is $\mathcal{N}(0,4)$
- Test statistic $\text{se}(\bar{x}) = 2/\sqrt{5}, t_0 = (9.5 - 8)/\text{se}(\bar{x}) = 1.68$, two-sided p-value = $2\times\big(1-\text{normcdf}(1.68)\big) = 0.0936$
  - 90% conf ($\alpha/2 = 5$): 1.68 > 1.645, 0.095 < 0.10, CI = [8.03, 10.97], reject
  - 95% conf ($\alpha/2 = 0.025$): 1.68 < 1.96, 0.094 > 0.05, CI = [7.75, 11.25], do not reject
- If the null hypothesis was true, then we would be wrong in one case

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

4

Example considering only one population

- We measuring sulfhydryl concentration (mmol) in blood samples
- Recall the numbers for 2-sided 95% confidence of the sample mean
  - n = 7, df = 6, t(0.025, 6) = 2.447 SE = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699, CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- A new patient has x_n = 2.10. Can we reject: patient comes from the without-RA population?
- The CI for the *mean* is the wrong measure

Without RA	
	1.84
	1.92
	1.94
	1.92
	1.85
	1.91
	2.07
Sample size	7
Sample mean	1.92143
Sample standard deviation	0.07559

5

Example

- Recall the numbers
  - n = 7, df = 6, $\bar{x}$ = 1.92143, t(0.025, 6) = 2.447, se = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- x_n = 2.10. H_0: new patient belongs to the without RA group
- We must account for
  - Variation of the new individual around the population mean
  - Uncertainty in our estimate of the mean
  - var$(x_n - \overline{x_1})$ = $0.07559^2 + 0.07559^2/7$, se = 0.0808, $t_o = \frac{2.10-\bar{x}}{0.0808}$ = 2.21, pvalue = 0.069
  - Inference: we fail to reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

7

Example

- Recall the numbers
  - n = 7, df = 6, $\bar{x}$ = 1.92143, t(0.025, 6) = 2.447, se = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- x_n = 2.10. H_0: new patient belongs to the without RA group
- We must account for
  - var$(x_n - \overline{x_1})$ = $0.07559^2 + 0.07559^2/7$, se = 0.0808, $t_o = \frac{2.10-\bar{x}}{0.0808}$ = 2.21, pvalue = 0.069
  - Inference: we fail to reject the null hypothesis
- Equivalent form: the 95% prediction interval: 1.92143 ± 2.447 × 0.0808 = (1.724, 2.119). The value 2.10 is inside it.

8

Significance Testing with Difference in Means

- $H_0: \mu_2 - \mu_1 = 0. H_1: \mu 2 - \mu 1 \neq 0$ α = 0.05

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

  - small values of n, do not have true variance
  - T-distribution, conservative df = min(n1−1, n2−1) = 5.
- Difference between means numbers
  - Diff = 2.42 – 1.92143 = 0.49857, se = sqrt(0.07559²/7 + 0.44049²/6) = 0.1821
  - $t_0 = 0.49857/0.1821 = 2.74$
  - P-value = 0.041
- We reject the null hypothesis since 0.04 < 0.05
  - Note 2.74 exceeds 2.571 = t(0.025, 5) = 2.571
  - CI for diff: (0.0304, 0.9668) does not contain 0

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

9

Review

- What is the essential difference between the information provided by a confidence interval and by a significance test?
- What is the most common form of null hypothesis?
- How does a null hypothesis typically relate to a belief, hunch, or research hypothesis that initiates a study?
- How can researchers try to demonstrate that effects or differences seen in their data are real?
- How does the alternative hypothesis typically relate to a belief, hunch, or research hypothesis that initiates a study?

10

Review: One-side alternatives

- We also use ***one-sided alternatives***
- Either: $H_1: \theta > \theta_0$ or $H_1: \theta < \theta_0$
  - when the research hypothesis specifies the direction of the effect
  - or more generally, when the investigators had good grounds for believing the true value of θ was on one particular side of θ_0 before the study began.
- Otherwise, a two-sided alternative $H_1: \theta \neq \theta_0$ is used.

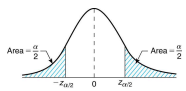
11

Type I Errors

- If the null hypothesis is true, and we decide to reject
  - at the 95% confidence level, we will make an error 5% of the time
  - At the 99% level, we would be less wrong
  - Never reject, we would never be wrong

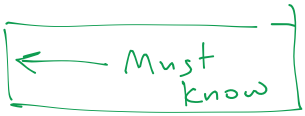
Decision Making

Decision made	Actual situation: H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error



12

Type I and Type II

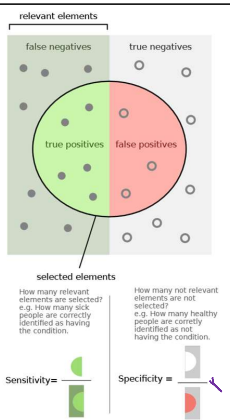


Decision Making		
	Actual situation	
Decision made	H_0 is true	H_0 is false
Do not reject H0	OK	Type II error
Reject H_0 as false	Type I error	OK

13

Sensitivity and specificity

- Sensitivity (true positive ratio): probability of a positive test result, conditioned on the individual truly being positive
  - "recall" in machine learning
- Specificity (true negative ratio): probability of a negative test result conditioned on the individual truly being negative
- The tradeoff is explored in ROC analysis
- Confusion matrix depends on the field of research



14

Type I and Type II errors

- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true
  - This calls for increasing sample size

The power of a test = 1 - probability of type II error

15

Review: Type I and Type II errors

- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true.
- Review:
  - We design the test in such a way that whenever H_0 is true, the probability of rejection based on the samples (type I error) is never greater than α, called the significance level of the test.
  - It is a parameter set in advance. Typical values of α are 0.05 or 0.01.

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Type II error

- What is the probability of failing to reject the null hypothesis H_0 when the true mean μ is **unequal** to μ_0?
- Define $\beta(\mu)$ to be the probability of not rejecting H_0

$$\beta(\mu) = P_\mu\left\{\left|\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}\right| \le z_{\alpha/2}\right\}$$
$$= P_\mu\left\{-z_{\alpha/2} \le \frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\}$$

$$\beta(\mu) = \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

The function $\beta(\mu)$ is called the **operating characteristic (OC) curve**. μ is hidden in the formula under the assumption that $\overline{X}$ is normal i.e., $Z = (\overline{X}-\mu)/(\sigma/\sqrt{n})$ is standard normal

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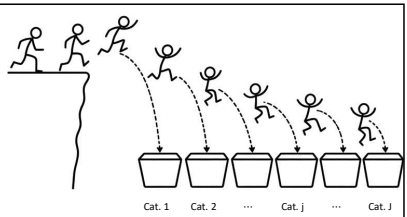
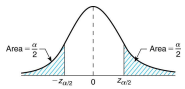
Agenda

- Earlier
  - Quantitative variables
  - Type II errors
- Today
  - Hypothesis testing for qualitative variables
  - Goodness-of-fit

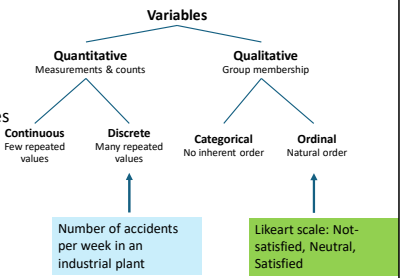




1

Types of variables

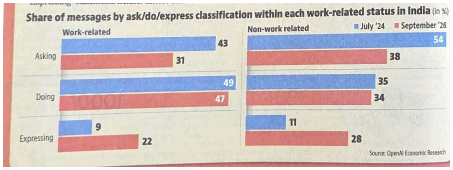
- Quantitative variables are measurements and counts
  - Variables with many repeated values are treated as discrete
  - Variables with few repeated values are treated as continuous.
- Qualitative variables describe group membership



2

Qualitative

- October 27, 2026 statistics from ChatGPT
- Blue 2024, Red 2026
  - Asking
  - Doing
  - Expressing



3

Example: Heart Attack Patients

ID	E	SYS	DIA	Blocked Arteries	STENT	TIME	Outcome	AGE	SMOKER	BETA Blockers	CHOL	SURGERY
	%	ml	ml	%	%	Months					mMol	
390	72	36	131	0	0	143	0	49	2	2	59	0
279	52	74	155	37	63	143	0	54	2	2	68	1
391	62	52	137	33	47	16	2	56	2	2	52	0
201	50	165	329	33	30	143	0	42	2	2	39	0

E: Percentage of blood in the left ventricle that is emitted (fraction)

Sys, Dia: Systolic diastolic volume

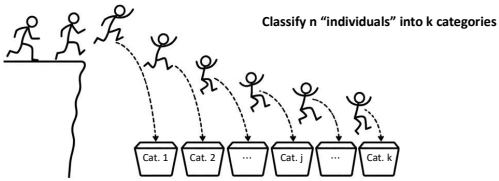
Stent: Percentage of arteries significantly lowered but not blocked

Time: Admitted until outcome (alive, sudden cardiac death, death within 30 days of heart attack …)

4

Table of counts

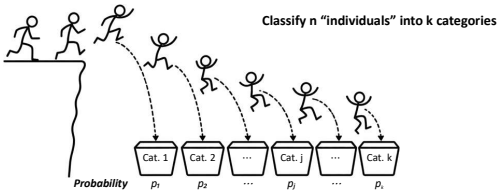
- It's usual to have a table of counts when we have one qualitative variables



5

Table of counts

- We want to test a hypothesis about the whole set of population proportions



6

Example

- Are these the rolls of a fair dice or not?
  - Fair dice: 35 for each case
- Do proportion of blood satisfy the Hardy-Weinberg model in genetics?
- Want a test statistic and thus a p-value to reject, or fail to reject our null-hypothesis

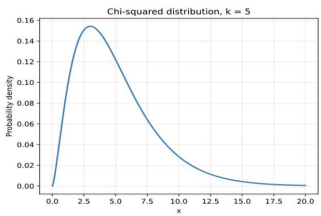
Three Blood Types				
	A	AB	B	Total
No. Observed	39	70	42	151
Proportion Observed	0.3	0.5	0.3	1.000

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

7

Chi-square Goodness of Fit Test

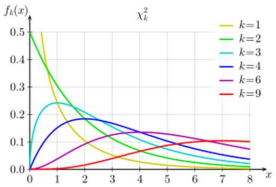
- Dice case: the test statistic is
  - $x_o^2 = \Sigma \frac{(\text{observed-expected})^2}{\text{expected}}$
  - We compute the p-value $\Pr(X^2 \geq x_0^2), X^2 \sim \chi^2(\text{df})$
  - We interpret the p-value as usual
- Degrees of freedom = k-1



8

Review: Chi-square distribution

- If $Z_1, Z_2, \ldots, Z_n$ are independent *standard* normal then
  - $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is said to have a chi-square distribution with n degrees of freedom and is denoted as $X \sim \chi_n^2$
- Example (n=1)
  - If $Z_1 \sim N(0,1)$, then $X = Z_1^2$ is χ_1^2



9

Example

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - df = 5
  - $x_0^2 = \frac{(26\quad)^2}{35} + \cdots + \frac{(38\quad)^2}{35} = 7.54$
  - P-value = 0.18
- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)

Intuition: Degree of freedom lost because $\Sigma p_i = 1$

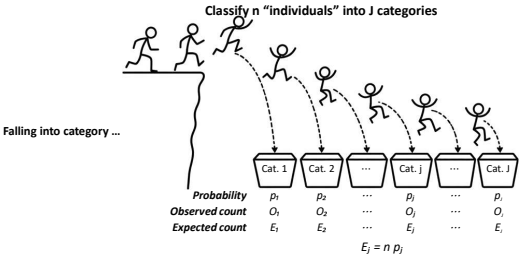
$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

10

Review

$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

- Look for observed and expected for the null hypothesis



11

Goodness of fit

- n independent variables $Y_i, \dots, Y_n$ each taking one of the values $1,2, \dots, k$
- Null hypothesis $H_0: p_i: i = 1, \dots, k$ is the pmf
- Alternate hypothesis $H_a: P(Y = i) \neq p_i$ for some $i = 1, \dots, k$
- Let $X_i, (i = 1, .., k)$ be the number of the Y_i that equal i
- Under $H_0, X_i \sim \text{Bin}(n, p_i)$
  - statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$
  - When n is large, if H_o is true, then $T \sim \chi^2(k-1)$
  - Reject H_o using p-value as usual

Intuition: Don't want numerator to be large compared to the denominator

Intuition: Degree of freedom lost because $\Sigma X_i = n$

12

GoF: Goodness of fit (k=2)

- statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$

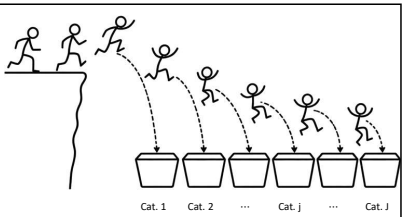
How did we get to the χ^2 distribution? Consider k=2

$$
\begin{aligned}
T &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \\
&= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1-p_1])^2}{n(1-p_1)} \\
&= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1-p_1)} \\
&= \frac{(X_1 - np_1)^2}{np_1(1-p_1)} \qquad \text{since} \qquad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}
\end{aligned}
$$

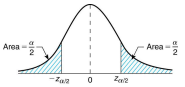
If X_1 is Bin, then we can recognizable the denominator as the variance. For large n, we can see the standard normal appearing (it has been squared)

13

Agenda



- Earlier
  - Qualitative variables
  - Hypothesis testing: Table of counts, Type II errors
- Today
  - Goodness-of-fit when probabilities are not fully specified
  - Poisson
  - Multivariate Gaussian



1

Review

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - chi-square distribution, df = 5
  - $x_0^2 = \frac{(26-35)^2}{35} + \cdots + \frac{(38-35)^2}{35} = 7.54$
  - p-value = 0.18

Intuition: Degree of freedom lost because $\Sigma p_i = 1$

```
octave:1> chi2cdf (7.54, 5)
ans = 0.8165
octave:2> 1- chi2cdf (7.54, 5)
ans = 0.1835
octave:3>
```

- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)

$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

2

Review: Goodness of fit

- n independent variables $Y_i, \dots, Y_n$ each taking one of the values $1,2, \dots, k$
- Null hypothesis $H_0: p_i: i = 1, \dots, k$ is the pmf
- Alternate hypothesis $H_a: P(Y = i) \neq p_i$ for some $i = 1, \dots, k$
- Let $X_i, (i = 1, .., k)$ be the number of the Y_i that equal i
- Under $H_0, X_i \sim \text{Bin}(n, p_i)$
  - statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$
  - When n is large, if H_o is true, then $T \sim \chi^2(k-1)$
  - Reject H_o using p-value as usual

Intuition: Don't want numerator to be large compared to the denominator

Intuition: Degree of freedom lost because $\Sigma X_i = n$

3

Example

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
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- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - df = 5
  - $x_0^2 = \frac{(26-35)^2}{35} + \dots + \frac{(38-35)^2}{35} = 7.54$
  - P-value = 0.18
- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)
- But the dice could be still be biased and we may be making a type-II error

Example: If there was a manufacturing defect, and the true probabilities was (0.25, 0.15, 0.15, 0.15, 0.15, 0.15)

4

Review: Type I and Type II errors

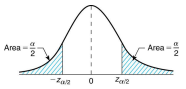
- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true
  - This calls for increasing sample size

Decision Making		
	Actual situation	
Decision made	H_0 is true	H_0 is false
Do not reject H0	OK	Type II error
Reject H_0 as false	Type I error	OK

The power of a test = 1 - probability of type II error

5

Example



- 5 signals are received by a transmitter on a particular day with the mean value of $\bar{X} = 9.5$ (signal could be corrupted)
  - Null hypothesis: signal is guessed to have $\mu_0 = 8$, noise is $\mathcal{N}(0,4)$, true mean is 10, not 8
  - Test statistic $t_0 = (9.5 - 8)/\text{se}(\bar{x}) = 1.68$, 1.68 < 1.96, do not reject
  - $(\mu_0 - \mu)/(2/\sqrt{5}) = -\sqrt{5}$
  - $\beta = \phi(-\sqrt{5} + 1.96) - \phi(-\sqrt{5} - 1.96) = 0.392$
- The null hypothesis is false, the probability of not rejecting is 39%
- Power of this method = 0.61

$$\beta(\mu) = \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

7

Type II error: Chi-Square Goodness of Fit

- H_0: $\mathbf{p} = \mathbf{p}_0 = (1/6,1/6,1/6,1/6,1/6,1/6)$.
- What is the probability of failing to reject the null hypothesis H_0at a significance level α when the true mean probability vector p is unequal to p_0?
- Define $\beta(\mathrm{p})$ to be the probability of not rejecting H_0

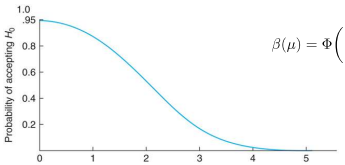
Not for endsem

Example: If there was a manufacturing defect, and the true probabilities was (0.25, 0.15, 0.15, 0.15, 0.15, 0.15)

8

Type II error

- The probability of not rejecting the null hypothesis H_0 drops as the difference between μ and μ_0 increases or n increases
- 1-$\beta(\mu)$ is called the **power function** of the test. It represents the probability of rejection of H_0 when the mean is μ



$$\beta(\mu) = \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

$$d = \left|\frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right|$$

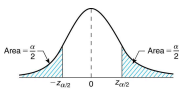
10

Choosing size

- What is the sample size n such that
  - the probability of not rejection H_0: μ=μ_0,
  - when the mean is actually μ_1,
  - is equal to some value β?
- We want to find n for which $\beta(\mu_1) = \beta$
  - but cannot solve in closed form for n in
  - $\beta = \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$
- An approximate solution is $n \approx \left(\frac{(z_{\alpha/2}+z_\beta)\sigma}{(\mu_1-\mu_0)}\right)^2$

11

Example



- An approximate solution is $n \approx \left(\frac{(z_{\alpha/2}+z_\beta)\sigma}{(\mu_1-\mu_0)}\right)^2$
- How many signals should be sent so that at the 95% conf level, the probability of rejection is at least 75%, if the true mean is 9.2? Assume, null hypothesis mean is 8, noise is normal with mean 0 and variance 4
  - $z_\beta = \text{norminv}(0.25) = 0.67$
  - $n \approx (1.96 + 0.67)^2 4/(9.2 - 8.0)^2 = 19.21$
- Hence we must send at least 20 signals. For this value
  - $\beta = \Phi\left(\frac{-1.2}{2/\sqrt{20}} + 1.96\right) - \Phi\left(\frac{-1.2}{2/\sqrt{20}} - 1.96\right) \approx 0.235$
  - We fail to reject 23.5% of the time

12

Underspecified parameters

13

GoF: Underspecified parameters

- In the dice example, the expected counts were clear. What if we do not have probabilities?
- Example from the book: Consider the number of accidents in an industrial plant over a 30 week period (mean is 95/30 = 3.17)
  - In such examples, we expect the rate λ of accidents to be constant
  - Ross suspects that this data comes from a Poisson distribution
- If the plant experiences an average of say, 3 accidents every 30 weeks, what is the probability that exactly 2 accidents occur during the next 30 weeks?

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

95 accidents over 30 weeks

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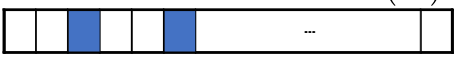
Binomial

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

- Plant experiences average of, say, 3 accidents/week (over 30 weeks)
- Divide each of the 30 weeks into n very small intervals
  - The probability of an accident in a particular interval is λ/n, where $\lambda = 3$
  - Accidents in different intervals occur independently.
  - The intervals are sufficiently short that the probability of two or more accidents in one interval is negligible
- The number of intervals containing an accident is then approximately

$$X_n \sim \text{Binomial}\left(n, \frac{\lambda}{n}\right)$$



15

Poisson

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

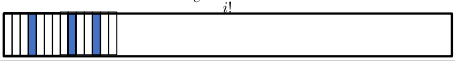
- # of intervals with accident approximately $X_n \sim \text{Bin}\left(n, \frac{\lambda}{n}\right)$
  - As n increases, the probability decreases, but expected number is constant
  - For exactly i accidents

$$P(X_n = i) = \binom{n}{i}\left(\frac{\lambda}{n}\right)^i\left(1-\frac{\lambda}{n}\right)^{n-i}$$

This is a discrete distribution

$$= \underbrace{\frac{n(n-1)\cdots(n-i+1)}{n^i}}_{\to 1}\frac{\lambda^i}{i!}\underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}}\underbrace{\left(1-\frac{\lambda}{n}\right)^{-i}}_{\to 1}$$

$$= e^{-\lambda}\frac{\lambda^i}{i!}$$



16

GoF: Underspecified parameters

- Consider the number of accidents in an industrial plant over a 30 week period (mean is 3.17)
  - Ross suspects that this data Y comes from a Poisson distribution
- Divide the number of accident into a **table of counts**
  - 0 accidents in a week $\Pr[Y=0] = e^{-\lambda} = e^{-3.17} = .042$
  - $T \approx 22.1$, df=3, $p \approx 0.00006$ Reject

$$= e^{-\lambda}\frac{\lambda^i}{i!}$$

df = k – 1 – 1 = 3

Accidents k	0	1	2 or 3	4 or 5	> 5
Observed O_i	6	5	8	6	5
Expected $E_i = 30\,p_i$	1.26	4.00	13.03	8.65	3.06
$(O_i - E_i)^2/E_i$	17.83	0.25	1.94	0.81	1.25

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

In general, df = k–1 –m with m underspecified parameters

17

Violated: Ross § 11.2 Each $np_i \geq 1$ and at least 80% of np_i should > 5

GoF: Illustrative data

- Poisson ($\hat{\lambda} = 3.0$) fit: $T \approx 2.73$, df=3, $p \approx 0.43$. Do not reject H_0
- For a Poisson distribution, mean = variance = λ, MLE = sample mean

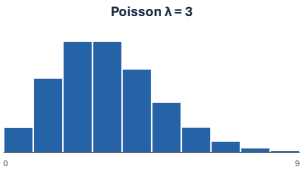
Accidents k	0	1	2 or 3	4 or 5	> 5
Observed O_i	3	4	12	7	4
Expected $E_i = 30\,p_i$	1.49	4.48	13.44	8.06	2.52
$(O_i - E_i)^2/E_i$	1.52	0.05	0.15	0.14	0.87

2	4	1	3	0	2	5	3	7	2
1	3	6	2	4	0	2	5	3	4
2	7	1	4	3	2	5	0	6	1

90 accidents over 30 weeks

19

Agenda



- Earlier
  - Poisson
  - Goodness-of-fit when probabilities are not fully specified
- Today
  - Multivariate Gaussian

1

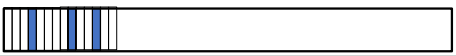
Review: Poisson

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

- # of intervals with accident $\approx X_n \sim \text{Bin}\left(n, \frac{\lambda}{n}\right)$, Rate $\lambda = 3.17$
  - For exactly i accidents per week
  - MGF $= e^{-\lambda} e^{\lambda e^t}$, $\mu = \sigma^2 = \lambda$

$$P(X_n = i) = e^{-\lambda} \frac{\lambda^i}{i!}$$

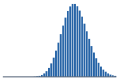
- Suppose we start with X ~ Bernoulli (p),
  - With CLT, sums $X_n = \sum X_i =$ Binomial (n, p), for large n, we get a normal
  - Here, as n increases, we get a Poisson
    - Crucial fact: the probability decreases, but np = constant,



2

Binomial Approximations

Poison: λ = 30:



- Fixed probability: 0 < p < 1
  - Both μ and σ^2 grow unbounded
  - Upon standardization approaches Z
  - As n increases, each B_i is the same
- Fixed $\lambda > 0$
  - No standardization
  - $B_{n,j}$ indep $\text{Ber}(\lambda/n), j = 1, \ldots, n$
    - As n increases, each B_j is different
- But! Large λ, the Poisson will tend to normal
- Many opportunities ≠ many actual events

$$X_n \sim \text{Bin}(n,p), \quad E[X_n] = np, \quad \text{Var}(X_n) = np(1-p)$$

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0,1)$$

$$p_n = \frac{\lambda}{n}, \quad E[X_n] = \lambda, \quad \text{Var}(X_n) = \lambda\left(1 - \frac{\lambda}{n}\right) \to \lambda.$$

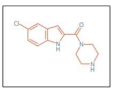
$$P(X_n = i) = \binom{n}{i}(\lambda/n)^i(1-\lambda/n)^{n-i} \to e^{-\lambda}\frac{\lambda^i}{i!}$$

$$X_n = \sum_j B_{n,j}$$

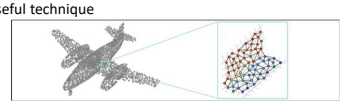
$$\frac{X - \lambda}{\sqrt{\lambda}} \xrightarrow{d} N(0,1) \quad (\lambda \to \infty)$$

3

Multivariate Normal



- Why multivariate? The world is full of interesting probability problems…
  - and many of them involve multiple random variables, being random together
  - ML course coming up
- Why Normal?
  - The central limit theorem generalizes
  - Visualization requires us to drop to 3 dimensions
    - Principal Component Analysis (PCA) is a useful technique
- OK, what is a multivariate normal?



4

Quantitative Scalar and Vector outcomes

- Earlier: Random Variable
  - One trial produces a scalar: Binomial, Poisson, Normal
  - Multiple trials produced more data, enabling, e.g., MLE estimates
- Also earlier:
  - One set of data produced vector outcomes also
  - $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
    - Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
    - $B = \arg\min_\beta \sum_{i=1}^{n} (Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij})^2$
    - estimates $\beta=(\beta_0,\ldots,\beta_k)^T$
- Vector (multivariate) random variables $B=(B_0,\ldots,B_k)^T$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

5

Vector (multivariate) random variables
$B=(B_0,\ldots,B_k)^T$

Are these “normal” multivariates?

What does it mean to say that this vector is normal?

6

Scalar Estimates

- The table below reports multiple samples, but the random variable is a scalar
- Sample mean and sample variance
  - $\bar{x} = 2.5, s^2 = (2.25 + 0.25 + 0.25 + 2.25)/3 = 5/3$

k	1	2	3	4
$\mathbf{u_k}$	1	2	3	4
$\mathbf{u_k - \bar{u}}$	−1.5	−0.5	0.5	1.5

7

Two variables

- We also looked at two random variables U and V
- The table below reports *paired* observations of the random vector of dimension d=2
  - $\bar{u} = \bar{v} = 2.5 \quad {s_U}^2 = \frac{2.25+0.25+0.25+2.25}{3} = \frac{5}{3}$ (as before)
  - but we also have the sample covariance $s_{UV} = \frac{0.75+0.75+0.75+0.75}{3} = 1$
  - We can organize these numbers in the form of a matrix

$$\Sigma = \begin{pmatrix} \frac{5}{3} & 1 \\ 1 & \frac{5}{3} \end{pmatrix}$$

k	1	2	3	4
$\mathbf{u_k}$	1	2	3	4
$\mathbf{v_k}$	2	1	4	3

8

Stacking and Linear Combinations

- Stacking results in a vector
  - One row: observations of one variable
  - One column: one stacked realization

$$X = \begin{bmatrix} U \\ V \end{bmatrix}$$

- One can also create a fixed linear combination of the rows
  - That will result in a scalar random variable
    - Example: W = 2U-V (Observations: 0 3 2 5)
  - Linear combination of *independent* (even non identical) normals is (exactly) normal
    - $U_i \sim \mathcal{N}(\mu_i, {\sigma_i}^2)\ W = \sum_{i=1}^{d} a_i\, U_i \sim \mathcal{N}\left(\sum_{i=1}^{d} a_i\, \mu_i, \sum_{i=1}^{d} {a_i}^2\, {\sigma_i}^2\right)$
- Multivariate normals will also involve linear combinations but this is not it

9

Covariance Matrix in d dimensions

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_d \end{bmatrix}$$

- In d-dimensions, we collect individual variances and pairwise covariances to form a $d \times d$ matrix called the covariance matrix
  - This will be central to the multivariate normal
  - We can write this as $\mathrm{Cov}(X) \equiv \mathrm{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^{\mathrm{T}}]$
  - The matrix is symmetric because covariance is symmetric
  - The matrix is also positive semi-definite
    - Why?

$$\mathbf{X}, \mathbf{a}, \boldsymbol{\mu} \in \mathbb{R}^{d \times 1} \quad \mathbf{C} \in \mathbb{R}^{d \times d}$$

$$W = \mathbf{a}^{\top}\mathbf{X} = \sum_{i=1}^{d} a_i X_i \quad W \in \mathbb{R}$$

$$\mathrm{Var}(W) = \mathbb{E}\left[\left(\mathbf{a}^{\top}(\mathbf{X} - \boldsymbol{\mu})\right)^2\right]$$

10

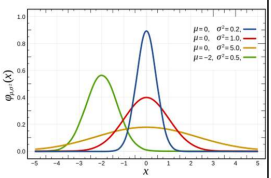
Covariance matrix

- The expression is true for any vector $\mathbf{a}$ so C is positive semi-definite
- We will also require C to be positive definite
  - $\mathbf{C} > 0 \Leftrightarrow \mathbf{a}^{\mathsf{T}}\mathbf{C}\mathbf{a} > 0 \quad \forall \mathbf{a} \neq \mathbf{0}$
  - The inverse exists and is also positive definite

$$
\begin{aligned}
\mathrm{Var}(W) &= \mathbb{E}\left[\left(\mathbf{a}^T(\mathbf{X}-\boldsymbol{\mu})\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{i=1}^{d} a_i(X_i-\mu_i)\right)^2\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j (X_i-\mu_i)(X_j-\mu_j)\right] \\
&= \sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j \mathbb{E}\left[(X_i-\mu_i)(X_j-\mu_j)\right] \\
&= \sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j \mathrm{Cov}(X_i, X_j) \\
&= \mathbf{a}^T\mathbf{C}\mathbf{a}
\end{aligned}
$$

11

$$g(x|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Density for multivariate Normal

When we wrote the definition of the univariate normal

- we first wrote the distribution with some parameters μ and σ^2
- Then showed that this was the mean and variance of the distribution of the random variable X

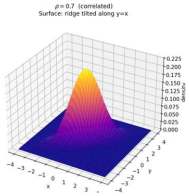
12

Multivariate normal density

$$\Delta^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x}-\boldsymbol{\mu})$$

- Parsing the symbols
  - Capital **X**, Small (**x**), **μ**: all vectors
  - **x** is the point at which the density is evaluated
  - C is a matrix, its inverse is deemed to exist, it is deemed to be positive definite
- Check out what happens when
  - d=1
  - X consists of *independent* standard normal Z_i

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$



$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-\frac{d}{2}} \exp\left(-\frac{1}{2}\sum_{i=1}^{d} x_i^2\right) = \prod_{i=1}^{d} g(x_i)$$

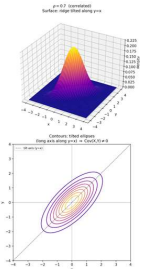
13

Geometry

$$\Delta^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x}-\boldsymbol{\mu})$$

- The numerator is the square of the Mahalanobis distance
  - The density is constant for all points on which the quadratic form
  - But the quadratic form is constant implies the iso-contour is an ellipse! (in 2D)
    - Direction of eigenvectors tells us the orientation of the ellipse
    - Magnitude of the eigenvector tells us the amount of variance

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Typo: eigenvalues

14

Geometry

$$P = (x_1, \ldots, x_p)^T, \quad Q = (y_1, \ldots, y_p)^T$$
$$d(P,Q) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_p - y_p)^2} = \sqrt{(x-y)^T(x-y)}$$

- Euclidean distance
  - Does not make sense when the axes refer to different features (example 3cm difference in screen size vs cost)
- Mahalanobis distance
  - 1D: We subtract the mean and scale by standard deviation
  - Inverse covariance for higher dimension

Phone	Screen (cm)	Price (₹)
A	15	20,000
B	18	23,000

$$\left(\frac{x_i - \mu_i}{\sigma_i}\right)^2$$

$$\begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} \qquad \begin{pmatrix} 25/9 & -20/9 \\ -20/9 & 25/9 \end{pmatrix}$$

$$r_M{}^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$
$$r_M{}^2(1,1) = \frac{10}{9} \qquad r_M{}^2(1,-1) = 10$$

15

Summary: Multivariate Normal

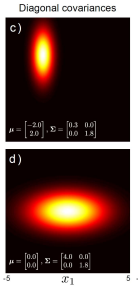
- Equivalent definitions
  1. Joint density of **X** as defined
     - The mean and covariance of random vector is **μ** and **C**
  2. Every linear combination of elements of **X** is normally distributed
  3. **X = AZ + b** for some i.i.d standard normal, an invertible A and a column vector b
- Definition: The (scalar) random variables in the stacked vector are said to be *jointly normal*

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

20

Agenda

$$\Sigma_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$



- Earlier
  - Goodness-of-fit when probabilities are not fully specified
  - Multivariate Normal
- Today
  - Meta content
  - Goodness of fit for exponential distribution
  - A word on PCA and t-SNE?
  - Epilogue ??

1

Review

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

- Can GoF: (Table of counts idea) work with *continuous* distribution?
  - Unlike discrete tables such as Poisson, there are no natural categories
- To motivate this:
  - We know how many accidents on a weekly basis but not when
  - If accidents occur at a constant rate (Poisson), what is the "waiting" time between successive accidents?
    - Accidents at a constant average rate of λ accidents per day (earlier per week)
    - Take the unit to be one day
    - N(t) = the number of accidents in a time window of length t days
    - T = the waiting time (in days) from a starting moment until the next accident

2

Exponential Distribution

- Waiting time, *T* random variable. Distribution == ?
- For the gap between accidents, start the clock at an accident
  - Same Binomial argument as before but with window of length t
  - $P(N(t) = k) = e^{-\lambda t}(\lambda t)^k/k!$
  - Therefore, the probability that the first accident occurred after time *u* is probability that there was no accident in the time interval *(0,u)*
    - $P(T > u) = P(N(u) = 0) = e^{-\lambda u}(\lambda u)^0/0! = e^{(-\lambda u)}$
    - $P(T \leq u) = 1 - P(T > u) = 1 - e^{-\lambda u}$
- This is the exponential distribution with **rate** λ with pdf $f_T(u) = \lambda e^{-\lambda u}, u \geq 0$
  - Continuous analog of waiting for the first "success" (geometric distribution)

3

$$P(T > s + u \mid T > s) = \frac{P(T > s + u)}{P(T > s)} = \frac{e^{-\lambda(s+u)}}{e^{-\lambda s}} = e^{-\lambda u} = P(T > u)$$

Table of counts (continuous)

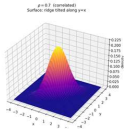
- 210 days of data. We know how many accidents on a weekly basis
  - We are now given finer record (timestamp of accidents): n = number of gaps 90, rate λ= 90/210 = 0.4286 (i.e., average "wait time" of 2.33 days)
  - Null H_0: Is T exponential?? Divide into 4 bins. Bin probabilities: $P(a < T <= b) = e^{-\lambda a} - e^{-\lambda b}$, $x_0^2 \approx 1.64$, 2 df, p−value ≈ 0.44. Do not reject.
- Memoryless property: If you have already waited s days with no accident, the chance of waiting at least u more days is the same as if you had just started waiting

We chop the real line into intervals for table of counts

Gap (days)	0–1	1–2	2–4	≥ 4
Observed O_i	29	25	19	17
Expected $E_i = 90\,p_i$	31.37	20.44	21.99	16.21
$(O_i - E_i)^2/E_i$	0.18	1.02	0.41	0.04

4

Review: Multivariate Normal



- Equivalent definitions
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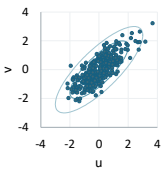
  3. **X = AZ + b** for some i.i.d standard normal, an invertible A and a column vector b
- Nomenclature: The (scalar) random variables in the stacked vector are said to be *jointly normal*

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

5

Caution: Independence not given

- Consider $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$
  - $U = Z_1 \;\; V = 0.8Z_1 + 0.6Z_2$
  - This is a linear combination and hence U is jointly normal



$$\mathbf{C} = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$

6

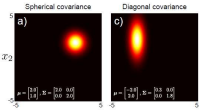
Diagonal Covariance = Independence

$$\Pr(x_1, x_2) = \frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}\begin{pmatrix} x_1 & x_2 \end{pmatrix} \boldsymbol{\Sigma}^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right]$$

$$\Pr(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{1}{2}\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} \sigma_1^{-2} & 0 \\ 0 & \sigma_2^{-2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right]$$

$$\Pr(x_1, x_2) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{x_1^2}{2\sigma_1^2}\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{x_2^2}{2\sigma_2^2}\right]$$

$$\Pr(x_1, x_2) = \Pr(x_1)\Pr(x_2)$$



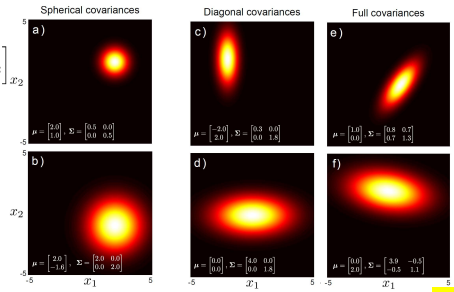
$$\boldsymbol{\Sigma}_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

7

Review

$$\boldsymbol{\Sigma}_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$\boldsymbol{\Sigma}_{\text{spher}} = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix}$$



$$\boldsymbol{\Sigma}_{\text{full}} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{21}^2 & \sigma_{22}^2 \end{bmatrix}$$

We will assume joint normal

8

Review: Marginal Distributions

We will assume joint normal

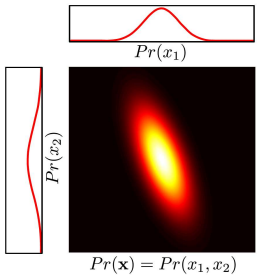
- Marginal distributions of a multivariate normal are also normal

$$\Pr(\mathbf{x}) = \Pr\left(\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right)$$

$$\Pr(\mathbf{x}) = \text{Norm}_{\mathbf{x}}\left[\begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{21}^T\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{bmatrix}\right]$$

$$\Pr(\mathbf{x}_1) = \text{Norm}_{\mathbf{x}_1}[\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}]$$

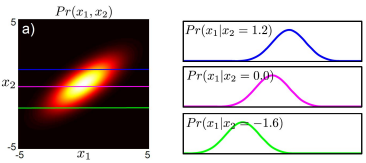
$$\Pr(\mathbf{x}_2) = \text{Norm}_{\mathbf{x}_2}[\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22}]$$



9

Conditional Distributions

- Conditional distributions of a multivariate normal are also normal



$$\Pr(\mathbf{x}) = \Pr\left(\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right) = \text{Norm}_{\mathbf{x}}\left[\begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}^T\\ \boldsymbol{\Sigma}_{12} & \boldsymbol{\Sigma}_{22}\end{bmatrix}\right]$$

We will assume joint normal

$$\Pr(\mathbf{x}_1 \mid \mathbf{x}_2) = \text{Norm}_{\mathbf{x}_1}\left[\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}^T\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}^T\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{12}\right]$$

$$\Pr(\mathbf{x}_2 \mid \mathbf{x}_1) = \text{Norm}_{\mathbf{x}_2}\left[\boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{11}^{-1}(\mathbf{x}_1 - \boldsymbol{\mu}_1), \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}^T\right]$$

10

Visualization: Projecting High Dimensional Data

- When a multivariate dataset has a large number of variables, analysis and interpretation of the data may be hard.
  - Too many variable pairs, so pairwise correlation may be hard to grasp.
- For convenient visualization and interpretation
  - Reduce the number of variables.
- How to reduce number of variables while capturing most of the information in the data
  - Information (variance) PCA, Factor analysis, Random Projection, t-SNE

11

T-distributed stochastic neighbourhood embedding

- Another data projection method, specifically designed for visualizing high dimensional data in two dimensions.
- Uses a probability distribution in latent space that has heavier tails
  - Eliminates the unwanted attractive forces between points that are relatively far in the high dimensional space.
  - Preserves local similarities and clusters better than PCA
  - Creates non-linear projection
  - Reference: Kevin Murphy Section 20.4.10 Probabilistic ML book

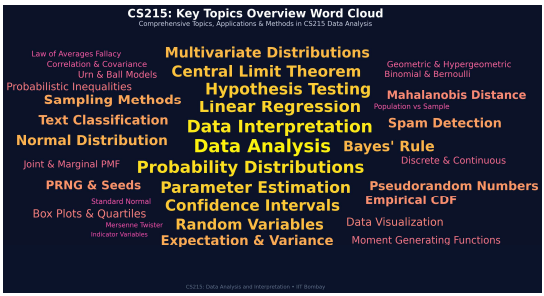
13

What topics have we discussed?

- Up to chapter 6 in the textbook (Sheldon Ross)
- What is the syllabus for the exam?
  - Coming up

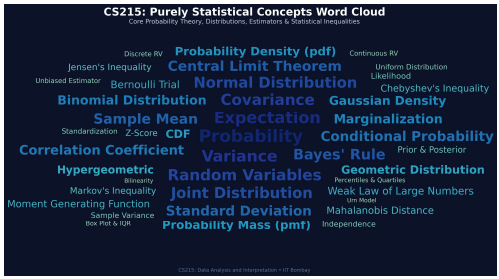
2

We have come a long way



3

We have come a long way



4

Let’s revisit the first week of classes

5

FOUNDATIONS

Population

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

- Suppose someone (an "oracle") told you that values are
  - Only (W= 0 or W= 2 or W= 4)
  - What's the mean and variance?
  - Expected value: $\mu = 2$
  - Variance as $\sigma^2 = 8/3$

Basic "first order" knowledge you must have

1

6

FOUNDATIONS

Population and Observations

POPULATION

The full process or distribution

Parameters: μ and σ^2
Fixed—but usually unknown

observe

SAMPLE

Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and S^2
Change from sample to sample

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

7

Basic "first order" knowledge you must have

Example

- The weights of men has a mean of 70 kg with a variance of 2 kg
- Want mean and variance of the *total of 3 randomly* sampled men?
  - Assume independence, weights are normally distributed, let X be the weight of a randomly sampled man and T the total weight
  - A student wrote: T = 3X
  - Using the properties discussed in class, $E[T] =?$ $\text{Var}[T] = ?$

$$X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

8

FOUNDATIONS

Population and Observations

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Parameters: μ and σ^2
Fixed—but usually unknown

observe

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Observed values: $x_1, \ldots, x_n$

Statistics: $\bar{x}$ and S^2
Change from sample to sample

$$\overline{x} = \frac{\sum x_i}{n} \qquad \frac{\sum (x_i - \overline{x})^2}{n-1}$$

Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

11

Focusing on the first column, how do we use the language of mathematics to express the first column

Experiments

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \bar{b})^2}{n-1}$	$\frac{\sum (x - \bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1$ = 0	0
0 2	1	$((0-1)^2+(2-1)^2)/1$ = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Does the first column represent
  - A: Two samples from one random variable W?
  - B: One sample each from two instances of the random variable W?

12

Focusing on the first column, how do we use the language of mathematics to express the first column

Experiments

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \bar{b})^2}{n-1}$	$\frac{\sum (x - \bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1$ = 0	0
0 2	1	$((0-1)^2+(2-1)^2)/1$ = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

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13

FOUNDATIONS

Population and Observations

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Parameters: μ and σ^2
Fixed—but usually unknown

observe

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Observed values: $x_1, \ldots, x_n$

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Preparing for later courses: Inference on the population from the data

Basic "first order" knowledge you must have

1

14

Focusing on the first column

Experiments

All possible samples of size 2	$\overline{b} = \frac{\sum x_i}{n}$	$\frac{\sum (x - \overline{b})^2}{n-1}$	$\frac{\sum (x - \overline{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1$ = 0	0
0 2	1	$((0-1)^2+(2-1)^2)/1$ = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Assume the sample was of size 1
- What is the statistician's view of the expected value of the random variable W?

15

The weak law of large numbers

- Let X_1, X_2, …, X_n be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

16

16

Experiment

Now focusing on the second column

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size 2:
  - What will be statistician view of the expected value of the random variable?

17

Experiment

Now focusing on the last two columns

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	((0-0)²+(0-0)²)/1 = 0	0
0 2	1	((0-1)²+(2-1)²)/1 = 2	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

18

CLT

- As stated, the random variables X_i are independent and identically distributed.
- Alternate: require independence but allow the random variables to belong to different distributions
  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
  - "take home message" … it works

○ Misconceptions

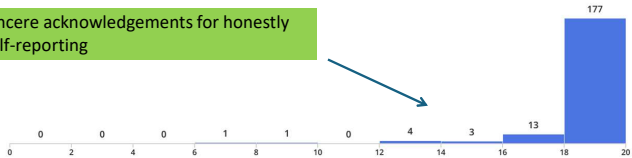
- "Lots" of random sampling "reveals" normal distribution
- Does not apply to random sampling of any variable
  - Applies to sampling distribution of mean
- Empirical rule of thumb (n=30) can lead to seriously flawed inferences

79

24

Part A results

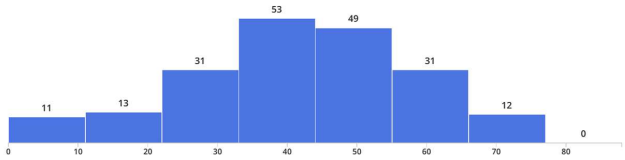
Sincere acknowledgements for honestly self-reporting



Minimum	Median	Maximum	Mean	Std Dev
6.0	20.0	20.0	19.01	1.92

2

Part B



Minimum	Median	Maximum	Mean	Std Dev
0.0	42.0	73.0	40.95	16.36

3

Part B Instructions

- You may assume any result in the Piazza-provided lecture handouts as "known", i.e., not to be proved.
- Conversely, you cannot assume results
  - that have not been presented in class or posted in the handouts.
  - that are implicit or explicit in homework solutions
  - Such results are to be derived if needed during the exam.
- All questions require precise and complete mathematical arguments to earn full credit

4

Agenda

- Midsemester Exam
  - Self evaluation
  - Menti
  - Question
  - Answers
    - Piazza: common mistakes, imprecise answers, incorrect answer, partial marking
  - Results
  - Display of answer papers
    - Regrade request process
    - Friday, Students with Low Marks,

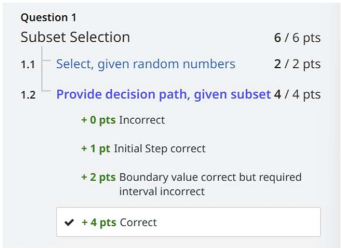
5

Subset Selection

- {B,D,A,E,C,F,G,H}
  - 0.52, 0.57, 0.66, 0.30, 0.80, 0.90, 0.25, 0.70
- Correct subset {D,E,G,H}
  - 4/8, 4/7 ~= 0.571, 3/6, 3/5, 2/4, 2/3
- Decision path for subset {D, E, F, H}
  - [0.50, 1): B Not selected, (0, 4/7): D: Selected, [0.5, 1) A not selected
  - (0, 3/5): E: Selected, [0.5, 1): C: Not Selected, (0, 0.66) F: Selected
  - [0.50, 1): G Not selected, H: Selected (Forced)

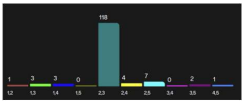
6

Grading Interface



7

Results



8

Attendance requirement

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)

Student	Classes attended
A	16
B	18
C	20

9

Attendance requirement

Student	Classes attended
A	16
B	18
C	20

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)
- Student A will not meet threshold if he attends 0 classes, or 1 classes, or 2 classes, and so on up to 17 classes since 36-16-2=18
- P(X <= 17) = 0.808 which is greater than the 4% threshold specified

10

Results

11

Red and Blue Balls in Urns

- One of two urns, A or B, is chosen with equal probability. Urn A contains 3 red balls and 1 blue ball, while urn B contains 1 red ball and 3 blue balls. Balls are drawn independently **with replacement** from the chosen urn.
- If the first two balls drawn are red, find the probability that urn A was chosen.
- Suppose that among the first n draws, exactly r are red. Find the condition on r and n under which urn A is more likely than urn B.

12

Results

13

Jenny and IPL franchise

- X and Y are independent normal with the same mean (70) and variance (64)
  - Want to understand S = X+Y and D=X-Y
- Find $\mathbb{E}[e^{uS+vD}]$ and simplify (*state assumptions you make*)
- Find $\mathrm{Cov}(S, D)$
- Find $\Pr(D \geq 8 | S > 150)$. Unjustified assumptions receive no credit

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Results

4: IPL	13.0 points	
4.1: Expected Value of e^(uS + vD)	6.0 points	65%
Click to Add Tags		
4.2: Cov(S,D)	3.0 points	79%
Click to Add Tags		
4.3: Probability of difference > 8 given sum > 150	4.0 points	0%
Click to Add Tags		

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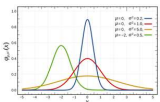
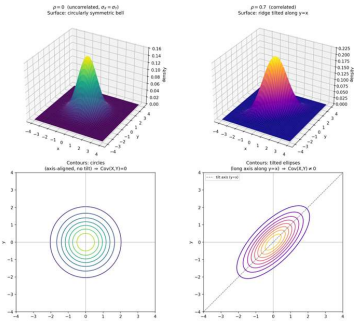
Joint distributions of S and D





In both cases, any slice of the "Mexican" hat will give us a nice Gaussian

16

Bounding probability of the binomial

- We know the pmf of the binomial, but what about the cdf?
- Simpler question: What is the probability that in *n* trials we get more than *np* successes (e.g. *nq*, with $0 < p < q < 1$)
  - Standard Markov $\Pr(S_n \geq nq) \leq \frac{\mathbb{E}[S_n]}{nq}$
  - With p = 0.2, q=0.5, direct Markov gives 0.4 (no dependance on n)
  - With $np < a < n$ and $q = \frac{a}{n}, p < q < 1$. Show
    - $P(S_n \geq nq) \leq \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n$
- This is the tightest Chernoff bound (in the example 0.8^n)

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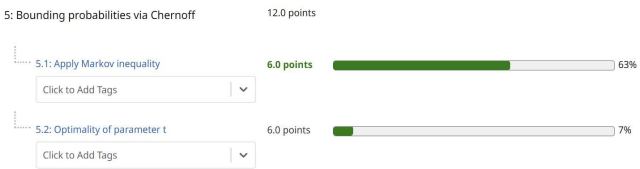
Results



18

Stick

- Stick broken into two: X = U(1-U)
  - Expectation, Variance, Covariance (X, U)

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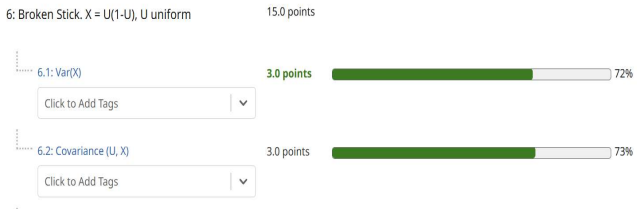
Results



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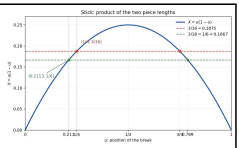
Stick



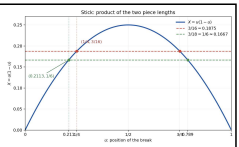
- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Umm X = f(U) that sounds suspiciously like not independent
    - Want $\Pr(X = a|U = u) \stackrel{?}{=} \Pr(X = a)$?
  - We want to look for events that knowing something about X makes U to be constrained, it can't be whatever it wants
    - Argue about the event $\Pr(X \geq 3/16|U < 1/4)$

21

Results

22

Stick



- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Wait, there is more! show that X and Y are independent
    - Y=1 if U < 0.5
    - This appears to be unexpected!

23

Results

6.4: Independence of X and Y **(only 99% graded)** 6.0 points 10%

Click to Add Tags

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Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathbb{E}[S]$
  - Distribution of N

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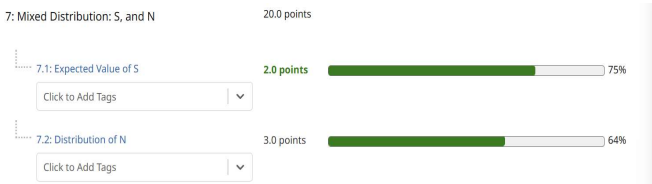
Results



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Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathrm{Cov}(S, N)$
  - $\mathbb{E}[S \mid \mathrm{N} = \mathrm{k}]$
  - $\mathrm{Var}(S|N = k)$

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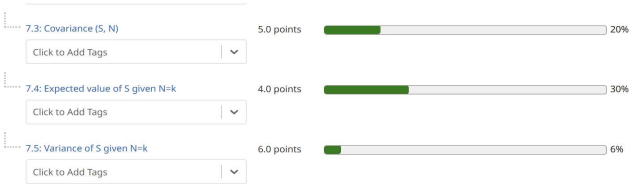
Results



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Chebychev

- X is random with expectation μ and finite variance
- Jensen's inequality
  - For a convex function, $f\big(\mathbb{E}(x)\big) \leq \mathbb{E}[f(x)]$
- For $0 < \theta < 1$, $\Pr(X \leq \theta\mu) \leq$?

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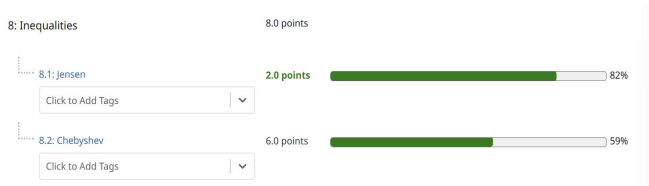
Results



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Agenda

- Midsemester Exam
  - Self evaluation
  - Menti
  - Question
  - Answers
    - Piazza: common mistakes, imprecise answers, incorrect answer, partial marking
  - Results
  - Display of answer papers
    - Regrade request process
    - Friday, Students with Low Marks,

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